

Matteo Matteucci

Curriculum Vitae et Studiorum

Name: Matteo Matteucci
Date/Place of Birth: 23/04/1974, Nuoro (Italy)
Affiliation: Politecnico di Milano
Dip. di Elettronica, Informazione e Bioingegneria
Address: Via Ponzio 34/5, I-20133, Milan (Italy)
E-mail: matteo.matteucci@polimi.it
Phone/Fax: +39 02 2399 3470 / 3411
H-Index / i10-index / Citations: 50 / 199 / 11109 ([Source: Google Scholar](#) 05/10/2025)
37 / -- / 6067 ([Source: Scopus.com](#) 05/10/2025)



One Pager	3
Biosketch	4
Education	4
Academic Life	5
Academic Positions	5
Academic Roles	5
Affiliations to Scientific Associations	6
Evaluation Committees	6
Prizes and Awards	7
Other	7
Teaching activity	7
Lecturer and Teaching Assistant (PhD, MS, BS)	7
Other Teaching Activities	9
Supervision and Revision of PhD Theses	10
Advised PhD Students	10
Co-Advised PhD Students	11
International Editorial Activities	12
International Committees	12
International Journal and Conference Editorial Role	12
International Conferences and Workshops Program Committees	12
Reviewer for International Journals and Magazines	13
Reviewer for International Conferences and Workshops	14
Session Chair and Co-Chair	14
Conference and Workshop Organization	15
Projects & Fundings	15
Research Projects	15
Industrial Projects and Contracts	17
Professional Activities	19
Research Statement	19
Research Achievements, Results, and Products	19
MRT: the Milan RoboCup Team	19
LURCH: an Autonomous Wheelchair with Multimodal Interfaces	20
Simultaneous Localization and Mapping in Autonomous Robots and Unmanned Vehicles	20

Benchmarking and Good Experimental Methodologies in Autonomous Robots	21
P300 + ErrP BCI: a Self-Correcting Brain-Computer Interface	21
Bioinformatics, Biosignal Interpretation, and Affective Computing	21
VeTRA: a Model-based Multi Camera Vehicle Tracking System	22
Geometry of Information in Estimation of Distribution Algorithms	22
Invited Talks and Papers	22
Full List of Publications	24
A. Articles in International Journals	24
B. Articles and Chapters in International Books	28
C. National and International Patents	29
D. Papers in Proceedings of International Conferences and Workshops	30
E. Articles in Collections	40
F. Articles in National Journals	41
G. Technical Reports	41
H. Accepted, Under Review, and arXiv preprints	42
Appendix: CV Summary & Funding Record	44

Autorizzo il trattamento dei miei dati personali presenti nel cv ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 “Codice in materia di protezione dei dati personali” e del GDPR (Regolamento UE 2016/679).

Date: 05/10/2025



In compliance with the Italian legislative Decree no. 196 “Codice in materia di protezione dei dati personali” dated 30/06/2003, and the GDPR (EU 2016/679). I hereby authorize you to use and process my personal details contained in this document.”

Date: 05/10/2025



My research focuses on the use of techniques and models from Pattern Recognition, Machine Learning, Signal Processing, and Artificial Intelligence to deal with uncertainty in autonomous physical & software systems perception and intelligent data analysis. Autonomous robots, unmanned vehicles, agricultural and assistive technologies have been the reference scenarios for developing models and algorithms to cope with uncertainty and incomplete knowledge.



H-Index/i10-index/Citations: 50 / 199 / 11109 ([Source: Google Scholar](#) 05/10/2025)
37 / -- / 6067 ([Source: Scopus.com](#) 05/10/2025)

Reviewed Publications: 60+ in Intl. Journals, 25+ in Intl. Books, 150+ in Intl. Conference Proceedings.

International Activity: Visiting Scholar for 1 year at Carnegie Mellon University graduating with a *Master of Science in Knowledge Discovery and Data Mining*. Active since its foundation in the Euron Special Interest Group on Good Experimental Methodologies & Benchmarking (GEM SIG) for the design of suitable benchmarks and experimental methodologies for intelligent systems. *Expert on Benchmarking* for the RoSta and on ELS (Ethical-Legal-Societal) issues for the euRobotics FP7 projects. Active in the IEEE RAS Standard Group in the "IEEE 1873-2015 Standard for Robot Map Data Representation" and "IEEE P2751 - 3D Map Data Representation for Robotics and Automation". *Member of IFAC Technical Committee* "7.5. Transportation and Vehicle Systems - Intelligent Autonomous Vehicles". Program Committee of more than 15 International Conferences and Workshops. Reviewer for 20 International Journals (8 IEEE Transactions) and more than 20 International Conferences & Workshops.

Project Funding: *Project Coordinator of FP6 European project RAWSEEDS, National Scientific Coordinator of PRIN 2009 project ROAMFREE, Co-coordinator of H2020 project AI-SPRINT, POLIMI Principal Investigator for FP7 project RoCKIn, POLIMI Principal Investigator for H2020 projects RoCKEU2, SciRoc, DeepField, and METRICS, POLIMI Principal Investigator & Project Technical Manager of AAL Joint Program project ALMA, and H2020 AI-SPRINT. Task Leader in the TEINVEIN project by Regione Lombardia. Coordinator of "Brain-Computer Interfaces in Everyday Applications" Politecnico di Milano & Regione Lombardia grant.*

Research achievements/products: Autonomous wheelchair to support disabled people mobility (awarded with Antonio D'Auria SIRI Prize, and Intesa Startup Initiative); benchmarking toolkit for simultaneous localization and mapping (used at international level); Brain-Computer Interface based on P300 and ErrP potentials (presented to general public and reported by national and international press); sensor fusion framework for unmanned vehicles. Co-author of "*General Guidelines for Robotics Papers Involving Experiments*" (Euron GEM SIG), "*Suggestion for a green paper on legal issues in robotics*" (euRobotics ESL SIG), "IEEE1873-2015 Standard for Robot Map Data Representation".

Technology Transfer: 8 patents applications (3 Italian + 1 European + 3 EU extension + 1 US extension), 1 spin-off company founded with a PhD student (Nova Labs), 1 start-up company founded by PhD students (Empatica), 2 tech-transfer award (Antonio D'Auria, Intesa SanPaolo "Startup Initiative"), several industrial contracts (e.g., Aerosekur, Indesit, Gaiotto, Infosolution, Noustat, Aermatica, Aster, ENI Progetti, ABB SpA, Tracetoo, Edison, Antemotion, Pirelli).

Awards and Prizes: Rotary Ambassadorial Scholarship; Dimitris N. Chorafas Foundation Award for best PhD thesis; the autonomous wheelchair project has been awarded with "Antonio D'Auria" and the Intesa San Paolo "Startup Initiative" prizes; advisor of Luigi Malagò awarded with the Dimitris N. Chorafas Foundation Award for best PhD thesis; tutor of the "Helios" project awarded as the best project of the first 6 years of Alta Scuola Politecnica, and of the "IRoKi" project, which was another of the 6 finalists. *Best papers at WACV 2016, CVPRW 2016, and CVPRW 2019.*

Lecturer (Grade): Soft Computing (PhD & MS), 3D Structure from Visual Motion (PhD), Advanced Topics in Deep Learning (PhD), Unmanned Autonomous Vehicles (PhD), Artificial Neural Networks and Deep Learning (MS), Robotics (MS & BS), Machine Learning (MS), Cognitive Robotics (MS), Pattern Analysis & Machine Intelligence (MS), Uncertainty in Artificial Intelligence (MS), Data Analysis for Smart Agriculture (MS), Knowledge Engineering (BS), Fondamenti Informatica (BS).

Multidisciplinary research: I pursue multidisciplinary both in the narrow sense of "within engineering" (i.e., Pattern Recognition, Machine Learning, Signal Processing, Robotics, Autonomous Driving), and in the broader sense of "involving other disciplines" (e.g., Neurosciences, Geology and Environmental Science, Medicine, Transportation Science, Bioengineering, Mathematics, Social Sciences, Genomics).

Matteo Matteucci ("Laurea" 1999, MS 2002, Phd 2003) is Associate Professor at Dipartimento di Elettronica Informazione e Bioingegneria of Politecnico di Milano, Italy. In 1999 he got a Laurea degree in Computer Engineering at Politecnico di Milano, in 2002 he received a Master of Science in Knowledge Discovery and Data Mining at Carnegie Mellon University (Pittsburgh, PA), and in 2003 he got a Ph.D. in Computer Engineering and Automation at Politecnico di Milano (Milan, Italy).

His main research topics are pattern recognition, machine learning, machine perception, robotics, computer vision and signal processing. His main research interest is in developing, evaluating and applying, in a practical way, techniques for adaptation and learning to autonomous systems interacting with the physical world. He has co-authored more than 50 (peer-reviewed) papers on international journals, 25 papers in International Books, and more than 150 (peer-reviewed) contributions to international conferences and workshops. He has been principal investigator in national and international funded projects on machine learning, autonomous robots, sensor fusion and benchmarking of autonomous and intelligent systems.

His research in robotics started with the Milan Robocup Team, a RoboCup team of six soccer robots equipped with custom panoramic vision sensors, adaptive color classification algorithms, and a conceptual model to integrate robot perception with information from teammates. Then he has worked on a complete 6DoF SLAM (i.e., Simultaneous Localization and Mapping) system based on multi-camera vision and hierarchical map decomposition through conditionally independent filtering. He is now working on sensor fusion in SLAM to fuse odometric estimates coming from different sensors (e.g., inertial measurements, laser range finder, single and multiple cameras, etc.) and semantic information coming from deep neural networks into an accurate and robust perception of the robot environment.

He has been involved in the Euron Special Interest Group on Good Experimental Methodologies and Benchmarking and he is in the IEEE RAS Standard Group on "Robot 3D Map Data Representation" for Navigation. He has been the Coordinator of the European project RAWSEEDS (<http://www.rawseeds.org>) a Specific Support Action for the development of a benchmarking toolkit for multi-sensor SLAM algorithms. He has been the National Scientific Coordinator of the ROAMFREE project (<http://roamfree.dei.polimi.it>) for the development of methods for the robust estimation of robot odometry by sensor fusion funded by the Italian Ministry for the University and the Research (MIUR) under the PRIN 2009 program. He has been the Politecnico di Milano Principal Investigator in the FP7 project RoCKIn (<http://rockinrobotchallenge.eu/>) for the development of 2 robot benchmarking competitions and Project Technical Manager of the European project ALMA (<http://www.alma-aal.org>), funded under the AAL Joint Program, for the realization of an Ambient Assisted Living system to support the autonomous mobility of the elderly. He has been Principal investigator for Politecnico di Milano of the H2020 project RockEU2 and for the development of the European Robotics League (https://www.eu-robotics.net/robotics_league/), H2020 project SciRoc (<https://sciroc.eu/>) for the development of a novel Smart City and Robotics competition, H2020 project DeepField for the study of deep learning techniques in field robotics, H2020 project METRICS (<https://metricsproject.eu/agri-food/>) for the development of novel robotics benchmarks and competitions, especially in the agricultural field. He has been co-coordinator of H2020 project AI-SPRINT (<https://www.ai-sprint-project.eu/>) for the development of tools for AI in the edge-cloud computing continuum.

Education

- 2000 - 2003 PhD in Computer Engineering and Automation at the Department of Electronics and Information of Politecnico di Milano (17/03/2003, Milan, Italy) [PhD Thesis title: *Evolutionary Learning of Adaptive Models within a Bayesian Framework*]. Advisor: Prof. Andrea Bonarini. Awarded with the Dimitris N. Chorafas Prize for the best PhD thesis by Chorafas Foundation.
- 2001 - 2002 Master of Science in Knowledge Discovery and Data Mining at "Center for Automatic Learning & Discovery" (08/2002, Carnegie Mellon University, Pittsburgh, PA) [MS Thesis title: *ELeaRNT: Evolutionary Learning of Rich Neural Network Topologies*]. Advisor: Prof. Manuela Veloso.
- 1993 - 1999 Laurea degree in Computer Engineering at Politecnico di Milano (19/04/1999, Milan, Italy)

Academic Positions

- Since 2021* Full Professor at the Department of Electronics Information and Bioengineering of Politecnico di Milano (Milan, Italy).
- 2015 - 2021* Associate Professor at the Department of Electronics Information and Bioengineering of Politecnico di Milano (Milan, Italy).
- 2008 - 2015* Assistant Professor (Ricercatore di ruolo confermato) at the Department of Electronics and Information of Politecnico di Milano (Milan, Italy).
- 2005 - 2008* Assistant Professor (Ricercatore di ruolo) at the Department of Electronics and Information of Politecnico di Milano (Milan, Italy).
- 2003 - 2005* Research Assistant (Assegnista di Ricerca) in the research program “Evoluzione tecnologica e nuove applicazioni delle basi di dati e dei sistemi informativi” at the Department of Electronics and Information of Politecnico di Milano (Milan, Italy).
- 2000 - 2003* PhD Student in the Computer Engineering and Automation program at the Department of Electronics and Information of Politecnico di Milano (Milan, Italy).
- 2001 - 2002* Visiting Student at the Center for Automatic Learning and Discovery at Carnegie Mellon University (Pittsburgh, PA - USA).
- 2000 - 2002* Research Assistant (Assegnista di Ricerca) in the research program “Modelli di apprendimento automatico in ambienti dinamici” at the Department of Electronics and Information of Politecnico di Milano (Milan, Italy).

Academic Boards

- Since 2025* Member of the Faculty Board of the “*Robotics and Intelligent Machines (DRIM)*” National PhD Program at Università di Genova
- Since 2025* Member of the Scientific and Technical Board of “Smart Eyewear Lab” a Joint Research Platform (JRP) between EssilorLuxottica and Politecnico di Milano
- Since 2019* Member of the Technical Board on “*Robust and Efficient mmWave Communication*” in the Joint Research Center (JRC) between Huawei and Politecnico di Milano
- Since 2018* Member of the Faculty Board of the “*Data Analytics and Decision Sciences*” PhD Program at Politecnico di Milano
- Since 2016* Segretario Consiglio Corso di Studi in Ingegneria Informatica (secretariat of the Computer Science and Engineering program) of Politecnico di Milano
- Since 2015* Reviewer for the Italian Ministry for University and Research registered to “REPRISE: Register of Expert Peer Reviewers for Italian Scientific Evaluation”
- Since 2015* Representative (and co-founder in 2014) of the Interaction between Driver, Road Infrastructure, Vehicle, and Environment (I.DRIVE) interdepartmental laboratory of Politecnico di Milano (<http://idrive.polimi.it>).
- Since 2013* Representative (and co-founder in 2012) of the Assistive Technology Group of Politecnico di Milano - Polo Regionale di Como (<http://atq.deib.polimi.it>).
- Since 2009* Member of “Commissione Orari” for the Computer Engineering track of Politecnico di Milano - Polo Regionale di Como.
- Since 2008* “Delegato SAT” (Struttura Accademica dei Tirocini) for the Computer Engineering track at Politecnico di Milano - Campus Leonardo.
- 2020 - 2024* Member of the “*Commissione Paritetica*” of the Bioinformatics for Computational Genomic joint Master program between Scuola di Ingegneria Industriale e dell’Informazione - Politecnico di Milano and Department of Biosciences Università degli Studi di Milano.

- 2018 - 2020 Scientific Director of the "Master di I livello in Artificial Intelligence and Machine Learning" offered by Politecnico di Milano and CEFRIEL
- 2010 - 2016 Member of "Commissione Didattica Laurea Specialistica" for the Computer Engineering track of Politecnico di Milano - Polo Regionale di Como.
- 2012 - 2015 Reviewer for the Italian Ministry of University and Research (Iscritto all'Albo Revisori MIUR).
- 2012 - 2013 National Action co-Leader for the "Health & Wellbeing" Action Line of the Italian Node of European Institute for Innovation and Technology (EIT).

Affiliations to Scientific Associations

- Since 2019 Member of the Istituto di Robotica e Macchine Intelligenti (I-RIM)
- Since 2018 Member of the Italian Association for Computer Vision, Pattern Recognition and Machine Learning (CVPL)
- Since 2007 Member of IEEE, the Institute of Electrical and Electronics Engineers (Computational Intelligence Society, Robotics and Automation Society).
- Since 2000 Member of the Artificial Intelligence and Robotics Lab of Politecnico di Milano.
- 2000 - 2005 Member of AI*IA, Associazione Italiana per l'Intelligenza Artificiale.

Board / Commission of Trust

- Since 2025 Elected Member of the Board of Director of I-RIM, the Italian Institute of Robotics and Intelligent Machines
- 2020 - 2025 IEEE RAS Standardization Committee for the "2751-2025 IEEE Standard 3D Map Data Representation for Robotics and Automation"
- 2015 - 2016 IEEE RAS Standardization Committee for the "1873-2015 - IEEE Standard for Robot Map Data Representation for Navigation"

Evaluation Committees

- 2020 Chair of the evaluation panel for PhD Research Scholarship from the Atlantic International Research Center (AIR Center) and the Portuguese Foundation for Science and Technology (FCT) on "Marine Litter: Aerial surveys of estuarine and beach plastic litter."
- 2020 Member of the evaluation panel for PhD Research Scholarship from the Atlantic International Research Center (AIR Center) and the Portuguese Foundation for Science and Technology (FCT) on "Marine robotics: Unmanned systems for marine litter detection and collection."
- 07/12/2020 Member of Final Examination Commission for the PhD program in Ingegneria Informatica at Dipartimento di Ingegneria Informatica, Automatica e Gestionale "A. Ruberti" of Università degli studi di Roma "La Sapienza"
- Oct. 2020 Member of the Evaluation Committee for a 09/H1 (ING-INF/05) "RTDa" (Temporary Researcher) position at Università degli studi di Roma "La Sapienza"
- 28/02/2020 Member of Final Examination Commission for the PhD program in Ingegneria Informatica at Dipartimento di Ingegneria Informatica, Automatica e Gestionale "A. Ruberti" of Università degli studi di Roma "La Sapienza"
- 19/02/2020 Member of Final Examination Commission for the PhD program in Informatics at Institute of Computational Science (ICS) of the Università della Svizzera Italiana
- 16/01/2020 Member of Final Examination Commission for the PhD program in Mathematics and Computer Science at Università della Calabria and Biomedical Engineering at Université Claude Bernard Lyon 1 (double degree program)
- 14/01/2019 Member of the Evaluation Committee for a 09/H1 (ING-INF/05) "Professore II fascia" position at Università degli studi di Roma "La Sapienza"
- 21/09/2018 Member of Final Examination Commission for the PhD program in Computer and Control Engineering of Politecnico di Torino
- 27/08/2018 Member of "Commissione di valutazione dell'attività didattica e di ricerca" for the renewal of a

- Temporary Researcher contract (Ricercatore 240/2010) for Università di Parma.
- 17/05/2018 Member of Final Examination Commission for the PhD program in Computer and Control Engineering of Politecnico di Torino
- 22/02/2017 Member of Final Examination Commission for the PhD in Computer Engineering at the Department of Electronics Information and Bioengineering of Politecnico di Milano
- 07/02/2017 Member of Final Examination Commission for the PhD in Mechanical Engineering at the Department of Mechanical Engineering of Politecnico di Milano
- 11/16-01/17 Member of of the Evaluation Committee for the 09/H1 (ING-INF/05) RTDa position (Temporary Researcher) for the “Progetto H2020 INTCATCH”, Dipartimento di Informatica, Università degli Studi di Verona
- 29/04/2016 Member of Final Examination Commission for the PhD in Computer Engineering at the Department of Electronics Information and Bioengineering of Politecnico di Milano
- 15/04/2014 Member of Final Examination Commission for the PhD in Meccatronica at Politecnico di Torino.

Prizes and Awards

- 2025 Awarded by NVIDIA within the NVIDIA Academic Grant Award with 4x RTX PRO 6000 Blackwell Max-Q Workstation Edition
- 2025 Leonardo Fibonacci Best Paper Award for the paper “Neuro-Symbolic Scene Graph Conditioning for Synthetic Image Dataset Generation” at the 11th International Conference on Machine Learning, Optimization, and Data Science.
- 2020 *BMVC 2020: Outstanding Reviewer Awards* for “significant contribution to the review process beyond normal expectations”.
- 17/06/2019 *Best Paper Award* for the paper “Asynchronous Convolutional Networks for Object Detection in Neuromorphic Cameras” at Second International Workshop on Event-based Vision and Smart Cameras in conjunction with IEEE CVPR 2019.
- 2018 Awarded with a 20K\$ Microsoft Azure sponsorship grant as a support for the PhD Course on “Deep Learning: Theory Techniques and Applications”.
- 2017 Awarded with a 20K\$ Microsoft Azure sponsorship grant for the experimentation of “Deep Learning machine on the cloud”
- 01/07/2016 *Best Paper Award* for the paper “ReSeg: A Recurrent Neural Network-based Model for Semantic Segmentation” at DeepVision 2016, International Workshop on Deep Learning in Computer Vision in conjunction with IEEE CVPR 2016.
- 10/03/2016 *Best Paper Award (people choice)* for the paper “Automatic 3D Reconstruction of Manifold Meshes via Delaunay Triangulation and Mesh Sweeping” at 2016 IEEE Winter Conference on Applications of Computer Vision (<http://wacv16.wacv.net/index.html?p=1225.html>)
- In 2015 Awarded by NVIDIA with a Titan X GPU within the Academic Hardware Grant
- In 2014 Awarded, with Dott. Martino Migliavacca and Prof. Andrea Bonarini, of “THE BUSINESS GRANT AND THE ACCELERATION PROGRAM” with the project R2P in the Working Capital Accelerator program (WCAP 2014) by Telecom Italia.
- In 2014 Winner, with Prof. Sara Comai (referente), of the “POLISOCIAL AWARD 2014” with the project proposal “MEP: Maps for Easy Paths”.
- In 2011 Winner, with the Industrial Partner Infosolution SpA, of “Premio Antonio D’Auria 2010 per progetti e prototipi di dispositivi meccatronici innovativi di ausilio a disabili motori” from Società Italiana di Robotica e Automazione (infosolution.it, corriere.it, robosiri.it).
- In 2011 Winner, with the Industrial Partner Infosolution SpA, of the Intesa San Paolo “Startup Initiative” and the Italian round of the “Global Social Venture Competition” with the RobyWheelChair project (infosolution.it, lobbyinnovazione.it).
- In 2003 Winner of the Dimitris N. Chorafas Foundation Award for the best PhD thesis.
- 2001 - 2002 Winner of the Rotary Foundation Ambassadorial Scholarship.

Other

- Since 2017 National Scientific Qualification ("Abilitazione Nazionale" art.16 of the law 30 Dec. 2010, n.240) "Professore I fascia" for sector 09/H1.
- Since 2013 National Scientific Qualification ("Abilitazione Nazionale" art.16 of the law 30 Dec. 2010, n.240) "Professore II fascia" for sector 09/H1.
- Since 2013 National Scientific Qualification ("Abilitazione Nazionale" art.16 of the law 30 Dec. 2010, n.240) "Professore II fascia" for sector 01/B1.
- In 2014 Review and *rapporteur* for the SIR 2014 (Scientific Independence of young Researchers) funding program by the Italian Ministry of University and Research (MIUR)
- In 2011 Reviewer for the PRIN 2010-2011 (Programmi di Ricerca Scientifica di Rilevante Interesse Nazionale) funding program by the Italian Ministry of University and Research (MIUR)

Teaching activity

Lecturer and Teaching Assistant (PhD, MS, BS)

- Since 2020 **Lecturer and coordinator** for the course "*Machine Learning*" of the Bioinformatics for Computational Genomics joint program, Scuola di Ingegneria Industriale e dell'Informazione - Politecnico di Milano and Department of Biosciences Università degli Studi di Milano.
- Since 2019 **Lecturer and coordinator** for the course "*Artificial Neural Networks and Deep Learning*" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano.
- Since 2016 **Lecturer and coordinator** for the course "*Robotics*" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano.
- Since 2004 **Teaching assistant** for the course "Soft Computing" in the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione (formerly Facoltà di Ingegneria Informatica), Politecnico di Milano - Campus Leonardo,
- 2020 **Co-coordinator** for the PhD course "*Machine Learning for Non-Matrix Data*" in the PhD program in Information Technology at Dipartimento di Elettronica Informazione e Bioingegneria of Politecnico di Milano (together with Giacomo Boracchi, Cesare Alippi, Mark Carman)
- 2019 **Lecturer and coordinator** for the PhD course "*Advances in Deep Learning with Applications in Text and Image Processing*" in the PhD program in Information Engineering at Dipartimento di Elettronica Informazione e Bioingegneria, Politecnico di Milano
- 2018 **Lecturer and coordinator** for the PhD course "*Deep Learning: Theory, Techniques and Applications*" in the PhD program in Information Engineering at Dipartimento di Elettronica Informazione e Bioingegneria del Politecnico di Milano
- 2017-2015-2012-2010-2007-2005-2003: **Lecturer** for the PhD course "Soft Computing: Theory, Techniques and Applications" in the PhD Program in Information Engineering at Dipartimento di Elettronica Informazione e Bioingegneria of Politecnico di Milano.
- 2017 - 2019 **Lecturer and coordinator** for the course "Cognitive Robotics" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2016 - 2019 **Lecturer and coordinator** for the course "Machine Learning" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2011 - 2016 **Lecturer and coordinator** of the Laurea Specialistica course "Pattern Analysis and Machine Intelligence" for the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione (formerly Scuola di Ingegneria dell'Informazione), Politecnico di Milano - Polo Regionale di Como
- 2014 - 2015 **Teaching assistant** for the course "Cognitive Robotics" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano.

- 2015 **Lecturer** on Data Mining for the course "Information Retrieval and Data Mining" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2014-2012-2010-2009: **Lecturer and coordinator** for the PhD course "3D Structure from Visual Motion: Novel Techniques in Computer Vision and Autonomous Vehicles" in the PhD Program in Information Engineering at Dipartimento di Elettronica Informazione e Bioingegneria of Politecnico di Milano.
- 2013-2014 **Teaching assistant** on Data Mining for the course "Information Retrieval and Data Mining" of the Computer Engineering degree, Scuola di Ingegneria Industriale e dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2009 - 2013 **Lecturer and coordinator** of the Laurea course "Knowledge Engineering" for the Computer Engineering degree, Facoltà di Ingegneria dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2006 - 2007 **Lecturer and coordinator** of the Laurea Specialistica course "Methods for Intelligent Systems" for the Computer Engineering degree, Facoltà di Ingegneria dell'Informazione, Politecnico di Milano - Polo Regionale di Como.
- 2003 - 2007 **Lecturer and coordinator** of the Laurea course "Fondamenti di Informatica" for the Electronics Engineering degree, Facoltà di Ingegneria dell'Informazione, Politecnico di Milano - Campus Leonardo.
- 2003 - 2007 **Teaching assistant** on Natural Computation for the Laurea course "Knowledge Engineering and Expert Systems" for the Computer Engineering degree, Facoltà di Ingegneria Informatica, Politecnico di Milano - Polo Regionale di Como.
- 1999 - 2001 **Teaching assistant** on C Language for the Laurea course "Fondamenti di Informatica (B)" for the Environmental Engineering degree, Facoltà di Ingegneria Civile, Ambientale e Territoriale, Politecnico di Milano - Campus Leonardo.

Other Teaching Activities

- Since 2003 I have been **advisor / co-advisor** for more than 100 Laurea thesis (Master Level) at Politecnico di Milano
- 2020 - 2021 **Tutor** for the Alta Scuola Politecnica (<http://www.asp-poli.it>) project "RA - SEI".
- 2019 - 2020 **Tutor** for the Alta Scuola Politecnica (<http://www.asp-poli.it>) project "Sprain' with Brain".
- 05/06/2020 **Lecturer** on "Deep learning for text analysis and documents classifications" (1.5h) in the "Corso di Formazione su AI", Deloitte - Politecnico di Milano
- 01/07/2020, 15/07/2020, 29/07/2020 **Lecturer and coordinator** for the course "Machine Learning" (6h) in the "[Nokia] Percorso formazione AI e ML Avanzato" program organized by CEFRIEL
- 16/06/2020, 30/06/2020, 21/07/2020 **Lecturer and coordinator** for the course "Machine Learning" (8h) in the "[Nokia] Percorso formazione AI & ML ed. 2020" program organized by CEFRIEL
- 05/05/2020, 12/05/2020, 26/05/2020 **Lecturer** on "Machine learning for Big Data" (12h) in the "[NOKIA] SW Development for Big Data Platform" program organized by CEFRIEL
- 2019 **Lecturer and coordinator** for the "Artificial Intelligence & Machine Learning" course (36h) in the "[Nokia] Percorso Artificial Intelligence e Machine Learning" organized by CEFRIEL
- 29/11/2019 **Lecturer** on "Machine Learning" (8h) in the COFIDIS - Strategic Data Management course organized by MIP, Politecnico di Milano Graduate School of Business.
- 16/07/2019 **Lecturer** on "Techniques and architectures for autonomous robot: autonomous driving and drones" (4h) in the "Master di I Livello in Strategic and Innovative O&M Management" for ENEL
- 04/11/2018 **Webinar** on "Data Analytics, Machine Learning & Artificial Intelligence" (1h) in the ABB "Digital Culture Webinars" series organized by CEFRIEL
- 24/10/2019, 16/10/2018, 12/01/2018 **Lecturer** on "Learning & Vision for Autonomous Machines" (2h) in the Advanced Master in Innovation and Entrepreneurship at MIP, Politecnico di Milano Graduate School of Business
- 27/03/2019 **Lecturer** on "Deep Learning for Image Understanding" (1h) in the AI@Deloitte day, Deloitte
- 2019 (April-May) **Lecturer and coordinator** of the course "Deep learning per l'analisi di immagini" (24h out

- of 80h) for AGS S.P.A under a Teaching Contract with the Department of Information Electronics and Bioengineering of Politecnico di Milano (resp. Prof. Matteo Matteucci and Giacomo Boracchi)
- 30/01/2019 **Lecturer** on “*Machine Learning*” (8h) for eGeos company as part of a Teaching Contract with the Department of Information Electronics and Bioengineering of Politecnico di Milano (resp. Prof. Chiara Francalanci)
- 10/12/2018 **Webinar** on “*Machine Learning and Artificial Intelligence*” (1h) in the ABB “Digital Culture Webinars” series organized by CEFRIEL
- 13/12/2018, 11/12/2018, 04/12/2018 **Lecturer** on “Artificial Intelligence” (3x8h) in the “[Edison] Formazione ICT” course at CEFRIEL.
- 29/03/2018, 05/04/2018, 12/04/2018, 18/04/2018 **Lecturer** on “*Advanced Machine Learning*” in the [Regione Lombardia] III Ed. Master Universitario in Alto Apprendistato e Ricerca on Analytics & Business Intelligence at CEFRIEL.
- 27/02/2018, 28/02/2018 **Lecturer** on “*Modulo Big Data*” in the [Regione Lombardia] Master Universitario in Alto Apprendistato e Ricerca on Industry 4.0 - BOSCH at CEFRIEL.
- 23/05/2016 **Invited lecturer** on “*Hidden Markov Models*” in the Laurea Specialistica course “*Machine Learning*” by Prof. Marcello Restelli in the Computer Science and Engineering Program of Politecnico di Milano
- 07/04/2016, 11/04/2016 **Invited lecturer** on “*Deep Learning and Neural Networks*” in the Laurea Specialistica course “*Machine Learning*” by Prof. Marcello Restelli in the Computer Science and Engineering Program of Politecnico di Milano
- 04/04/2017, 17/04/2015, 08/05/2014, 09/04/2013, 04/05/2012, 29/04/2011 **Invited lecturer** on “*Brain-Computer Interfaces @AIRLab*” in the Laurea Specialistica course “*Accessibility*” by Prof. Licia Sbattella in Computer Engineering degree of Politecnico di Milano - Polo Regionale di Como
- 15/04/2015 **Invited lecturer** on “*An Introduction to Computer Vision*” in the ISEP - 8th International Week - “*Inspiring Engineering*” at the Instituto Superior de Engenharia do Porto, Porto (Portugal).
- 2014 - 2015 **Lecturers** on “*Robotica*” in the IFTS 56628286 “Tecnico superiore per l’innovazione di prodotto e del processo in ambito manifatturiero e delle energie rinnovabili”
- 2010 - 2012 **Tutor** for the Alta Scuola Politecnica (<http://www.asp-poli.it>) project “*TSC4MiTo: The Social Computing for Milano and Torino*”.
- 25/05/2012 **Invited lecturer** on “*Robotics benchmarking from the inside: The RAWSEEDS Experience*” for the PhD course on “*Computing and Science*” by Prof. Viola Schiaffonati and Prof. Francesco Amigoni from the Dipartimento di Elettronica e Informazione del Politecnico di Milano
- 2009 - 2011 **Tutor** for the Alta Scuola Politecnica project “WillChair: reinventing the wheels ... into wills!” from which the project “Clever” has won the “[Lifeability Award](#)” in the Bioengineering category
- 2006 - 2008 **Tutor** for the Alta Scuola Politecnica project “SenSoBot: Sensors and control for Societal Robots”
- 2005 - 2007 **Tutor** for the Alta Scuola Politecnica project “*WoMan: Windows On Man*”, from which the project “*Helios*” has been awarded as the best multidisciplinary project for the first 6 years of Alta Scuola Politecnica during the event “[Riconoscere e premiare l'eccellenza e l'innovazione - una sfida per l'Italia che crede nel futuro](#)” (ASP + Confindustria)
- 17/11/2006 **Lecturer** on Neural Networks for the course “*Cybernetics*” in the “Sapere a tutto campo” program at Università Bocconi
- 2005 - 2006 **Tutor** for the Alta Scuola Politecnica project “*IRoPa: Intelligent Robotic Partners*”
- 2005 - 2006 **Tutor** for the Alta Scuola Politecnica project “*AMoRoSA: Autonomous Mobile Robots for Service Applications*”. The team “*leRoKi*” (<http://www.ieroki.it/>) has been selected in 2012 among the 6 best ASP project in the 2006-2012 years (award won by the “*Helios*” project, 2005-2007 period)
- 06/09/2006 **Lecturer** on “Soft Computing: Neural Networks Theory and Applications” Continuous Education course for an Italian Company (company name undisclosed for NDA clause)
- 26/05/2005 **Lecturer** on “Dal filtraggio alla Kalman ai filtri a particelle” for the Laurea course “*Robotica*”

complementi” for the Computer Science degree, Facolta' di Scienze Fisiche e Naturali, Università degli Studi di Milano-Bicocca.

20/05/2005 **Lecturer** on “Tecniche di filtraggio Bayesiano” for the Laurea course “Robotica complementi” for the Computer Science degree, Facolta' di Scienze Fisiche e Naturali, Università Degli Studi Milano-Bicocca..

26/04/2004 **Invited lecturer** on “Perchè oggi non servono (ancora) le tre leggi della robotica?” for the thematic session “Dall’Intelligenza Artificiale ai robot” for the Master Program in Scientific Communication at SISSA, Trieste.

04/2000 - 06/2000 **Tutoring activity** for the Laurea courses of “Fondamenti di Informatica” for the Engineering degrees of Politecnico di Milano, Campus Bovisa.

1997 - 1999 **Tutoring activity** (Computer Science, Calculus and Geometry) for the freshmen at Politecnico di Milano, a.a. 97/98 e a.a 98/99.

Supervision and Revision of PhD Theses

Advised PhD Students

Cristian Sbrilli

Since 2021 Nico Catalano

Since 2021 Alberto Archetti

Since 2020 Stefano Samele

Since 2020 Paolo Cudrano

Since 2020 Eugenio Lomurno

Since 2020 Lorenzo Cazzella

Since 2020 Leonardo Di Perna (E&Y PhD Executive Student): Bio-Data Analysis, Deep Social Analytics

Since 2019 Riccardo Bertoglio (PhD Student): Agricultural Robotics

Since 2019 Matteo Frosi (PhD Student): 3D SLAM with Priors

Since 2018 Enrico Piazza (PhD Student, joint program with IST Lisbon): Robot Benchmarking

2018 - 2022 Marco Cannici, “Learning Efficient and Effective Representations for Event-based Cameras”

2018 - 2022 Francesco Lattari, “Learning to Observe Earth from Satellite Synthetic Aperture Radar Sensorsthrough Deep Learning Methodologies” (in cooperation with TRE-Altamira)

2017 - 2022 Alessandro Brusaferri “Machine learning for optimization of energy intensive industrial processes” (CNR PhD Executive Student)

2017 - 2022 Simone Mentasti (PhD Student): “Embedded AI and sensor fusion for autonomous vehicles” (inter-department scholarship with the Mechanical department of Politecnico di Milano)

2016 - 2021 Alessandro Gabrielli, “Studying uses acceptance of autonomous driving in the wild”

2016 - 2021 Marco Ciccone, “On iterative and conditional computation for visual representation learning”

2015 - 2020 Gianluca Bardaro, “Models, Code Generation and Abstractions - A Triple Approach to Enhance Robot Software Development”. Post-doc at Open University, UK.

2012 - 2017 Andrea Romanoni, “Incremental Large-Scale Visual 3D Mesh Reconstruction”. Applied scientist at Amazon.

2012 - 2017 Francesco Visin, “Deep recurrent neural networks for visual scene understanding”. Research scientist at Google DeepMind (<https://www.linkedin.com/in/francescovicin/>)

2011 - 2014 Davide Cucci, “A General Sensor-fusion and Parameters Self-calibration Framework with Applications in Mobile Robotics”. Post-doc at EPFL (<http://personnes.epfl.ch/211269>) consultant at Mind Earth (<https://www.mindearth.org/>)

2010 - 2013 Simone Ceriani, “Conditionally independent visual slam with integrated bundle adjustment”. Scientific Project Officer at the European Commission Joint Research Centre ([linkedin](#)).

- 2008 - 2011 Luigi Malagò, “*On the geometry of optimization based on the exponential family relaxation*”. (**Thesis award:** Dimitris N. Chorafas Foundation). Principal Investigator and Group Leader of the Machine Learning and Optimization Group, Romanian Institute of Science and Technology - RIST ([linkedin](#)).
- 2006 - 2009 Davide Antonio Migliore, “*Monocular Simultaneous Localization and Mapping with Bearing-Only Tracking*”. Project Manager on Embedded Computer Vision Solutions at Prophesee ([linkedin](#)).
- 2005 - 2009 Bernardo Dal Seno, “*Toward an Integrated P300-And ErrP-Based Brain-Computer Interface*”. Site Reliability Engineer at Google ([linkedin](#)).

Co-Advised PhD Students

- Since 2018 Linda Greta Dui (advised by Prof. Simona Ferrante at Politecnico di Milano), working on Quantitative Evaluation of Writing Difficulties
- Since 2017 Ava Vali (advised by Prof. Sara Comai at Politecnico di Milano), working on HyperSpectral Images Analysis
- 2014 - 2017 Ludovico Russo (advised by Prof. Basilio Bona at Politecnico di Torino): working on Robot Visual Navigation and Cloud Robotics (Telecom Italia grant).
- 2013 - 2016 Vahid Jalili (advised by Marco Masseroli at Politecnico di Milano): working on data analysis of Next Generation Sequencing (NGS) data for the Genomic Computing project.
- 2010 - 2014 Martino Migliavacca (advised by prof. Andrea Bonarini at Politecnico di Milano), “*The R2P framework for robot prototyping: methodological approach, hardware modules, and software components*”. Post-doc at the AIRLab, Politecnico di Milano ([linkedin](#)).
- 2008 - 2011 Maurizio Garbarino (advised by Prof. Andrea Bonarini at Politecnico di Milano), “*Modeling emotional interaction in affective computing experiments: a study on affect recognition in videogames*”. Chief Science Officer at Empatica S.r.l ([linkedin](#)).
- 2007 - 2010 Simone Tognetti (advised by Prof. Andrea Bonarini at Politecnico di Milano), “*A methodological framework for physiology based affective computing: definition and evaluation*”. CTO & co-founder of Empatica S.r.l ([linkedin](#)).
- 2006 - 2009 Daniele Marzorati (advised by Prof. Domenico G. Sorrenti at Università degli Studi di Milano-Bicocca) “*Uncertainty Modeling in 3D Vision-based SLAM*”. AI Application Engineer at Infosolution SpA ([linkedin](#)).
- 2003 - 2006 Melchiorre Caterina (advised by Prof. Angelo Cavallin at Università degli Studi di Milano-Bicocca) “*Progettazione ed applicazione di modelli quantitativi basati su reti neurali artificiali per la cartografia della suscettibilità connessa a movimenti di versante*”. Researcher at Uppsala University ([linkedin](#)).

PhD Thesis Reviewer

- Dec 2019 Reviewer for the thesis titled “*A hierarchy of graph-based methods to study the behavior of immune cells in vivo*” by Diego Ulisse Pizzagalli, PhD program in Informatics, Università della Svizzera Italiana
- May 2018 Reviewer for the thesis titled “*Physical Interaction of Autonomous Robots in Complex Environments*” by Giorgio Toscana, PhD program in Computer and Control Engineering of Politecnico di Torino.
- Jan. 2017 Reviewer for the thesis titled “*Robust Feature-based LIDAR Localization and Mapping in Unstructured Environments*” by Fabjan Kallasi, PhD program in Tecnologie dell’Informazione from Università degli Studi di Parma.
- May 2017 Review for the thesis titled “*A General Approach to Map Merging on Pose Graphs*” by Tiago Maria Buonanni, PhD program in Engineering in Computer Science, Sapienza Università di Roma
- Dec. 2016 Reviewer for the thesis titled “*Metrological Analysis of Time-Of-Flight Cameras Performance for*

Multipurpose 3D Reconstruction” by Silvio Giancola, PhD program in Measurement and Experimental Techniques, from Politecnico di Milano.

May 2014 Reviewer for the thesis titled “Instrumenting and Mining Smart Spaces” by Francesco Leotta, PhD program in Engineering in Computer Science, Sapienza Università di Roma

International Editorial Activities

International Committees

- IEEE RAS Standards Development Working Group “3D-MDR - Robot 3D Map Data Representation”
- IEEE Standardization Committee for the “P1873/D1 Standard for Robot Map Data Representation for Navigation”
- IFAC Technical Committee “7.5. Transportation and Vehicle Systems - Intelligent Autonomous Vehicles”

International Journal and Conference Editorial Role

- Review Editor on the Editorial Board of Frontiers in Computer Vision (specialty section of Frontiers in ICT and Frontiers in Computer Science).
- Associate Editor for the Special Session on “Research Reproducibility and Benchmarking of Intelligent Robots” of 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2018).
- Associate Editor of 2017 IEEE International Conference on Robotics and Automation (ICRA 2017)

International Conferences and Workshops Program Committees

- AAAI 2021/2022: Program Committee of Thirty-Fifth AAAI Conference on Artificial Intelligence (AAAI-21)
- IJCAI 2021/2022: Senior Program Committee member
- ICARSC 2021: Program Committee of 21st IEEE International Conference on Autonomous Robot Systems and Competitions
- IAS-16 2020: IAS 2014: Program Committee Member of the 16th International Conference on Intelligent Autonomous Systems.
- I-RIM 2020: Program Committee of 2a Conferenza Italiana di Robotica e Macchine Intelligenti
- ECAI 2020: Senior Program Committee member of the European Conference on Artificial Intelligence
- ECMR 2019: Technical Program Committee of the European Conference on Mobile Robots
- EBHAIS 2019: Program Committee of Workshop on Evaluation and Benchmarking of Human-Centered AI Systems
- EUSIPCO 2018: Technical Program Committee of the 26th European Signal Processing Conference, Area Chair for the “Signal Processing for Robotics” track
- CPSBench 2018: Program Committee of the 1st Workshop on Benchmarking Cyber-Physical Networks and Systems (CPSBench)
- ICARSC 2017: Program Committee of the 17th International Conference on Autonomous Robot Systems and Competitions)
- MOD 2017: Program Committee of 3rd Int.l Workshop on Machine learning, Optimization and big Data
- IKT2015: 7th International Conference on Information and Knowledge Technology
- MVS 2014: Program Committee of the 3rd IFAC Workshop on Multivehicle Systems
- IAS 2014: Program Committee of the 13th International Conference on Intelligent Autonomous Systems.
- IFAC 2014: Program Committee for International Federation of Automatic Control World Conference.
- IAV 2013 - IAV 2010: Program Committee of IFAC Symposium on Intelligent Autonomous Vehicles.
- CETC 2013: Program Committee of Conference on Electronics, Telecommunication and Computers.
- PPSN 2012: Program Committee of International Conference on Parallel Problem Solving from Nature.

- VS-Games 2011: Program Committee of International Conference in Games and Virtual Worlds for Serious Applications.
- ACII 2011: Program Committee of International Conference on Affective Computing and Intelligent Interaction.
- ISNN 2011: Program Committee of International Symposium on Neural Networks.
- ICANN 2010: Program Committee of International Conference on Artificial Neural Networks.
- ICDSC 2009: Program Committee of ACM/IEEE International Conference on Distributed Smart Cameras.
- RoboPer 2008: Program Committee of International Workshop on Robotic Perception.
- RoboVis 2007: Program Committee of International Workshop on Robot Vision.

Reviewer for International Journals and Magazines

- AICom (Artificial Intelligence Communication)
- Automatica
- Computers and Electronics in Agriculture
- Computer Methods and Programs in Biomedicine
- Data and Knowledge Engineering Journal
- Evolutionary Computation
- IEEE Transactions on System Man and Cybernetics - Part C
- IEEE Transactions on Automation Science and Engineering
- IEEE Transactions on Evolutionary Computation
- IEEE Transactions on Emerging Topics in Computational Intelligence
- IEEE Transactions on Instrumentation and Measurement
- IEEE Transactions on Neural Networks
- IEEE Transactions on Fuzzy Systems
- IEEE Transactions on Robotics (IEEE TRO)
- IEEE Robotics and Automation Magazine (IEEE RAM)
- IEEE Robotics and Automation Letters (IEEE RA-L)
- IEEE Systems Journal
- International Journal of Advanced Robotic Systems
- Computers in Biology and Medicine
- Sensors
- Robotica

Reviewer for International Conferences and Workshops

- ICRA 2021, 2019, 2018, 2016, 2015, 2014, 2011, 2010, 2008: Reviewer for the IEEE International Conference on Robotics and Automation
- SSRR 2020: Reviewer for IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)
- BMVC 2020: Reviewer for the British Machine Vision Conference
- IROS 2020, 2017, 2015, 2014, 2013, 2012, 2010: Reviewer for the IEEE/RSJ International Conference on Intelligent RObots and Systems
- ITSC 2020, 2019: Reviewer for the IEEE Intelligent Transportation Systems Conference
- ICUAS 2020: Reviewer for the International Conference on Unmanned Aircraft Systems
- ECMR 2019: Reviewer for the European Conference on Mobile Robots
- AUTOMOTIVE 2019, 2018: Reviewer for the International Conference of Electrical and Electronic Technologies for Automotive
- MOD 2017: Reviewer for the 3rd Int. Workshop on Machine learning, Optimization and big Data
- RSS 2016: Reviewer for the Robotics: Science and System Conference
- IECON 2016: Reviewer for the 42nd Annual Conference of the IEEE Industrial Electronics Society

- AIRO 2016: Reviewer for the 3rd Italian Workshop on Artificial Intelligence and Robotics
- IJCAI 2015: Reviewer for the International Joint Conference on Artificial Intelligence
- IAS 2014, 2008, 2004, Reviewer for the International Conference on Intelligent Autonomous Systems.
- IFAC WC 2014, 2011, 2008, Reviewers for the International Federation of Automatic Control World Conference.
- CETC 2013, Reviewer for the Conference on Electronics, Telecommunication and Computers.
- ISIEA 2012, Reviewer for the IEEE Symposium on Industrial Electronics and Applications.
- PPSN 2012, Reviewer for the 12th International Conference on Parallel Problem Solving From Nature.
- IJCNN 2012, 2011, 2010, 2009, 2007, Reviewer for the International Joint Conference on Neural Networks.
- MED 2012, Reviewer for the 20th Mediterranean Conference on Control and Automation.
- TAROS 2011: Reviewer for the 12th Conference Towards Autonomous Robotic Systems.
- ISNN 2011, Reviewer for the International Symposium on Neural Networks.
- ACII 2011, Reviewer for the 4th biannual International Conference on Affective Computing and Intelligent Interaction.
- RoboCup 2011, 2010, 2009, 2008, 2007, 2006, 2005 Reviewer for the RoboCup International Symposium.
- ICANN 2010, 2009 Reviewer for the International Conference on Artificial Neural Networks.
- IAV 2010, Reviewer for the 7th Symposium on Intelligent Autonomous Vehicles.
- IEEE CIG 2010, Reviewer for the IEEE Conference on Computational Intelligence and Games.
- CITSA 2009 Reviewer for the 6th International Conference on Cybernetics & Information Technologies, Systems & Applications.
- IEEE WCCI 2008, Reviewers for the IEEE World Congress on Computational Intelligence.
- CIMSA 2008, Reviewer for Computational Intelligence for Measurement Systems and Applications conference.
- WILF 2007, Reviewer for the International Workshop on Fuzzy Logic and Applications.

Session Chair and Co-Chair

- IROS 2018, IEEE/RSJ International Conference on Intelligent Robots and Systems, Special Session: "Research Reproducibility and Benchmarking of Intelligent Robots"
- IAS-15 2018, 15th International Conference on Intelligent Autonomous Systems. Session: "3D Sensing"
- IAV 2016, IFAC Intelligent Autonomous Vehicles Symposium 2016. Session "Localization and Mapping"
- IAV 2016, IFAC Intelligent Autonomous Vehicles Symposium 2016. Session "Applications"
- IAS-13 2014, 13th International Conference on Intelligent Autonomous Systems. Session: "Robot Development I"
- IROS 2012 Workshop on Progress, Challenges and Future Perspectives in Navigation and Manipulation Assistance for Robotic Wheelchairs. Session: "Autonomous wheelchair navigation and environment modelling".
- IFAC 2011, World Congress of the International Federation of Automatic Control. Session: "Mechatronics, robotics and components".
- IFAC 2011, World Congress of the International Federation of Automatic Control. Session: "Mission planning and decision making".
- ICRA 2007, IEEE International Conference on Robotics and Automation. Session: "Monocular SLAM".
- IAV 2004, 5th IFAC Symposium on Intelligent Autonomous Vehicles. Session: "Architectures".

Conference and Workshop Organization

- 1st ACRE Workshop [online <https://metricsproject.eu/agri-food/1st-acre-workshop/>]
- Local Chair for the 25th International Conference on Pattern Recognition (ICPR 2020), Milan, Italy, September 2020 (moved online to January 2021)

- ERF 2018 Workshop on “*Research Reproducibility in Robotics*” (part of the Competitions and Benchmarking) co-organized with Fabio Bonsignorio, Tampere, Finland, 12th March 2018
- ERF 2017 Workshop on “*Robotics Competitions and Challenges*” co-organized with Agostino De Santis, Edinburgh, Scotland, 24th March 2017
- IROS 2012 Workshop on “Progress, Challenges and Future Perspectives in Navigation and Manipulation Assistance for Robotic Wheelchairs”, Vilamoura, Algarve, Portugal, October 7th to 12th, 2012.
- ICAR 2009 International Workshop on “Benchmarking in Mobile Robotics – State of the Art, Open Challenges, and Research Roadmap –” at the 14th International Conference on Advanced Robotics, Munich, Germany June 22nd to 26th, 2009.

Projects & Fundings

Research Projects

- 2023 - 2027 **Principal Investigator** for Politecnico di Milano for the DIGITAL EUROPE project “AgrifoodTEF: *Test and Experiment Facilities for the Agri-Food Domain*” developing a network of test and validation infrastructures in Europe that supports Agri-Food technology companies to do near product development of their AI and Robotics solutions in real-world facilities.
- 2021 - 2023 **Co-Principal Investigator** for Politecnico di Milano (Coordinator) for the H2020 project “AI-SPRINT: *Artificial Intelligence in Secure PRivacy-preserving computing coNTinuum*” developing a framework for AI applications in computing continua, enabling a finely-tuned tradeoff between performance and AI model accuracy, while providing security and privacy guarantees. (Principal Investigators for Politecnico di Milano prof. Danilo Ardagna and Matteo Matteucci DEIB)
- 2020 - 2022 **Investigator** for Politecnico di Milano (Partner) for the H2020 project “ESSENCE: *Empathic platform to personally monitor, Stimulate, enrich, and aSsist Elders aNd Children in their Environment*” for the development of AI-enabled technologies for elderly and children homecare. (Principal Investigator for Politecnico di Milano prof. Simona Ferrante DEIB)
- 2020 - 2022 **Principal Investigator** for Politecnico di Milano (Partner) in the H2020 European project “METRICS: *Metrological evaluation and testing of robots in international competitions*”. In particular POLIMI is in charge of the design and implementation of the Agri-food Competition for Robot Evaluation (ACRE).
- 2020 **Principal Investigator** for Politecnico di Milano (Partner) for the EIT Manufacturing Activity A2001 in the EIT-20119 project “LoViCoSpec” for the development of a industrial inspections systems based on hyperspectral cameras. In particular we are in charge of using machine/deep learning techniques for image analysis.
- 2019 - 2022 **Principal Investigator** for Politecnico di Milano (Partner) in the H2020 European project “DEEPFIELD: *Deep Learning in Field Robotics - from conceptualization towards implementation*”. In particular, POLIMI is in charge of the application scenario related to Deep Learning in the precision agriculture and agricultural robotics fields.
- 2019 - 2020 **Principal Investigator** for Politecnico di Milano (Partner) for the “BEAST: *Benchmark-Enabling Active Shopping Trolley*” Financial Support for Third Parties (FSTP) action within the H2020 project EUROBENCH. The goal of the FSTP action is the design of a benchmark for autonomous robots based on a sensorized and actuated trolley to be pushed by humanoid robots and to be deployed at the EUROBENCH benchmarking facility.
- 2019 - 2020 **Principal Investigator** for Politecnico di Milano (Partner) for the “MADROB: *Modular Active Door for RObot Benchmarking*” Financial Support for Third Parties (FSTP) action within the H2020 project EUROBENCH. The goal of the FSTP action is the design of a benchmark for autonomous robots based on a sensorized and actuated door to be operated by humanoid robots and to be deployed at the EUROBENCH benchmarking facility.

- 2019 - 2020 **Principal Investigator** for Politecnico di Milano (Partner) for the “*MATCH: Machine Learning Methods for SAR-derived Time Series Trend Change Detection*” funded by the European Space Agency (ESA) with the aim of using Deep Learning techniques in detecting changes in ground displacement time series measured via SAR interferometry.
- 2018 - 2021 **Principal Investigator** for Politecnico di Milano (Partner) in the H2020-ICT-780086 European project “*SciRoc: European Robotics League plus Smart Cities Robot Competitions*”. In particular POLIMI is in charge of the benchmarking activities of the European Robotics League continuation and for the Smart City Competitions.
- 2017 - 2021 **Investigator** for Politecnico di Milano (Partner) for the H2020 project “*L4MS: Logistics for Manufacturing SMEs*” for the development of an integration platform to speed up the deployment of robotics logistics in manufacturing SMEs. (Principal Investigator for Politecnico di Milano prof. Luca Fumagalli DIG)
- 2018 - 2019 **Principal Investigator** for Politecnico di Milano (Partner) for the “*Plug&Bench*” Integrated Targeted Project (cascade funding schema) within the H2020 project RobMoSys. The goal of the ITP is to design a meta-model for a robot benchmarking component within the RobMoSys Meta-Models Ecosystem.
- 2017 - 2020 **Investigator** for Politecnico di Milano (Partner) of the project “*TEINVEIN: Tecnologie per veicoli intelligenti*” funded by Regione Lombardia for the development of technologies for autonomous driving. In particular I am involved in the development of sensor fusion technologies. (Principal Investigator for Politecnico di Milano, prof. Franco Zappa)
- 2018 **Principal Investigator** for Politecnico di Milano (Partner) for the EIT Activity A1803 in the EIT-18065 project “*Drone112*” for the development of a cloud service for the deployment of drones in search and rescue scenarios. In particular I have been involved in the development of machine learning algorithms for the video analytics.
- 2016 - 2018 **Principal Investigator** for Politecnico di Milano (Partner) in the H2020-ICT-688441 European project “*RockEU2: Robotics Coordination Action for Europe Two*”. In particular POLIMI is in charge of the benchmarking activities of the European Robotics League (https://www.eu-robotics.net/robotics_league/).
- 2016 - 2018 **Investigator** for Politecnico di Milano (Partner) in the “*GRAPE: Ground Robot for vineyard monitoring and Protection*” Experiment in agricultural robotics of the FP7-ICT601116 European project *ECHORD++* (<http://www.grape-project.eu/>). (Principal Investigator for Politecnico di Milano, Prof. Luca Bascetta).
- 2017 **Principal Investigator** for Politecnico di Milano (Partner) for the EIT Activity A1703 in the EIT-17118 project “*Cloud4Drones*” for the development of a cloud service for drones. In particular I have been involved in the development of cloud service for 3D reconstruction.
- 2013 - 2015 **Principal Investigator** for Politecnico di Milano (Partner) in the FP7-ICT-601012 European project “*RoCKIn: Robot Competitions Kick Innovation in Cognitive Systems and Robotics*”. (<http://www.rockinrobotchallenge.eu/>).
- 2013 - 2015 **Project Technical Manager** and **Principal Investigator** for Politecnico di Milano (Partner) in the AAL Joint Program European project “*ALMA: Aging without Losing Mobility and Autonomy*”. (<http://www.alma-aal.org>).
- 2014 - 2015 **Investigator** and member for Politecnico di Milano (Partner) of the **Technical Board** of OR3 activities (Comfort Manager) of project SHELL (Ecosistemi domestici condivisi ed interoperabili per ambienti di vita sostenibili, confortevoli e sicuri) funded by MIUR as part of the research program “*Tecnologie per gli ambienti di vita*”. (Scientific director for Politecnico di Milano, Prof. Letizia Tanca).
- 2012 - 2014 **Principal Investigator** (together with Prof. Marco Lovera) for the Department of Electronics and Information in the project SINOPIAE “*Sistema prototipale multisorgente Integrante tecniche di Osservazione multispettrale da satellite, aeromobile e a terra per il monitoraggio multi-scala della variazione di Indicatori ambientali legata ai costituenti Atmosferici e dispersione Energetica*” funded by the Italian Ministry of University and Research and Regione Lombardia in the program “*Progetti di Ricerca Industriale e Sviluppo Sperimentale per i settori*

strategici di Regione Lombardia". (Scientific director for Politecnico di Milano Prof. Raffaella Brumana).

- 2012 - 2014 **Principal Investigator** for the Department of Electronics and Information in the project "Un Masterplan innovativo, aperto e digitale per gestire concretamente progetto Città Studi Campus Sostenibile" funded by Politecnico di Milano under the "5 per 1000" program. (Scientific director Prof. Alberto Longo).
- 2011 - 2013 **National Coordinator** of PRIN 2009 project "ROAMFREE: Robust Odometry Applying Multisensor Fusion to Reduce Estimation Errors" funded by the Italian Ministry of University and Research. (<http://roamfree.dei.polimi.it>).
- 2006 - 2009 **Project Coordinator** of the the sixth framework (FP6) project "RAWSEEDS: Robotics Advancement through Web-publishing of Sensorial and Elaborated Extensive Data Sets". European Project FP6-045144. (<http://www.rawseeds.org>).
- 2007 - 2008 **Principal Investigator** grantee of "Brain-Computer Interfaces in Everyday Applications" grant by Politecnico di Milano and Regione Lombardia (within the joint program "Grant di Avvio alla Ricerca - Accordo di Collaborazione tra il Politecnico di Milano e la Regione Lombardia").
- 2006 - 2008 **Investigator** in the Workpackage "Robotic Companion Exploiting Affective Feedback for Modeling Emotional State of the Patient and Adapting the Rehabilitation Treatment" of the IIT Funded Project on Rehabilitation within the Politecnico di Milano IIT Unit. (Scientific director of the workpackage Prof. Andrea Bonarini).
- 2003 - 2005 **Investigator** in the PRIN project MADSys "Sviluppo di metodologie e strumenti per lo sviluppo di comunità di agenti robotici" funded by the Italian Ministry of University and Research (MIUR). (Project coordinator Prof. Andrea Bonarini).
- 2002 - 2003 **Principal Investigator** (Grantee) of "Adattatività in ambienti dinamici tramite transductive learning e boosting" grant funded by the "Young Researcher" initiative at Politecnico di Milano.

Industrial Projects and Contracts

- 2021 **Principal Investigator** in the research project "Tigheder - Prototipo" with WebRatio Srl
- 2021 **Principal Investigator** in the research project "Analisi Risposta Spettrale Estensimetri-Fase 1" with SNAM SpA
- 2021 **Co-Principal Investigator** in the research project "LiDAR algorithms for Tram Obstacle Detection" with Thales Italia SpA
- 2021 **Co-Principal Investigator** in the research project "Supporto per la realizzazione di un algoritmo per la classificazione e l'identificazione automatica delle non conformità di serbatoi caricatori a partire da misurazioni ottiche" with MADE S.C.A.R.L. and Meccanica del Sarca SpA
- 2021 **Principal Investigator** in the research project "Ricostruzione automatica di modelli urbani 3D da dati multisensore" with Antemotion Srl
- 2020 **Principal Investigator** in the research project "Ricerca e sviluppo su algoritmi avanzati di navigazione autonoma applicati sull'AGV Jobot finalizzata, secondo gli obiettivi del progetto J4S finanziato dalla Regione Emilia-Romagna" with EUTRONICA Srl
- 2020 **Principal Investigator** in the research project "Sviluppo prototipale di un algoritmo per l'identificazione di tag olografici anti contraffazione tramite intelligenza artificiale" with Tracetoo Srl
- 2020 **Investigator** in the research project "EB-MIMO: Evidence Based MIMO Channel Estimation for Mobility Scenarios" with Huawei (Principal Investigator Prof. Umberto Spagnolini)
- 2019 **Co-Principal Investigator** for the teaching contract "Deep learning per l'analisi di immagini" with Advanced Global Solutions SpA (AGS)
- 2019 **Co-Principal Investigator** in the research project "Algoritmi per stimare automaticamente l'omografia che mappa il piano del campo dallo spazio 3D ad un video acquisito da uno smartphone" with MOXOFF SpA.

- 2018 - 2019 **Principal Investigator** in the research project "Sistema di Acquisizione di Diagrammi Unifilari: definizione di architettura e specifiche del sistema di riconoscimento" with ABB SpA.
- 2018 - 2019 **Principal Investigator** in the research project "Studio sistema navigazione indoor basato su dispositivo DJI guidance" with ENI SPA and ENIProgetti (formerly Tecnomare).
- 2018 - 2019 **Investigator** in the research project "3D MADNex - 3D Multidimensional Analysis Nexus" with ENI SPA. (Principal investigator Prof. Stefano Tubaro)
- 2018 **Principal Investigator** in the research project "Studio ed identificazione delle architetture software per sistemi robotici, con particolare attenzione a ROS e OROCOS" with ENI Progetti (formerly Tecnomare).
- 2017 - 2018 **Investigator** in the research project "Studio di fattibilità per lo sviluppo di un drone per la ricerca di emissioni fuggitive" with ENI Progetti (formerly Tecnomare). (Principal investigator Prof. Marco Lovera)
- 2017 - 2018 **Investigator** in the research project "Attività di R&D sull'analisi e lo sviluppo di algoritmi e modelli predittivi per l'anticipazione di fenomeni associati a Non Productive Time nelle attività di perforazione Eni" with ENI SPA. (Principal investigator Prof. Luigi Piroddi)
- 2017 - 2018 **Principal Investigator** in the research project "Supporto nella progettazione e nello sviluppo di un sistema di guida autonoma per un veicolo elettrico autonomo" with Elettronica ASTER SPA.
- 2013 - 2018 **Principal Investigator** for Politecnico di Milano in the "Gara per l'affidamento di un appalto pre-commerciale ai sensi dell'art.19 co.1 lett. F) del D.LGS.N. 163/2016 relativo a servizi di ricerca industriale e sviluppo sperimentale funzionali alla realizzazione di un nuovo sistema universale per traino dei letti di degenza" with ARCA (Agenzia Regionale Centrali Acquisti) Regione Lombardia.
- 2016 **Investigator** in the research project "Fornitura di un software all'interno di una cella robotica e specifica assistenza alla Fiera Automatica di Monaco" with COMAU SpA. (Principal investigator Prof. Luca Bascetta)
- 2015 - 2016 **Principal Investigator** in the research project "Realizzazione di una interfaccia cervello-computer basata su potenziale P300" with Infosolution SpA in the project "STELE: telemonitoraggio e autonomia nell'assistenza domiciliare dei pazienti di patologie neurodegenerative" funded by FILAS SpA.
- 2012 - 2013 **Principal Investigator** in the research project "Studio fattibilità software di programmazione off-line" with Gaiotto Automation SpA.
- 2010-2013 **Investigator** in the research contract "QUADRIVIO" with AEROSEKUR in the project "QUADRIVIO" funded by FILAS SpA. (Scientific director Prof. Gianantonio Magnani).
- 2010-2013 **Principal Investigator** in the research project "Realizzazione di una interfaccia cervello-computer basata su potenziale P300" with Infosolution SpA in the project "ON: monitoraggio e autonomia nella assistenza domiciliare dei pazienti affetti da SLA" funded by FILAS SpA.
- 2009-2013 **Investigator** in the workpackage "Intelligenza di prodotto" of project Industria 2015 n. EE01_00015 "Studio, progettazione e sviluppo di una nuova gamma di elettrodomestici caratterizzata da tecnologie innovative mirate a una notevole riduzione dei consumi energetici e dell'impatto ambientale" funded by Ministero dello Sviluppo Economico. (Scientific director of the work package prof. Andrea Bonarini).
- 2009-2012 **Investigator** in the research project "Metodi statistici e di machine learning per lo sviluppo automatico di ontologie di dominio" with Noustat S.r.l. (Scientific director prof. Andrea Bonarini).
- 2009-2010 **Investigator** in the research contract "Valutazione dell'efficacia della politica regionale di assegnazione agli E.E.L.L. di risorse finanziarie, in forme concertate, per interventi di sicurezza stradale e analisi dell'efficienza funzionale" funded by Regione Lombardia (Scientific director Prof. Lorenzo Mussone).

Professional Activities

04/2007 - 09/2007 Research Contract for the development of a Neural tool for sport engine diagnosis for an

- Italian Company (company name undisclosed for NDA clause).
- 12/2006 - 05/2007 Research Contract for consultancy on Emotion detection from biosignals for an Italian Research Center (company name undisclosed for NDA clause).
- 06/2000 - 09/2000 Special Research Contract: Ottimizzazione di paradigmi neurali per l'elaborazione di immagine, Politecnico di Milano, Department of Electronics and Information (Milan, Italy).
- 03/2000 - 04/2000 Special Research Contract: Progetto e prototipazione di un'architettura behavior based per robot autonomi, Politecnico di Milano, Department of Electronics and Information (Milan, Italy).
- 10/1999 - 11/1999 Special Research Contract: Sviluppo sistema per l'analisi di algoritmi di apprendimento , Politecnico di Milano, Department of Electronics and Information (Milan, Italy).

Research Statement

Autonomous systems are systems able to plan, perceive, and act, so to affect their surrounding environment without the need for human intervention. Classical examples are autonomous robots, unmanned vehicles, but also (ambient) assistive technologies where system autonomy is used to supply user deficits being either physical or cognitive. Out of the three activities of an autonomous system, perception is the main limiting factor for autonomous systems aiming at being deployed in the real world in a robust, adaptive, and reliable way.

My research in autonomous perception has focused on the use of techniques and models from Pattern Recognition, Machine Learning, Signal Processing, and Dynamic State Estimation for dealing with uncertainty in perception so to allow autonomous systems to perceive their environment effectively (i.e., being able to cope with sensor uncertainty and incomplete knowledge). In doing this, three real-world scenarios have been investigated: environment perception from an autonomous robot, user state/intent perception by and autonomous system, perception of the environment by a set of distributed sensors.

In the following the results of my research in autonomous system perception and intelligent data analysis, also known as machine learning, are described together with their current a future research perspective; selected references to relevant publications are given while the full list of my publications is reported afterwards.

Research Achievements, Results, and Products

MRT: the Milan RoboCup Team

From 2001 to 2009, after graduation, I participated to the Milan Robocup Team (MRT - <http://robocup.elet.polimi.it/MRT/>) a Robocup (<http://www.robocup.org/>) team of six autonomous soccer robots; my contribution in terms of algorithms and models was related to custom panoramic vision sensors [B12], adaptive color classification algorithms [B11][B7], and a conceptual model to integrate robot perception with information from teammates based on fuzzy logic [A8][A5]. The behavior of the robots was implemented using BRIAN, a system able to manage the interaction among fuzzy behavioral modules, which I designed at the beginning of my PhD [A2]. This research gave me the opportunity to develop and verify "on the field" the technologies and methodologies I developed to make perception algorithms able to deal with dynamic and non stationary environments. To cope with the complexity of designing and maintaining a team of robots, I worked at the design of the message oriented middleware "Device Communities Development Toolkit" (DCDT) [B13][D5], which has been used to program the MRT robots. After the RoboCup experience, and because of it, I pursued research in developing modular and flexible tools and methodologies for the rapid prototyping of low-cost (service) robots. The last result is Rapid Robot Prototyping (R2P), a framework for the programming and integration of embedded systems in robotics (e.g., motor control board, inertial

measurement unit, sonar sensors, etc.), which sports a publish/subscribe communication paradigm integrated over the CAN fieldbus [A27]. Beside being an effective design methodology, as testified by the different robot realized with it, R2P is currently at a mature stage and we are now planning to present it as a commercial open source project. In 2015, based on the outcome of the research on the Robot Rapid Prototyping framework a Spin-off has been founded, namely Nova Labs, for the commercialization and the customization of the Robot Rapid Prototyping tools. My research on modular and component based robotics has now moved to the software aspects of complex systems leveraging the tool of model based development [A46].

LURCH: an Autonomous Wheelchair with Multimodal Interfaces

In 2007 I started the LURCH project for the development of an autonomous wheelchair, which could help people unable to move autonomously [B17]. This effort has resulted in an autonomous wheelchair, which sports the “shared autonomy” design paradigm: depending on the needs of the user, the robot supplies the needed autonomy from simple safety enhancer up to fully autonomous vehicle [D100]. Having the user at the center of the robot design, beside the capabilities needed for autonomous operation, I decided to design several human computer interfaces for the autonomous wheelchair ranging from the simple ones (e.g., touch screen GUI or electromyographic sensor) to the most innovative ones (e.g., a Brain Computer Interface based on Event Related Potentials) [D56]. The project has reached the maturity of an industrial prototype and this has been recognized by the two prizes (Antonio D’Auria SIRI Prize and Intesa San Paolo “Startup Initiative” award) obtained by its RobyWheelChair industrial counterpart developed together with the industrial partner Infosolution SpA. The competences built during the development of LURCH prototype have been the base for the project “ALMA: Aging without Losing Mobility and Autonomy” aimed at deploying two autonomous wheelchairs in real scenarios and at finalizing an industrial prototype (funded under the European Ambient Assisted Living (AAL) Joint Program). The LURCH prototype is now become a more mature product named Personal Mobility Kit (PMK) demonstrated in several public events.

Simultaneous Localization and Mapping in Autonomous Robots and Unmanned Vehicles

In the field of incremental building of maps and localization for autonomous robots in unknown environments I worked, in cooperation with Università degli Studi di Milano - Bicocca, on the development of a complete SLAM (i.e., Simultaneous Localization and Mapping) system in six degrees of freedom based on trinocular and monocular vision sporting hierarchical map decomposition [D46][D34][D35]. In particular, I applied my background in statistics and modeling to the various aspects of localization, simultaneous localization and mapping, and data association with specific interests in error modeling [D88][D71][D61]. Then I worked on a Visual SLAM system with multiple cameras able to build large scale visual maps (hundreds of meters, thousands of features) in real-time based on the Conditional Independence SLAM technique [A28]. This Visual SLAM system has then been combined with other odometric estimates through the ROAMFREE framework, a methodology we developed to perform multi odometric sensor fusion (e.g., inertial measurement units, laser range finder, single and multiple camera systems, etc.), to obtain an accurate and robust perception of the robot path [D104]. The resulting system has been deployed on the indoor robots at the AIRLab (e.g., the autonomous wheelchair LURCH), and in outdoor unmanned vehicles such as the QUADRIVIO all terrain robot at MERLIN (MERLIN (MEchatronics and Robotics Laboratory for INnovation - Politecnico di Milano) [D99].

Benchmarking and Good Experimental Methodologies in Autonomous Robots

The development of novel methods and techniques in intelligent systems needs a proper methodology for the evaluation of real advancements in research; however, the field of Benchmarking and Good Experimental Methodologies in Robotics is still in its infancy. I have been actively involved in the design of suitable benchmarks and good experimental methodologies for intelligent systems, in particular, for the

benchmarking of Simultaneous Localization and Mapping systems through the coordination of the FP6-045144 European project RAWSEEDS for the creation of a Multi Sensor Benchmark for Simultaneous Localization and Mapping (www.rawseeds.org) [A10][B18][D38]. Since its constitution I have been part of the Euron Special Interest Group on Good Experimental Methodologies and Benchmarking (GEM SIG) and I worked on the writing of the “General Guidelines for Robotics Papers Involving Experiments” for the chapter regarding papers on “Simultaneous Localization And Mapping” (www.heronrobots.com). I participated as External Expert to the FP7 RoSta (Robot Standards) project in the benchmarking workpackage (wiki.robot-standards.org), and I have been Politecnico di Milano Principal Investigator for the FP7-ICT-601012 European project RoCKIn (www.rockinrobotchallenge.eu) for the design and execution of two international competitions for the benchmarking of autonomous robots in the home environment (RoCKIn@Home) and at work (RoCKIn@Work). I have been part of the IEEE RAS Standard Group which has defined IEEE P1873/D1 Standard for Robot Map Data Representation for Navigation [A49], and founder member of the euRobotics AISBL (the Private counterpart of the Robotics PPP being the European Commission the Public one) Topic Group on “Evaluation of Research Results: Result Replication, Benchmarking, Challenges and Competitions”. This research field on good experimental methodologies in robotics has recently been at the center of the Horizon 2020 ICT Call on Robotics and I have been invited by a EC Project Officer to give a talk at a Workshop at the European Robotics Forum on “*Benchmarking in Robotics: Challenges and Visions*” to share my past experience.

P300 + ErrP BCI: a Self-Correcting Brain-Computer Interface

While developing the autonomous wheelchair LURCH, lot of effort has been directed to the development of effective human robot interfaces, the most challenging one being an Event Related Brain-Computer Interface. In the years from 2007 to 2009 I studied an interface able to capture the user intent from brain activity so to transfer this intent to the autonomous wheelchair or to enable people suffering severe pathologies (e.g., Amyotrophic Lateral Sclerosis) to communicate. The result of this research activity has been the first Brain Computer Interface integrating the classical P300 event related potential and the ErrP error potential [A25][A13]. In the winter of 2009 this BCI was used to control the autonomous wheelchair and this had quite resonance on the press (<http://airwiki.elet.polimi.it/index.php/MediaCoverage>). To evaluate the effectivity of this novel self-correcting paradigm I have worked on a novel metric for the performance evaluation of a Brain-Computer Interface [A16][A32], which could be applied as a methodology for the evaluation of pattern recognition based interfaces used as assistive devices (e.g., gesture recognition), and to be used within the design cycle of a machine-user interface based on pattern recognition. In the past I was already involved in the development of communication systems for disabled people applying machine learning techniques for developing symbolic language prediction models for Alternative and Augmentative Communication [B8][B5] so I have applied these techniques also to the Brain-Computer Interface field in order increase the effectiveness of a Motor Imagery Brain-Computer Interface [A22].

Bioinformatics, Biosignal Interpretation, and Affective Computing

Beside clinical and end-user involvement (e.g., Ospedale S. Camillo di Venezia, Policlinico di Roma), my research activity on Brain-Computer Interfaces has been conducted in collaboration with researchers from the (former) Bioengineering Department of Politecnico di Milano and this gave birth to a series of fruitful collaborations on biosignal interpretations too. In particular I have contributed with my expertise in machine learning and pattern recognition to the development of algorithms for the detection of Obstructive Sleep Apnea [A14][A12] and the classification of sleep stages [A15] from ECG signals. My experience on EEG signal analysis (from Brain-Computer Interface research) has been applied also to the development of an automated technique for the detection of Cyclic Alternating Patterns in sleep [A23][A20]. Out of these collaborations, came my interest in Affective Computing, i.e., the study and development of systems and devices that can recognize, interpret, process, and simulate human affects [D92][D77]. The research on this field has led two of my PhD students in starting up a company, Empatica S.r.l. (www.empatica.com), which produces a portable device (wristband) for the detection of user stress and emotions. In this field I have

been mostly interested in the methodological aspects of experiment design and validation in affective computing experiments [D90]. In the last few years, I have been involved by the research group on Genomic Computing of Politecnico di Milano led by prof. Stefano Ceri in applying machine learning techniques to genomic data, our research has led to international publications on top venues on statistical analysis [A36], signal processing [A42], data visualization [A45], efficient data handling, indexing, and retrieval [A44][A52].

VeTRA: a Model-based Multi Camera Vehicle Tracking System

Since 2008 I have been interested in visual tracking and user behaviour analysis. Beside specific advancements in background subtraction [A29][D36], the most relevant result is the development of VeTRA, a visual tracking system composed of multiple synchronized cameras for the analysis and the 3D reconstruction of vehicle traffic in roundabout intersections [A30]. VeTRA is able to reconstruct the 3D trajectory of vehicles, their type and their dimensions; the system has been evaluated on real data collected during a field survey for the evaluation of the roundabout effectiveness on behalf of Regione Lombardia. VeTRA is one of the tools developed within a methodology for the evaluation of the effectiveness of different road intersections in a transport system design [A26][A24].

Geometry of Information in Estimation of Distribution Algorithms

Because of my background in statistics I have always used tools from Bayesian inference in autonomous systems perception and intelligent data analysis. I also applied those tools to Genetic Algorithms by proposing a Bayesian extension to Learning Classifier Systems in dealing with uncertainty [D32]. That is why I immediately found myself intrigued by the Estimation of Distribution Algorithm (EDA) approach in evolutionary computation. Because of their use of directed and undirected graphical models from Statistics EDAs allow a principled theoretical analysis of the way optimization is performed in this class of evolutionary algorithms. With a colleague from the Mathematics Department of Politecnico di Torino and a co-tutored PhD Student we have proved some theorems on the (Information) Geometry of Estimation Distribution Algorithms based on models from the exponential family [D72][D81] and their relationship with natural gradient descent and Gibbs samplers [D106]. This highly theoretical work has nevertheless lead to novel techniques in the Estimation of Distribution Algorithm field capable of learning the models to be exploited in the optimization process by leveraging on their Information Geometry properties [D97][D93].

Invited Talks and Papers

- Invited Talk titled “Benchmarking via Challenges & Competitions” at CHIST-ERA “Workshop Challenge Call & ERA-NET”, <https://www.chistera.eu/workshop-challenge-call-era-net>, 23rd November, 2020
- Interview about “Industrial IoT” at Class CNBC show “Italia 4.0”, 28th October, 2020.
- Invited paper titled “Agricoltura 4.0, i dati al centro del nuovo sviluppo: come raccogliarli e gestirli” on Agenda Digitale, online on 26 October 2020
<https://www.agendadigitale.eu/cultura-digitale/agricoltura-4-0-i-dati-al-centro-del-nuovo-sviluppo-come-raccogliarli-e-gestirli>
- Invited talk titled “Benchmarking through Competition” at the Gerhard K. Kraetzschmar Memorial Symposium, at Sankt Augustin, Germany, 21-25th September, 2020
- Invited talk titled “Robotics and Farming: Technologies & Innovations for an Agriculture 4.0” at the 2nd Annual Automation & Robotics for Non-Road Applications, Gothia Towers, Gothenburg, 13th of December 2019
- Invited talk titled “AI applied to 5G: Learning Methods for Channel Estimation” at the Huawei Workshop on High Frequency Technologies for 5G, Politecnico di Milano, Milan, 18th of November 2019
- Invited talk titled “Intelligent robots are at the door ... but can they open it?” at the Sciroc-AI4EU joint Workshop on Evaluation and Benchmarking of Human-Centered AI Systems, Milton Keynes, 20th

September 2019

- Invited talk titled “Dai Big Data ai Sistemi di Autoapprendimento” hosted by Fondazione Faraggiana, Novara, 15th of October, 2019.
- Invited talk titled “Intelligenza artificiale: ambiti applicativi e impatto sui sistemi industriali” hosted by ANIE as part of “Master ANIE per Industria 4.0”, Milano, 11th of June, 2019.
- Invited talk titled “*Intelligenza artificiale: ambiti applicativi e impatto sui sistemi industriali*” hosted by APITech as part of the event “INTELLIGENZA ARTIFICIALE E SISTEMI INDUSTRIALI: Applicazioni nel contesto Impresa 4.0”, Lecco, 4th of February, 2019.
- Invited talk titled “*Cosa intendiamo per IA ... tutto ciò che avreste voluto sapere e non avete mai osato chiedere.*” at Convegno “INTELLIGENZA ARTIFICIALE E RICERCA CLINICA”, Milan, 18th of September 2018 <https://www.advicepharma.com/intelligenza-artificiale-e-ricerca-clinica/>
- Invited talk titled “*A che punto siamo con le tecnologie dell’AI e il loro futuro possibile*” at Convegno “AI TECH CONFERENCE: Artificial Intelligence per le imprese e per un mondo migliore”, Milan, 5th of July 2018
- Invited talk titled “*Sul senso (comune) dei Big Data*” at Convegno “*Big Data e Robotica nella fabbrica digitale connesso*”, Milan, 30th of November 2017
- Invited talk titled “*AI and Cognitive Technology: Why, What, How ...*” at the Cognitech Conference, Milan, 29th of March 2017
- Invited talk titled “*The RAWSEEDS Project: How we did it!*” at the CHIST-ERA Conference on “Object recognition and manipulation by robots: Data sharing and experiment reproducibility”, Krakov, 21st of June 2017
- Invited talk titled “*Modular development of service robots: from Rapid Robot Prototyping to Nova Core*” in the Sessione Specialistica “Robot & Co.” at A&T 2016, Torino, 21st of April 2016
- Invited talk titled “*Do Big Data make (common) sense?*” at Convegno “L’automazione nell’era dei Big Data: scenari e prospettive”, at MECSPE, Parma, 18th of March 2016
- Invited talk titled “*Do we really need to replicate experiments?*” in the “Open forum on evaluation of results, replication of experiments and benchmarking in robotics research” at the 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2015), on 28th of September 2015
- Invited talk titled “*The RAWSEEDS Project: How we did it!*” at the 2015 IEEE-RAS Summer School on Experimental Methodology, Performance Evaluation and Benchmarking in Robotics on 15th of September 2015
- Invited talk titled “*Benchmarking and Competitions*” at the 2015 IEEE-RAS Summer School on Experimental Methodology, Performance Evaluation and Benchmarking in Robotics on 15th of September 2015
- Invited presentation titled “*RoCKIn: Driving Robotics Forward through Collaboration, Benchmarking and Competition*” in the Forum on “Robotics Challenges and Competitions” at the 2014 IEEE International Conference on Robotics and Automation (ICRA 2014), on 4th of June 2014.
- Invited presentation titled “*Benchmarking in Robotics: Challenges and Visions*” in the Workshop “Shared infrastructures, benchmarking and performance evaluation” at the 2014 European Robotics Forum (ERF) on 12th of March 2014.
- Invited presentation titled “*Benchmarking through competitions*” to the EuRoC Challenge Design Workshop on 23rd of January 2014.
- Invited presentation titled “*The Utility Metric*” to the Workshop on “BCI Performance Metrics” held during the 5th International BCI Meeting from June 3rd - June 7th, 2013. Submitted to the Journal of Neural Engineering as part of the contribution “Performance Measurement for Brain-Computer or Brain-Machine Interfaces: A Tutorial”.
- Invited seminar (26/01/2012) “*Integrating P300 and Error Potentials in a Single BCI: Algorithms, Techniques & Performance Assessment*” in the seminar program “Bioengineering Seminar Series” at the Department of Bioengineering of Politecnico di Milano
- Invited paper at the 18th World Congress of International Federation of Automatic Control “*On Feature Parameterization for EKF-based Monocular SLAM*”, in the Special Session on SLAM
- Invited paper at the 5th Symposium on Intelligent Autonomous Vehicle “*A model to manage data*”

Full List of Publications

A. Articles in International Journals¹

- A58. Ava Vali, Sara Comai, M. Matteucci. “*Deep Learning for Land Use and Land Cover Classification Based on Hyperspectral and Multispectral Earth Observation Data: A Review*”. Remote Sensing 12(15), 2495. doi:10.3390/rs12152495. 2020. **Q1**
- A57. Alessandro Brusafferri, Matteo Matteucci, Stefano Spinelli, Andrea Vitali. “*Learning behavioral models by recurrent neural networks with discrete latent representations with application to a flexible industrial conveyor*”. Computers in Industry, *in press*. 2020. **Q1**
- A56. Alessandro Brusafferri, Matteo Matteucci, P Portolani, A Vitali. “*Bayesian deep learning based method for probabilistic forecast of day-ahead electricity prices*”. Applied Energy 250, p. 1158-1175. 2019. **Q1**
- A55. Andrea Romanoni, Matteo Matteucci. “*Mesh-based camera pairs selection and occlusion-aware masking for mesh refinement*”. Pattern Recognition Letters 125, p. 364-372. 2019. **Q1**
- A54. Meysam Basiri, Enrico Piazza, Matteo Matteucci, Pedro Lima. “*Benchmarking Functionalities of Domestic Service Robots Through Scientific Competitions*”. KI - Künstliche Intelligenz 33, 33, pages 357-367, doi: 10.1007/s13218-019-00619-9, 2019.
- A53. Francesco Lattari, Borja Gonzalez Leon, Francesco Asaro, Alessio Rucci, Carlo Prati, Matteo Matteucci. “*Deep Learning for SAR Image Despeckling*”. Remote Sensing 11 (13), 1532. 2019. **Q1**
- A52. Vahid Jalili, Matteo Matteucci, Jeremy Goecks, Yashar Deldjoo, Stefano Ceri. “*Next Generation Indexing for Genomic Intervals*”. IEEE Transactions on Knowledge and Data Engineering, doi: 10.1109/TKDE.2018.2871031, 2018. **Q1**
- A51. Gianluca Bardaro, Luca Bascetta, Marcello Farina, Matteo Matteucci. “*MPC-based control architecture of an autonomous wheelchair for indoor environments*”. Control Engineering Practice, vol.78, p.160-174. 2018. **Q1**
- A50. Francesco Amigoni, Wonpil Yu, Torsten Andre, Dirk Holz, Martin Magnusson, Matteo Matteucci, Hyungpil Moon, Masashi Yokozuka, Geoffrey Biggs, Raj Madhavan. “*A Standard for Map Data Representation: IEEE 1873-2015 Facilitates Interoperability Between Robots*”. IEEE Robotics and Automation Magazine, 25(1):65-76, ISSN: 1070-9932, doi: 10.1109/MRA.2017.2746179. 2018. **Q2/Q1**
- A49. Fabio Veronese, Andrea Masciadri, Sara Comai, Matteo Matteucci, Fabio Salice. “*Behavior Drift Detection Based on Anomalies Identification in Home Living Quantitative Indicators*”. Technologies, vol. 6, p. 1-15, ISSN: 2227-7080, doi: 10.3390/technologies6010016. 2018.
- A48. Enrico Piazza, Andrea Romanoni, Matteo Matteucci. “*Real-Time CPU-Based Large-Scale Three-Dimensional Mesh Reconstruction*”. IEEE Robotics and Automation Letters, vol. 3, p. 1584-1591, ISSN: 2377-3766, doi: 10.1109/LRA.2018.2800104. 2018.
- A47. Daniel Sinkonde, Leonard Mseli, Nima Shidende, Sara Comai, Matteo Matteucci. “*Developing an Intelligent PostGIS Database to Support Accessibility Tools for Urban Pedestrians*”. Urban Science, 2(3):52, doi: 10.3390/urbansci2030052. 2018.
- A46. Gianluca Bardaro, Andrea Semperebon, Matteo Matteucci. “*AADL for robotics: a general approach for system architecture modeling and code generation*”. Journal of Software Engineering in Robotics, vol. 8, p. 32-44, ISSN: 2035-3928. 2017.
- A45. Vahid Jalili, Matteo Matteucci, Marco Masseroli, Stefano Ceri. “*Explorative visual analytics on interval-based genomic data and their metadata*”. BMC BIOINFORMATICS, vol. 18, p. 1-15, ISSN: 1471-2105, doi: 10.1186/s12859-017-1945-9. 2017. **Q1**
- A44. Vahid Jalili, Matteo Matteucci, Marco Masseroli, Stefano Ceri. “*Indexing Next-Generation Sequencing data*”. Information Sciences, vol. 384, p. 90-109, ISSN: 0020-0255, doi: 10.1016/j.ins.2016.08.085.

¹ I report the SJR (Scimago Journal & Country Rank <https://www.scimagojr.com/>) quartiles for the year of publication of the paper, or the year before if not yet available. When the journal has multiple subjects the quartile(s) of the subject(s) appropriate for the paper topic is(are) reported.

2017. **Q1**

- A43. Sara Comai, Emanuele De Bernardi, Matteo Matteucci, Fabio Salice. *"Maps for Easy Paths (MEP): Accessible Paths Tracking and Reconstruction"*. EAI Endorsed Transactions on Internet of Things, vol.3, p.1-10, ISSN: 2414-1399, doi: 10.4108/eai.31-8-2017.153050. 2017.
- A42. Vahid Jalili, Matteo Matteucci, Marco J. Morelli, Marco Masseroli. *"MuSERA: Multiple sample enriched region assessment"*. Briefings in Bioinformatics, vol.18, p.367-381, ISSN: 1467-5463, doi: 10.1093/bib/bbw029. 2017. **Q1**
- A41. Fabio Veronese, Andrea Masciadri, Anna A. Trofimova, Matteo Matteucci, Fabio Salice. *"Realistic human behaviour simulation for quantitative ambient intelligence studies"*. Technology and Disability, vol. 28, p. 159-177, ISSN: 1055-4181, doi: 10.3233/TAD-160453. 2017.
- A40. Giulio Vitale, Andrea Bonarini, Matteo Matteucci, Luca Bascetta. *"Toward vocational robotics"*. IEEE Robotics and Automation Magazine, vol. 23, p. 73-81, ISSN: 1070-9932, doi: 10.1109/MRA.2016.2571998. 2016. **Q2/Q1**
- A39. Simone Mangano, Hassan Saidinejad, Fabio Veronese, Sara Comai, Matteo Matteucci, Fabio Salice. *"Bridge: Mutual Reassurance for Autonomous and Independent Living"*. IEEE INTELLIGENT SYSTEMS - ISSN:1541-1672 vol. 30 (4), pp.31-38. DOI:10.1109/MIS.2015.58. 2015. **Q2/Q1**
- A38. Francesco Amigoni, Emanuele Bastianelli, Jakob Berghofer, Andrea Bonarini, Giulio Fontana, Nico Hochgeschwender, Luca Iocchi, Gerhard K. Kraetzschmar, Pedro Lima, Matteo Matteucci, Pedro Miraldo, Daniele Nardi, Viola Schiaffonati. *"Competitions for Benchmarking. Task and Functionality Scoring Complete Performance Assessment"*. IEEE Robotics and Automation Magazine ISSN:1070-9932 vol. 22 (3) pp.53-61. DOI:10.1109/MRA.2015.2448871. 2015. **Q1**
- A37. Andrea Romanoni, Domenico G. Sorrenti, Matteo Matteucci. *"Backward-Simulation Particle Smoother with a hybrid state for 3D vehicle trajectory, class and dimension simultaneous estimation"*. Machine Vision and Applications, Volume 26(2-3):369-385, Springer Berlin Heidelberg, 2015. DOI:[10.1007/s00138-015-0668-z](https://doi.org/10.1007/s00138-015-0668-z) **Q2/Q1**
- A36. Jalili Vahid, Matteo Matteucci, Marco Masseroli, Marco J. Morelli. *"Using combined evidence from replicates to evaluate ChIP-seq peaks"*. BIOINFORMATICS - ISSN:1367-4803 vol. 31 (17), pp. 2761-2769. DOI:10.1093/bioinformatics/btv293. 2015. **Q1**
- A35. Adnan Tahirovic, Matteo Matteucci, Luca T Mainardi. *"An Averaging Technique for the P300 Spatial Distribution"*. Methods of Information in Medicine, Volume 54(3):215-220, Schattauer Publishers, 2015. DOI:[10.3414/ME13-02-0037](https://doi.org/10.3414/ME13-02-0037)
- A34. Gianpaolo Cugola, Alessandro Margara, Matteo Matteucci, Giordano Tamburrelli. *"Introducing Uncertainty in Complex Event Processing: Model, Implementation, and Validation"*. Computing, Volume 97(2):103-144, Springer Vienna, 2015. (First published online 4th May 2014). DOI:[10.1007/s00607-014-0404-y](https://doi.org/10.1007/s00607-014-0404-y)
- A33. Andrea Romanoni, Lorenzo Mussone, Davide Rizzi, Matteo Matteucci. *"A comparison of two Monte Carlo algorithms for 3D vehicle trajectory reconstruction in roundabouts"*, Pattern Recognition letters, Volume 51:79-85, January 2015, DOI: 10.1016/j.patrec.2014.09.003 **Q1**
- A32. Davide Antonio Cucci, Matteo Matteucci. *"On the Development of a Generic Multi-Sensor Fusion Framework for Robust Odometry Estimation"*. Journal of Software Engineering for Robotics (JOSER), Special Issue on "Best Practice in Robotic Software Development", Volume 5(1):48-62, ISSN 2035-3928, 2014.
- A31. D. E. Thompson, L. R. Quitadamo, L. Mainardi, K. U. Laghari, S. Gao, P. J. Kindermans, J. D. Simeral, R. Fazel-Rezai, Matteo Matteucci, T. H. Falk, L. Bianchi, C. A. Chestek, J. E. J. Huggins. *"Performance measurement for brain-computer or brain-machine interfaces: a tutorial"*. Journal of Neural Engineering. Volume 11(3):035001, June 2014. DOI: [10.1088/1741-2560/11/3/035001](https://doi.org/10.1088/1741-2560/11/3/035001) **Q1**
- A30. Lorenzo Mussone, Matteo Matteucci, Marco Bassani, and Davide Rizzi. *"An innovative method for the analysis of vehicle movements in roundabouts based on image processing"*. Journal of Advanced Transportation 47(6):581-594, 2013. DOI: [10.1002/atr.184](https://doi.org/10.1002/atr.184) **Q1**
- A29. Andrea Romanoni, Matteo Matteucci, Domenico G Sorrenti. *"Background subtraction by combining Temporal and Spatio-Temporal histograms in the presence of camera movement"*. Machine Vision and Application, DOI:[10.1007/s00138-013-0587-9](https://doi.org/10.1007/s00138-013-0587-9). 25(6):1573-1584. Springer Berlin Heidelberg, 2014

(Published online digitally on 08 December 2013, ISSN 1432-1769) **Q2**

- A28. Simone Ceriani, Daniele Marzorati, Matteo Matteucci and Domenico G Sorrenti. "*Single and Multi Camera Simultaneous Localization and Mapping Using the Extended Kalman Filter*". Journal of Mathematical Modelling and Algorithms in Operations Research, Volume 13(1):23-57, ISSN: 2214-2487, Springer Netherlands, 2014 (Published online in 2013). DOI 10.1007/s10852-013-9219-7
- A27. Andrea Bonarini, Matteo Matteucci, Martino Migliavacca, Davide Rizzi. "*R2P: An open source hardware and software modular approach to robot prototyping*". Robotics and Autonomous Systems, Volume 62(7):1073-1084, ISSN 0921-8890, Elsevier, 2014. (Published online digitally on 29th August 2013). DOI 10.1016/j.robot.2013.08.009. **Q1**
- A26. Matteo Matteucci, Lorenzo Mussone. "*An ant colony system for transportation user equilibrium analysis in congested networks*". Swarm Intelligence, Volume 7(4):255-277, ISSN 1935-3812, Springer, 2013. DOI 10.1007/s11721-013-0083-x. **Q2**
- A25. Mauro Marchetti, Francesco Onorati, Matteo Matteucci, Luca Mainardi, Francesco Piccione, Stefano Silvoni and Konstantinos Priftis. "*Improving the Efficacy of ERP-Based BCIs Using Different Modalities of Covert Visuospatial Attention and a Genetic Algorithm-Based Classifier*". PloS one 8(1: e53946):1-10, 2013. **Q1**
- A24. Lorenzo Mussone, Matteo Matteucci. "*OD Matrices Network Estimation from Link Counts by Neural Networks*". Journal of Transportation Systems Engineering and Information Technology 13(4):84 - 92, 2013.
- A23. Sara Mariani, Elena Manfredini, Valentina Rosso, Andrea Grassi, Martin Mendez, Alfonso Alba, Matteo Matteucci, Liborio Parrino, Mario Terzano, Sergio Cerutti, Anna Bianchi. "*Efficient automatic classifiers for the detection of A phases of the cyclic alternating pattern in sleep*". Medical and Biological Engineering and Computing. 50(4):359-372. 2012. **Q2**
- A22. Tiziano D'Albis, Rossella Blatt, Roberto Tedesco, Licia Sbattella, Matteo Matteucci. "*A predictive speller controlled by a brain-computer interface based on motor imagery*". ACM Transactions on Computer-Human Interaction. Volume 19(3):1-25, ISSN 1073-0516, ACM New York, 2012. DOI 10.1145/2362364.2362368. **Q1**
- A21. Giancarlo Ferrigno, Guido Baroni, Federico Casolo, Elena De Momi, Giuseppina Gini, Matteo Matteucci and Alessandra Pedrocchi. "*Medical Robotics*". IEEE PULSE 2(3):55-61, 2011.
- A20. Sara Mariani, Elena Manfredini, Valentina Rosso, Martin O Mendez, Anna M Bianchi, Matteo Matteucci, Mario G Terzano, Sergio Cerutti, and Liborio Parrino. "*Characterization of A phases during the Cyclic Alternating Pattern of Sleep*". Clinical Neurophysiology 122(10):2016-2024, 2011. **Q1**
- A19. Caterina Melchiorre, EA Castellanos, CJ van Westen, Matteo Matteucci. "*Evaluation of Prediction Capability, Robustness, and Sensitivity in Non Linear Landslide Susceptibility Models, Guantánamo, Cuba*". Computers & Geosciences 37(4):410-425, 2011. **Q2/Q1**
- A18. Federico Maggi, Matteo Matteucci, Stefano Zanero. "*Detecting Intrusions through System Call Sequence and Argument Analysis*". IEEE Transactions on Dependable and Secure Computing. Volume 7(4):381-395, ISSN 1545-5971, 2010. DOI 10.1109/TDSC.2008.69. **Q1**
- A17. Juha M Kortelainen, Martin O Mendez, Anna Maria Bianchi, Matteo Matteucci, and Sergio Cerutti. "*Sleep Staging based on Signals Acquired through Bed Sensor*". IEEE Transactions on Information Technology in Biomedicine 14(3):776-785, 2010. **Q2**
- A16. Bernardo Dal Seno, Matteo Matteucci, Luca T Mainardi. "*The Utility Metric: A Novel Method to Assess the Overall Performance of Discrete Brain-Computer Interfaces*". IEEE Transactions on Neural Systems and Rehabilitation Engineering, Volume 18(1):20-28, ISSN 1534-4320, 2010. DOI 10.1109/TNSRE.2009.2032642. **Q2**
- A15. Martin O Mendez, Matteo Matteucci, Vincenza Castronovo, Luigi Ferini-Strambi, Sergio Cerutti, and Anna Maria Bianchi. "*Sleep staging from Heart Rate Variability: time-varying spectral features and Hidden Markov Models*". International Journal on Biomedical Engineering and Technology 3(3/4):246-263, 2010.
- A14. Martin O Mendez, J Corthout, S Van Huffel, Matteo Matteucci, Thomas Penzel, Sergio Cerutti, and Anna Maria Bianchi. "*Automatic screening of obstructive sleep apnea from the ECG based on empirical mode decomposition and wavelet analysis*". Physiological Measurement 31:273-289, 2010.

Q2

- A13. Bernardo Dal Seno, Matteo Matteucci, and Luca T Mainardi. "Online Detection of P300 and Error Potentials in a BCI Speller". Computational Intelligence and Neuroscience (Article ID 307254), 2010.

Q1

- A12. Martin O Mendez, Anna Maria Bianchi, Matteo Matteucci, Sergio Cerutti, and Thomas Penzel. "Sleep Apnea Screening by Autoregressive Models From a Single ECG Lead". IEEE Transactions on Biomedical Engineering 56(12):2838-2850, ISSN 0018-9294, 2009. DOI 10.1109/TBME.2009.2029563. **Q2**
- A11. Federico Maggi, Matteo Matteucci and Stefano Zanero. "Reducing false positives in anomaly detectors through fuzzy alert aggregation". Information Fusion, Volume 10(4):300--311, ISSN 1566-2535 Elsevier, 2009. DOI j.inffus.2009.01.004. **Q1**
- A10. Simone Ceriani, Giulio Fontana, Alessandro Giusti, Daniele Marzorati, Matteo Matteucci, Davide Migliore, Davide Rizzi, Domenico G Sorrenti, Pierluigi Taddei. "Rawseeds ground truth collection systems for indoor self-localization and mapping". Autonomous Robots, Volume 27(4):353--371, ISSN 1573-7527, 2009. DOI 10.1007/s10514-009-9156-5. **Q2**
- A9. Caterina Melchiorre, Matteo Matteucci, A Azzoni, and A Zanchi. "Artificial neural networks and cluster analysis in landslide susceptibility zonation". 2, Volume 94(3-4):379-400, ISSN 0169-555X February, Elsevier, 2008. DOI j.geomorph.2006.10.035. **Q1**
- A8. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. "Problems and solutions for anchoring in multi-robot applications". Journal of Intelligent and Fuzzy Systems 8(3):245-254, 2007.
- A7. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. "Learning Fuzzy Classifier Systems: Architecture and Exploration Issues". International Journal on Artificial Intelligence Tools 16(2):269--289, 2007.
- A6. Valentino DA Corino, Matteo Matteucci, and Luca T Mainardi. "Analysis of Heart Rate Variability to Predict Patient Age in a Healthy Population". Methods of Information in Medicine (2010):191-195, 2007. **Q1**
- A5. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. "Concepts and Fuzzy Models for Behavior-Based Robotics". International Journal of Approximate Reasoning, Volume 41(2):110-127, ISSN 0888-613X Jan., 2006. DOI 10.1016/j.ijar.2005.06.017. **Q1**
- A4. Valentina D A Corino, Matteo Matteucci, Luca Cravello, Ettore Ferrari, Antonio A Ferrari, and Luca T Mainardi. "Long-term heart rate variability as a predictor of patient age". Computer Methods and Programs in Biomedicine 82(3):248-257, 2006.
- A3. Matteo Matteucci, and Dario Spadoni. "Evolutionary learning of rich neural networks in the Bayesian model selection framework". International Journal of Applied Mathematics and Computer Science 14(3):423-440, 2004.
- A2. Andrea Bonarini, Giovanni Invernizzi, Thomas Halva Labella, and Matteo Matteucci. "An architecture to coordinate fuzzy behaviors to control an autonomous robot". Fuzzy sets and systems, Volume 134:101-115, ISSN 0165-0114, February, Elsevier, 2003. DOI 10.1016/S0165-0114(02)00232-4 **Q2**
- A1. Andrea Bonarini, Claudio Bonacina, Matteo Matteucci. "An approach to the design of reinforcement functions in real-world, agent-based applications". IEEE Transactions on Systems, Man, and Cybernetics, Part B, Volume 31(3):288--301, ISSN 1083-4419 June, 2001. DOI 10.1109/3477.931510.

Q1

B. Articles and Chapters in International Books

- B29. Ferran Roure, Luca Bascetta, Marcel Soler, Matteo Matteucci, Davide Faconti, Jesus-Pablo Gonzalez, Daniel Serrano. "Lessons Learned in Vineyard Monitoring and Protection from a Ground Autonomous Vehicle". In: Grau A., Morel Y., Puig-Pey A., Cecchi F. (eds) Advances in Robotics Research: From Lab to Market. Springer Tracts in Advanced Robotics, vol 132, pages 81-105, Springer, Cham, doi: 10.1007/978-3-030-22327-4_5, 2020.
- B28. Luca Morreale, Andrea Romanoni, Matteo Matteucci. "Predicting the Next Best View for 3D Mesh Refinement". In: Strand M., Dillmann R., Menegatti E., Ghidoni S. (eds) Intelligent Autonomous

- Systems 15. IAS 2018. Advances in Intelligent Systems and Computing, vol 867. p.760-772, Springer, Cham. doi: https://doi.org/10.1007/978-3-030-01370-7_59. 2019
- B27. Fabrizio Frasca, Matteo Matteucci, Marco J. Morelli, Marco Masseroli. *"Exposing and Characterizing Subpopulations of Distinctly Regulated Genes by K-Plane Regression"*. In: Raposo M., Ribeiro P., Sério S., Staiano A., Ciaramella A. (eds) Computational Intelligence Methods for Bioinformatics and Biostatistics. CIBB 2018. Lecture Notes in Computer Science, vol 11925. Springer, Cham. doi: 10.1007/978-3-030-34585-3_20, 2018.
- B26. Sara Comai, Emanuele De Bernardi, Andrea Masciadri, Matteo Matteucci, Fabio Salice, Fabio Veronese. *"ALMA: An Indoor Localization and Navigation System for the Elderly"*. In: (a cura di): Barbara Guidi Laura Ricci Carlos Calafate Ombretta Gaggi Johann Marquez-Barja, Smart Objects and Technologies for Social Good. p. 82-91, Springer, ISBN: 978-3-319-76110-7, doi: 10.1007/978-3-319-76111-4_9. 2018
- B25. Andrea Masciadri, Anna A. Trofimova, Matteo Matteucci, Fabio Salice. *"Human Behavior Drift Detection in a Smart Home Environment"*. In: (a cura di): Peter Cudd Luc de Witte, Studies in Health Technology and Informatics. p. 199-203, Nieuwe Hemweg 6B:IOS Press, ISBN: 9781614997979, doi: 10.3233/978-1-61499-798-6-199. 2017
- B24. Fabio Veronese, Andrea Masciadri, Sara Comai, Matteo Matteucci, Fabio Salice. *"Quantitative Indicators for Behaviour Drift Detection from Home Automation Data"*. In: (a cura di): P. Cudd L. de Witte, Studies in Health Technology and Informatics. p. 208-215, Nieuwe Hemweg 6B:IOS Press, ISBN: 9781614997979, doi: 10.3233/978-1-61499-798-6-208. 2017
- B23. Ludovico Biagi, Sara Comai, Raffaella Mangiarotti, Matteo Matteucci, Marco Negretti, Secil Ugur Yavuz. *"Enriching Geographic Maps with Accessible Paths Derived from Implicit Mobile Device Data Collection"*. In: (a cura di): Shin'ichi Konomi (University of Tokyo Japan) and George Roussos (University of London UK), Enriching Urban Spaces with Ambient Computing, the Internet of Things, and Smart City Design. p. 89-113, IGI Global, ISBN: 9781522508274, doi: 10.4018/978-1-5225-0827-4.ch005. 2016.
- B22. Sara Comai, Daniel Kayange, Raffaella Mangiarotti, Matteo Matteucci, Secil Ugur Yavuz, Francesco Valentini. *"Mapping City Accessibility: Review and Analysis"*. In: (a cura di): Cecilia Sik-Lányi Evert-Jan Hoogerwerf Klaus Miesenberger Peter Cudd, Assistive Technologies. p. 325-331, IOS Press, ISBN: 978-1-61499-565-4, doi: 10.3233/978-1-61499-566-1-325. 2015
- B21. Fabio Veronese, Daniele Proserpio, Sara Comai, Matteo Matteucci, Fabio Salice. *"SHARON: a Simulator of Human Activities, ROutines and Needs"*. In Studies in Health Technology and Informatics, Vol.17, pp.560-566. ISBN:978-1-61499-565-4. DOI: [10.3233/978-1-61499-566-1-560](https://doi.org/10.3233/978-1-61499-566-1-560). 2015.
- B20. Davide Eynard, Marco Alberto Javarone, Matteo Matteucci. *Clustering Algorithms*, In R. Alhajj, J. Rokne (eds.), Encyclopedia of Social Network Analysis and Mining, pp. 101-114, 2014. DOI: [10.1007/978-1-4614-6170-8_138](https://doi.org/10.1007/978-1-4614-6170-8_138)
- B19. Giulio Fontana, Matteo Matteucci, Domenico G Sorrenti. *Rawseeds: Building a Benchmarking Toolkit for Autonomous Robotics*. Methods and Experimental Techniques in Computer Engineering, pp. 55-68, Springer International Publishing, 2014. DOI: [10.1007/978-3-319-00272-9_4](https://doi.org/10.1007/978-3-319-00272-9_4)
- B18. Paolo Ciuccarelli, Giorgia Lupi, Davide Eynard, F. Manfredini, Matteo Matteucci, Giorgio Caviglia, P. Dilda, Fabio Marfia, M. Azzi. *"Telltale: Visualizing the Use and Perception of Cities Through Digital Traces"*. In Morello, E., & Piga, B.E.A. (eds.), EAEA11 2013. Envisioning Architecture: Design, Evaluation, Communication. Edizioni Nuova Cultura, Roma. ISBN 9788868121365, pp. 383-392, 2013. DOI: 10.4458/1365. <http://books.google.com/books?id=Z8fWqlxwot0C>
- B17. Andrea Bonarini, Simone Ceriani, Giulio Fontana, Matteo Matteucci. *"On the development of a multi-modal autonomous wheelchair"*. HandBook of Research on ICTs for Healthcare and Social Services. pp. 727-748, 2013.
- B16. Davide Eynard, Matteo Matteucci, Fabio Marfia. *"A Modular Framework to Learn Seed Ontologies from Text"*. Semi-Automatic Ontology Development: Processes and Resources. Editors Maria Teresa Pazienza, and Armando Stellato. pp. 22-47. Information Science Reference. 2012.
- B15. Rossella Blatt, Andrea Bonarini, and Matteo Matteucci. *"Pattern Classification Techniques for Lung Cancer Diagnosis by an Electronic Nose"*. Computational Intelligence in Healthcare 4: Advanced

Methodologies, Studies in Computational Intelligence 309:397-423, 2010.

- B14. Davide Migliore, Matteo Matteucci and Pier Paolo Campari. *"Improving Geodesic Invariant Descriptors through Color Information"*. Computer Vision and Computer Graphics. Theory and Applications 24:148-161, 2009.
- B13. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"MRT: Robotics off-the-shelf with the modular robotic toolkit"*. Software Engineering for Experimental Robotics Springer Tracts in Advanced Robotics 30/2007:345-364, April, 2007.
- B12. Alberto Colombo, Matteo Matteucci, Domenico G Sorrenti. *"On the Calibration of Non Single Viewpoint Catadioptric Sensors"*. RoboCup 2006: Robot Soccer World Cup X 4434:194-205, June, 2007.
- B11. Federico Anzani, Daniele Bosisio, Matteo Matteucci, Domenico Giorgio Sorrenti. *"On-Line Color Calibration in Non-stationary Environments"*. RoboCup 2005: Robot Soccer World Cup IX LNCS 4020/2006:396-407, June, 2006.
- B10. Andrea Bonarini, Daniele Lavatelli, and Matteo Matteucci. *"A Composite System for Real-Time Robust Whistle Recognition"*. RoboCup 2005: Robot Soccer World Cup IX LNCS 4020/2006:130-141, June, 2006.
- B9. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"Concepts and Fuzzy Models for Behavior-Based Robotics"*. Fuzzy Logic and Applications 2955/2006:72-79, January, 2006.
- B8. Nicola Gatti, and Matteo Matteucci. *"CABA2L a Bliss Predictive Composition Assistant for AAC Communication Software"*. Enterprise Information Systems VI pp. 277-284, July, 2006.
- B7. Erio Grillo, Matteo Matteucci, and Domenico Giorgio Sorrenti. *"Getting the Most from Your Color Camera in a Color-Coded World"*. RoboCup 2004: Robot Soccer World Cup VIII LNCS 3276/2005:221-235, March, 2005.
- B6. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"Filling the Gap among Coordination, Planning, and Reaction Using a Fuzzy Cognitive Model"*. RoboCup 2003: Robot Soccer World Cup VII 3020/2004:662-669, August, 2004.
- B5. Nicola Gatti, Matteo Matteucci, Licia Sbattella. *"An Adaptive and Predictive Environment to Support Augmentative and Alternative Communication"*. Computers Helping People with Special Needs LNCS 3118:983-990, 2004.
- B4. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"A framework for robust sensing in multi-agent systems"*. RoboCup 2001: Robot Soccer World Cup V LNCS 2377/2002:329-351, January, 2002.
- B3. Andrea Bonarini, Giovanni Invernizzi, Fabio Marchese, Matteo Matteucci, Marcello Restelli, Domenico Sorrenti. *"Fun2Mas: The Milan Robocup Team"*. RoboCup 2001: Robot Soccer World Cup V LNCS 2377/2002:123-125, January, 2002.
- B2. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"Concepts for Anchoring in Robotics"*. AI*IA 2001: Advances in Artificial Intelligence LNCS 2175/2001:327-332, January, 2000.
- B1. Andrea Bonarini, Claudio Bonacina, and Matteo Matteucci. *"Fuzzy and Crisp Representations of Real-Valued Input for Learning Classifier Systems"*. Learning Classifier Systems LNCS 1813/2000:107-124, January, 2000.

C. National and International Patents

- C8. Antonello Antoniazzi, Gabriele Perrone, Matteo Matteucci "A computer-implemented method for reading electrical graphical diagrams". European Patent Application 19188378.4 - 1207. 2019
- C7. M. Matteucci, G. Fontana, D. Sorrenti, G. De Salvo, M. Penco. *"Device for assisted omnidirectional movement of hospital beds and other omnidirectionally mobile loads"*. US Patent Application. 16/308,552. 2019
- C6. M. Matteucci, G. Fontana, D. Sorrenti, G. De Salvo, M. Penco. *"Device for assisted omnidirectional movement of hospital beds and other omnidirectionally mobile loads"*. PCT/EP2017/063623, Politecnico di Milano, Università degli studi di Milano-Bicocca, InfoSolution S.r.l., 2017.
- C5. M. Matteucci, Giulio Fontana, Matteo Matteucci, Domenico G. Sorrenti, Giovanni De Salvo, Marco Penco. *"Dispositivo per la movimentazione omnidirezionale assistita di letti da ospedale e altri carichi"*

- mobili omnidirezionalmente*". Patent Pending ITUA20164282A1, Politecnico di Milano, Università degli studi di Milano-Bicocca, Info-Solution S.p.A.
- C4. Italo Belmonte, Matteo Bianchi, Basilio Bona, Andrea Bonarini, Luca Carlone, Davide Girlando, Andrea Mangone, Matteo Matteucci. "*Robotized Lighting Apparatus and Control Method*". European Patent EP2443388A1, 25th April, 2012.
 - C3. Italo Belmonte, Matteo Bianchi, Basilio Bona, Andrea Bonarini, Luca Carlone, Davide Girlando, Andrea Mangone, Matteo Matteucci. "*Apparato di illuminazione robotizzato e metodo di comando*". Patent IT1394314B1, 16th June, 2009.
 - C2. Cesare Alippi, G D'Angelo, Matteo Matteucci, Giorgio Pasquettaz, Vincenzo Piuri, Fabio Scotti. "*A system and method for monitoring laser welds and giving an indication of the quality of welding*". European Patent EP1371443, 2006.
 - C1. Cesare Alippi, G D'Angelo, Matteo Matteucci, Giorgio Pasquettaz, Vincenzo Piuri, Fabio Scotti. "*A real time quality monitoring system for laser welding*". PN TO2002A000508, 14th June, 2002.

D. Papers in Proceedings of International Conferences and Workshops²

- D161. Andrea Romanoni, Matteo Matteucci. "TAPA-MVS: Textureless-Aware PATCHMATCH Multi-View Stereo" In Proceedings of the IEEE International Conference on Computer Vision, pages 10413-10422, IEEE, 2019. **Class 1 (A++)**
- D160. Alessandro Brusaferri, Lorenzo Fagiano, Matteo Matteucci, Andrea Vitali. "Day ahead electricity price forecast by NARX model with LASSO based features selection". In Proceedings of 2019 IEEE 17th International Conference on Industrial Informatics (INDIN), pages 1051-1056, IEEE, doi: 10.1109/INDIN41052.2019.8972263, 2019. **Class 3 (B-)**
- D159. Alessandro Brusaferri, Matteo Matteucci, Pietro Portolani, Stefano Spinelli. "Nonlinear system identification using a recurrent network in a Bayesian framework". In Proceedings of 2019 IEEE 17th International Conference on Industrial Informatics (INDIN), pages 319-324, IEEE, doi: 10.1109/INDIN41052.2019.8972113, 2019. **Class 3 (B-)**
- D158. Gianluca Bardaro, Mohamed El-Shamouly, Giulio Fontana, Ramez Awad, Matteo Matteucci. "Toward model-based benchmarking of robot components". In Proceedings of 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pages 1682-1687, IEEE, doi: 10.1109/IROS40897.2019.8967682, 2019. **Class 1 (A+)**
- D157. Peter Detzner, Tim Pose, Luca Fumagalli, Matteo Matteucci. "Towards a Plug and Play Architecture for a Materialflow Handling System". In Proceedings of 2019 IEEE Conference on Open Systems (ICOS), pages 28-33, IEEE, doi: 10.1109/ICOS47562.2019.8975705, 2019
- D156. Simone Mentasti and Matteo Matteucci. "Image Segmentation on Embedded Systems via Superpixel Convolutional Networks". In Proceedings of European Conference on Mobile Robots, Prague, September, IEEE, doi: 10.1109/ECMR.2019.8870967, 2019.
- D155. Simone Mentasti, Matteo Matteucci. "Multi-layer occupancy grid mapping for autonomous vehicles navigation". In proceedings of AEIT International Conference of Electrical and Electronic Technologies for Automotive (AEIT AUTOMOTIVE), pages 1-6, IEEE, doi: 10.23919/EETA.2019.8804556, 2019
- D154. Luca Morreale, Andrea Romanoni, Matteo Matteucci. "*Dense 3D Visual Mapping via Semantic Simplification*". In Proceedings of IEEE International Conference on Robotics and Automation (ICRA). Montreal, pages 6891-6897, Canada. Doi: 10.1109/ICRA.2019.8793256, 2019. **Class 2 (A)**
- D153. Marco Cannici, Marco Ciccone, Andrea Romanoni, Matteo Matteucci. "*Asynchronous Convolutional Networks for Object Detection in Neuromorphic Cameras*". In Proceeding of IEEE CVPR Second International Workshop on Event-based Vision and Smart Cameras (CVPRW), pages 1-10, June 17, Long Beach. 2019.
- D152. Gianluca Bardaro, Andrea Semprebon, Agnese Chiatti, Matteo Matteucci. "*From Models to Software Through Automatic Transformations: An AADL to ROS End-to-End Toolchain*". In Proceedings of the

² I report the class -- Class 1 (A++,A+), Class 2 (A, A-), or Class 3 (B,B-) -- from the GII-GRIN-SCIE Rating (<http://valutazione.unibas.it/gii-grin-scie-rating/ratingSearch.jsf>) for the venue of the paper.

Third International Workshop on New Frontiers in Computational Robotics (NFCR '19) in conjunction with The Third IEEE International Conference on Robotic Computing (IRC), p. 580-585, Napoli, doi: 10.1109/IRC.2019.00118, 2019.

- D151. Marco Cannici, Marco Ciccone, Andrea Romanoni and Matteo Matteucci. "*Attention Mechanisms for Object Recognition with Event-Based Cameras*". In Proceedings of IEEE Winter Conference on Applications of Computer Vision (WACV 2019), p. 1127-1136, Hawaii, January. doi: 10.1109/WACV.2019.00125, 2019. **Class 2 (A)**
- D150. Pietro Astolfi, Alessandro Gabrielli, Luca Bascetta, Matteo Matteucci. "*Vineyard Autonomous Navigation in the Echord++ GRAPE Experiment*". In proc. of 16th IFAC Symposium on Information Control Problems in Manufacturing INCOM 2018 by Elsevier BV in IFAC-PapersOnLine, Volume 51, pp 704-709; doi:10.1016/j.ifacol.2018.08.401, 2018.
- D149. Fabrizio Frasca, Matteo Matteucci, Marco Morelli, Marco Masseroli. "Unveiling gene expression histonic regulative patterns by hyperplanes clustering". In Proceedings of 15th International Meeting on Computational Intelligence Methods for Bioinformatics and Biostatistics (CIBB 2018), p. 1-6, Caparica, PT, September 6-8, 2018.
- D148. Gianluca Bardaro, Andrea Semprebon, Matteo. Matteucci. "*A use case in model-based robot development using AADL and ROS*". In Proceedings of 1st International Workshop on Robotics Software Engineering, p. 9-16, Göteborg, 2018.
- D147. Andrea Romanoni, Matteo Matteucci. "*A Data-driven Prior on Facet Orientation for Semantic Mesh Labeling*". In proceedings of International Conference on 3D Vision (3DV 2018). Verona 2018.
- D146. Fabrizio Frasca, Matteo Matteucci, Marco Masseroli, Marco Morelli. "*Modeling gene transcriptional regulation by means of hyperplanes genetic clustering*". In: Ludermir TB, editor. The 2018 IEEE International Joint Conference on Neural Networks (IJCNN), Jul 8-13, 2018; Rio de Janeiro, Brasile. [CD-ROM] ISBN 978-1-5090-6014-6. Piscataway, NJ: IEEE; 2018. p. 5628-5635. **Class 3 (B)**
- D145. Fausto Ferreira, Gabriele Ferri, Yvan Petillot, Xingkun Liu, Marta Palau Franco, Matteo Matteucci, Francisco Javier Pérez Grau, Alan Ft Winfield. "*Scoring robotic competitions: Balancing judging promptness and meaningful performance evaluation*". In Proceedings of IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC), pages 179-185. Doi: 10.1109/ICARSC.2018.8374180. 2018.
- D144. Ferran Roure, Germán Moreno, Marcel Soler, Davide Faconti, Daniel Serrano, Pietro Astolfi, Gianluca Bardaro, Alessandro Gabrielli, Luca Bascetta, Matteo Matteucci. "*GRAPE: Ground robot for vineyard monitoring and protection*". In Advances in Intelligent Systems and Computing. vol. 693, p. 249-260, Springer Verlag, ISBN: 978-3-319-70832-4, Sevilla, doi: 10.1007/978-3-319-70833-1_21. 2018
- D143. Davide Antonio Cucci, Phillip Clausen, Jan Skaloud, Matteo Matteucci. "*A general approach to time-varying parameters in pose-graph optimization*". In Proceedings of 2017 European Navigation Conference, ENC 2017. p. 265-271, Institute of Electrical and Electronics Engineers Inc., ISBN: 9781509059225, doi: 10.1109/EURONAV.2017.7954217 2017
- D142. Daniele Ruscio, Luca Bascetta, Alessandro Gabrielli, Matteo Matteucci, Dedy Ariansyah, Monica Bordegoni, Giandomenico Caruso, Lorenzo Mussone. "*Collection and comparison of driver/passenger physiologic and behavioural data in simulation and on-road driving*". In proceedings of 5th IEEE International Conference on Models and Technologies for Intelligent Transportation Systems (MT-ITS). p. 403-408, ISBN: 978-1-5090-6484-7, Napoli, Italia, doi: 10.1109/MTITS.2017.8005705. 2017.
- D141. Andrea Romanoni, Daniele Fiorenti, Matteo Matteucci. "*Mesh-based 3D Textured Urban Mapping*". In Proceedings of 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), p. 1-7, Vancouver. 2017. **Class 1 (A+)**
- D140. Eugenio Ceravolo, Mauro Gabellone, Marcello Farina, Luca Bascetta, Matteo Matteucci. "*Model Predictive Control of an autonomous wheelchair*". In Proceedings of the 20th World Congress of the International Federation of Automatic Control, 9-14 July 2017, Toulouse, France.. vol. 50, p. 9821-9826, Elsevier B.V., Tolosa, Francia, doi: 10.1016/j.ifacol.2017.08.894. 2017
- D139. Andrea Romanoni, Marco Ciccone, Francesco Visin, Matteo Matteucci. "*Multi-View Stereo with Single-View Semantic Mesh Refinement*". In Workshop of the IEEE International Conference on Computer Vision (ICCV). p. 706-715, Venezia. 2017

- D138.Fabio Veronese, Sara Comai, Simone Mangano, Matteo Matteucci, Fabio Salice. *"PIR probability model for a cost/reliability tradeoff unobtrusive indoor monitoring system"*. In Lecture Notes of the Institute for Computer Sciences, Social-Informatics and Telecommunications Engineering, LNICST. LECTURE NOTES OF THE INSTITUTE FOR COMPUTER SCIENCES, SOCIAL INFORMATICS AND TELECOMMUNICATIONS ENGINEERING, p. 61-69, Springer Verlag, ISBN: 9783319619484, ISSN: 1867-8211, ita, 2016, doi: 10.1007/978-3-319-61949-1_7. 2017
- D137.Pedro U. Lima, Daniele Nardi, Gerhard K. Kraetzschmar, Rainer Bischoff, Matteo Matteucci. *"RoCKIn and the European robotics league: Building on RoboCup best practices to promote robot competitions in Europe"*. In: Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). vol. 9776, p. 181-192, Springer Verlag, ISBN: 9783319687919, deu, 2016, doi: 10.1007/978-3-319-68792-6_15. 2017
- D136.Gianluca Bardaro, Matteo Matteucci. *"Using AADL to model and develop ROS-based robotic application"*. In Proceedings of the 1st IEEE International Conference on Robotic Computing, IRC 2017. p. 204-207, Institute of Electrical and Electronics Engineers Inc., ISBN: 9781509067237, twt, 2017, doi: 10.1109/IRC.2017.59. 2017
- D135.Gianluca Bardaro, Luca Bascetta, Francesco Casella, Matteo Matteucci. *"Using Modelica for advanced Multi-Body modelling in 3D graphical robotic simulators"*. In Proceedings of the 12th International Modelica Conference. vol. 132, p. 887-894, ISBN: 9789176855751, Prague, doi: 10.3384/ecp17132887. 2017.
- D134.Gianluca Bardaro, Luca Bascetta, Francesco Casella, Matteo Matteucci. *"Advancement in multi-body physics modeling for 3D graphical robot simulators"*. In: Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics). p. 189-195, Springer Verlag, ISBN: 9783319476049, Roma, Italia, 2016, doi: 10.1007/978-3-319-47605-6_15
- D133.Andrea Romanoni, Amaël Delaunoy, Marc Pollefeys, Matteo Matteucci. *"Automatic 3D Reconstruction of Manifold Meshes via Delaunay Triangulation and Mesh Sweeping"*. In Proceedings of 2016 IEEE Winter Conference on Applications of Computer Vision. p. 1-8, ISBN: 978-1-5090-0641-0, Lake Placid, NY, 07/03/2016-09/03/2016, doi: 10.1109/WACV.2016.7477650. 2016. **Class 2 (A)**
- D132.Luca Bascetta, Davide Antonio Cucci, Matteo Matteucci. *"Kinematic trajectory tracking controller for an all-terrain Ackermann steering vehicle"*. In: 9th IFAC Symposium on Intelligent Autonomous Vehicles IAV 2016. p. 13-18, Elsevier B.V., Leipzig, Germania, doi: 10.1016/j.ifacol.2016.07.600. 2016.
- D131.Luca Bascetta, Gianni Ferretti, Matteo Matteucci, Marco Bossi. *"LFT-based MPC Control of an Autonomous Vehicle"*. In: 9th IFAC Symposium on Intelligent Autonomous Vehicles IAV 2016. p. 7-12, Elsevier B.V., Leipzig, Germania, doi: 10.1016/j.ifacol.2016.07.597. 2016
- D130.Francesco Visin, Marco Ciccone, Adriana Romero, Kyle Kastner, Kyunghyun Cho, Yoshua Bengio, Matteo Matteucci, Aaron Courville. *"ReSeg: A Recurrent Neural Network-Based Model for Semantic Segmentation"*. In Proceedings of CVPR 2016 DeepVision Workshop. Las Vegas, 2016
- D129.Cesare Svelto, Matteo Matteucci, Roberto Resmini, Alexey Pniov, Lorenzo Pedotti, Fabio Giordano. *"Semi-and-Automatic Wheel Measurement System for Prick Test Digital Imaging and Analysis"*. In Proceedings of 2016 IEEE International Conference on Imaging Systems and Techniques (IST) p. 482-486, ISBN: 978-1-5090-1817-8, Chania, Crete Island, Greece, 04/10/2016 - 06/10/2016. 2016.
- D128.G. Postica, A. Romanoni, M. Matteucci. *"Robust Moving Objects Detection in Lidar Data Exploiting Visual Cues"*. In Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 1093-1098, Daejeon, Korea, 2016. **Class 1 (A+)**
- D127.E.L. D'Amelio, Luca Bascetta, Davide Antonio Cucci, Matteo Matteucci, Gianluca Bardaro. *"A Modelica simulator to support the development of the control system of an autonomous All-Terrain mobile robot"*. In Proceedings of the Vienna Conference on Mathematical Modelling - ISBN:978-3-901608-46-9, pp.274-279. DOI:10.1016/j.ifacol.2015.05.017. 2015.
- D126.Sara Comai, Daniel Kayange, Raffaella Mangiarotti, Matteo Matteucci, Secil Yavuz Ugur, Francesco Valentini. *"Mapping City Accessibility: Review and Analysis"*. In proceedings of In Assistive Technologies, ISBN:978-1-61499-565-4, pp.325-331. DOI:10.3233/978-1-61499-566-1-325. 2015.
- D125.Gianluca Bardaro, Ava Vali, Sara Comai, Matteo Matteucci. *"Accessible Urban Routes Reconstruction by Fusing Mobile Sensors Data"*. In proceedings of 13th International Conference on Advances in

- Mobile Computing and Multimedia (MoMM2015), pp.84-92. DOI:10.1145/2837126.2837181, ISBN:978-1-4503-3493-8, 2015.
- D124. Ludovico Biagi, Sara Comai, Raffaella Mangiarotti, Matteo Matteucci, M. Negretti, Secil Ugur, M.G. Visconti. "The MEP project: Map for Easy Paths". In Proceedings of the FOSS4G Europe 2015 Symposium (Free and Open Source For Geospatial Europe Conference). 2015
- D123. Andrea Romanoni, Amaël Delaunoy, Marc Pollefeys, Matteo Matteucci. "Automatic 3D Reconstruction of Manifold Meshes via Delaunay Triangulation and Mesh Sweeping". In WACV 2016: IEEE Winter Conference on Applications of Computer Vision, pp.1-8, 2016. **Class 2 (A)**
- D122. Andrea Romanoni, Matteo Matteucci. "Reconstruction of Urban Environments by Edge-Points Delaunay Triangulation". Proceedings of the 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2015), pp 4473--4479, DOI: 10.1109/IROS.2015.7354012, IEEE, 2015. **Class 1 (A+)**
- D121. Andrea Romanoni, Matteo Matteucci. "Efficient moving point handling for incremental 3d manifold reconstruction". In Image Analysis and Processing (ICIAP 2015). LNCS 9279, pp 489-499, DOI: 10.1007/978-3-319-23231-7_44, Springer, 2015. **Class 3 (B)**
- D120. Matteo Matteucci, Martino Migliavacca, Andrea Bonarini. "Practical Applications of the R2P Embedded Framework for Robot Rapid Development" in Proceedings of the 7th Annual IEEE International Conference on Technologies for Practical Robot Applications (TePRA). ISBN 978-1-4799-8757-3. 2015.
- D119. E.L. D'Amelio, Luca Bascetta, Davide Antonio Cucci, Matteo Matteucci, Gianluca Bardaro. "A Modelica simulator to support the development of the control system of an autonomous All-Terrain mobile robot". 8th Vienna Conference on Mathematical Modelling. DOI: 10.3182/20150218-3-AU-30250//978-3-902823-71-70021, pp. 274-279, Vienna Austria, February 2015.
- D118. Vahid Jalili, Matteo Matteucci, Marco Masseroli, Marco J. Morelli. "Using replicates to evaluate ChIP-seq peaks". In BITS 2014: 11th Annual Meeting of the Bioinformatics Italian Society, pp.150-151. 2014.
- D117. Fabio Veronese, Sara Comai, Matteo Matteucci, Fabio Salice. "*Method, Design and Implementation of a Multiuser Indoor Localization System with Concurrent Fault Detection*". In Proceedings of 11th International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services. P 100-109. DOI:10.4108/icst.mobiquitous.2014.258215. 2014. **Class 3 (B)**
- D116. Fabio Veronese, Daniel Soleimani Pour, Sara Comai, Matteo Matteucci and Fabio Salice. "*Method, Design and Implementation of a Self-Checking Indoor Localization System*". In Joint Proceedings of 8th International Conference on Ubiquitous Computing & Ambient Intelligence (UCAmI 2014) & 6th International Work-conference on Ambient Assisted Living (IWAAL 2014), DOI:[10.1007/978-3-319-13105-4_29](https://doi.org/10.1007/978-3-319-13105-4_29), LNCS 8868, pp. 187-194, Springer International Publishing, 2014.
- D115. Gianluca Bardaro, Davide Antonio Cucci, Luca Bascetta, Matteo Matteucci. "A Simulation Based Architecture for the Development of an Autonomous All Terrain Vehicle". In Proceedings of 4th Simulation, Modeling, and Programming for Autonomous Robots International Conference, Lecture Notes in Computer Science Volume 8810, DOI: 10.1007/978-3-319-11900-7_7, pp 74-85, Springer International Publishing, Bergamo, Italy, 2014.
- D114. Davide Antonio Cucci, Martino Migliavacca, Andrea Bonarini and Matteo Matteucci. "Development of Mobile Robots Using Off-The-Shelf Open-Source Hardware and Software Components for Motion and Pose Tracking". In Proceedings of the 13rd International Conference on Intelligent Autonomous Systems (IAS-13), pp 1-14, Padova, Italy, 2014.
- D113. Davide Antonio Cucci, Matteo Matteucci. "*Position tracking and sensors self-calibration in autonomous mobile robots by Gauss-Newton optimization*". In Proceedings of IEEE International Conference on Robotics and Automation (ICRA), pp. 1269-1275. Hong Kong, China, 2014. DOI: [10.1109/ICRA.2014.6907016](https://doi.org/10.1109/ICRA.2014.6907016) **Class 2 (A)**
- D112. Tom De Smedt, Fabio Marfia, Matteo Matteucci, Walter Daelemans. "*Using Wiktionary to Build an Italian Part-of-Speech Tagger*". In Proceedings of 19th International Conference on Applications of

Natural Language to Information Systems, NLDB 2014, Montpellier, France, June 18-20, 2014.
DOI:[10.1007/978-3-319-07983-7_1](https://doi.org/10.1007/978-3-319-07983-7_1)

D111 Licia Sbattella, Luca Colombo, Carlo Rinaldi, Roberto Tedesco, Matteo Matteucci, Alessandro Trivilini "Extracting emotions and communication styles from prosody," in Proceedings of the international conference on physiological computing systems (PhyCS), edited by H. P. da Silva, A. Holzinger, S. Fairclough, and D. Majoe, LNCS 8908, pp. 20-41. Ed., Springer-Verlag Berlin Heidelberg, 2014.

D110. Sven Schneider, Frederik Hegger, Iman Awaad, Rhama Dwiputra, Gerhard Kraetzschmar, Aamir Ahmad, Pedro Lima, Francesco Amigoni, Andrea Bonarini, Giulio Fontana, Matteo Matteucci, Viola Schiaffonati, Jakob Berghofer, Rainer Bischoff, Luca Iocchi, Daniele Nardi. *"The RoCKIn@Home Challenge"*. In Proceedings of ISR/Robotik 2014 - 41st International Symposium on Robotics and 6th German Conference on Robotics. 06/02/2014 - 06/03/2014 at München, Germany, pp 321-327. 2014

Class 3 (B)

D109. Rhama Dwiputra, Jakob Berghofer, Rainer Bischoff, Amir Ahmad, Pedro Lima, Iman Awaad, Frederik Hegger, Nico Hochgeschwender, Gerhard Kraetzschmar, Sven Schneider, Francesco Amigoni, Andrea Bonarini, Giulio Fontana, Matteo Matteucci, Viola Schiaffonati, Luca Iocchi, Daniele Nardi. *"The RoCKIn@Work Challenge"*. In Proceedings of ISR/Robotik 2014 - 41st International Symposium on Robotics and 6th German Conference on Robotics. 06/02/2014 - 06/03/2014 at München, Germany, pp. 328-333. 2014 **Class 3 (B)**

D108. Matteo Matteucci, Davide Rizzi, Andrea Romanoni, Lorenzo Mussone. 3D image processing of vehicular trajectories in roundabouts. Selected Proceedings of World Conference on Transport Research (WCTR2013), pp. 1-20, Rio de Janeiro, Brazil, July, 2013.

D107. Luigi Malagò, Matteo Matteucci. *"Robust Estimation of Natural Gradient in Optimization by Regularized Linear Regression"*. Geometric Science of Information LNCS 8085:861-867, 2013. **Class 2 (A-)**

D106. Luigi Malagò, Matteo Matteucci and Giovanni Pistone. *"Natural gradient, fitness modelling and model selection: A unifying perspective"*. In Proceedings of 2013 IEEE Congress on Evolutionary Computation (CEC) pp. 486--493, 2013. **Class 2 (A-)**

D105. Martino Migliavacca, Andrea Bonarini and Matteo Matteucci. *"RTCAN: a Real-Time CAN-Bus Protocol for Robotic Applications"*. Proceedings of 11th International Conference on Informatics in Control, Automation and Robotics (ICINCO), pp. 353-360, 2013.

D104. Davide Cucci, Matteo Matteucci. *"A Flexible Framework for Mobile Robot Pose Estimation and Multi-Sensor Self-Calibration"*. Proceedings of 11th International Conference on Informatics in Control, Automation and Robotics (ICINCO). pp. 361-368, 2013.

D103. Martino Migliavacca, Andrea Bonarini and Matteo Matteucci. *"Modular Development of Mobile Robots with Open Source Hardware and Software Components"*. Proceedings of 17th annual RoboCup International Symposium. On CD. 2013.

D102. Fabio Veronese, Hassan Saidinejad, Matteo Matteucci, Fabio Salice. *"Communication Oriented Brain Computer Interface in a Remote Monitoring System for Amyotrophic Lateral Sclerosis"*. Assistive Technology: From Research to Practice (AAATE 2013). pp. 483-488, 2013.

D101. Tiziano D'Albis, Rossella Blatt, Roberto Tedesco, Licia Sbattella, Matteo Matteucci. *"A predictive speller controlled by a brain-computer interface based on motor imagery"*. Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems (CHI), Paris (France), April, 2013.

Class 1 (A++)

D100. Andrea Bonarini, Simone Ceriani, Giulio Fontana, Matteo Matteucci. *"Introducing LURCH: a Shared Autonomy Robotic Wheelchair with Multimodal Interfaces"*. In Proceedings of IROS 2012 Workshop on Progress, Challenges and Future Perspectives in Navigation and Manipulation Assistance for Robotic Wheelchairs. pp. 1-6, 2012.

D99. Luca Bascetta, Davide Cucci, Gianantonio Magnani, Matteo Matteucci, Dinko Osmankovic, and Adnan Tahirovic. *"Towards the Implementation of a MPC-based Planner on an Autonomous All-Terrain Vehicle"*. In Proceedings of Workshop on Robot Motion Planning: Online, Reactive, and in Real-time (IEEE/RJS IROS 2012) pp. 1-7, 2012.

D98. Augusto L Ballardini, Axel Furlan, Alberto Galbiati, Matteo Matteucci, Francesco Sacchi, Domenico G

- Sorrenti. "An effective 6DoF motion model for 3D-6DoF Monte Carlo Localization". In Proceedings of 4th Workshop on Planning, Perception and Navigation for Intelligent Vehicles (IEEE/RJS IROS 2012). pp. 1-6, 2012.
- D97. Davide Cucci, Luigi Malagò, Matteo Matteucci. "Variable Transformations in Estimation of Distribution Algorithms". Parallel Problem Solving from Nature - PPSN XII. LNCS 7491:428-437, 2012. **Class 2 (A-)**
- D96. Adnan Tahirovic, Matteo Matteucci, Luca T Mainardi. "An Averaging Technique for the P300 Spatial Distribution". Proceedings of 7th International BioSignal Interpretation Workshop (BSI2012). pp. 319-322, 2012.
- D95. Andrea Bonarini, Matteo Matteucci, Martino Migliavacca, Davide Rizzi. "R2P: an Open Source Modular Architecture for Rapid Prototyping of Robotics Applications". Proceedings of 1st IFAC Conference on Embedded Systems, Computational Intelligence and Telematics in Control (CESCIT'12). pp. 68-73, April, 2012.
- D94. Emanuele Corsano, Davide Cucci, Luigi Malagò, Matteo Matteucci. "Implicit Model Selection based on Variable Transformations in Estimation of Distribution". Proceedings of 6th Learning and Intelligent Optimization Conference (LION 6) pp. 1-6, January, 2012.
- D93. Gabriele Valentini, Luigi Malagò, Matteo Matteucci. "Optimization by l1-constraining a Markov Fitness Modelling". Proceedings of 6th Learning and Intelligent Optimization Conference (LION 6) pp. 1-15, January, 2012.
- D92. Simone Tognetti, Maurizio Garbarino, Matteo Matteucci and Andrea Bonarini. "The affective triad: stimuli, questionnaires and measurements". In proceedings of Affective Computing and Intelligent Interaction (ACII) pp. 101-110, 2011. **Class 3 (B-)**
- D91. Maurizio Garbarino, Simone Tognetti, Matteo Matteucci, and Andrea Bonarini. "Learning General Preference Models from Physiological Responses in Video Games: How Complex is it?". In proceedings of Affective Computing and Intelligent Interaction (ACII) pp. 517-526, 2011. **Class 3 (B-)**
- D90. Maurizio Garbarino, Matteo Matteucci, and Andrea Bonarini. "Affective Preference from Physiology in Videogames: A Lesson Learned from the TORCS Experiment". In proceedings of Affective Computing and Intelligent Interaction (ACII) pp. 528-537, 2011. **Class 3 (B-)**
- D89. Andrea Bonarini, Fiammetta Costa, Maurizio Garbarino, Matteo Matteucci, Maximiliano Romero, Simone Tognetti. "Affective videogames: the problem of wearability and comfort". In proceedings of 14th International Conference on Human-Computer Interaction (HCI) pp. 649-658, 2011.
- D88. Simone Ceriani, Daniele Marzorati, Matteo Matteucci, Davide Migliore, and Domenico G Sorrenti. "On Feature Parameterization for EKF-based Monocular SLAM". In proceedings of 18th World Congress of the International Federation of Automatic Control (IFAC) pp. 6829--6834, August, 2011.
- D87. Lorenzo Mussone, Matteo Matteucci, Marco Bassani, and Davide Rizzi. "Traffic Analysis in Roundabout Intersections by Image Processing". In proceedings of 18th World Congress of the International Federation of Automatic Control (IFAC) pp. 14922--14927, August, 2011.
- D86. Andrea Bonarini, Matteo Matteucci, Martino Migliavacca, Roberto Sannino, and Daniele Caltabiano. "Modular Low-Cost Robotics: What Communication Infrastructure?". In proceedings of 18th World Congress of the International Federation of Automatic Control (IFAC) pp. 917--922, August, 2011.
- D85. Luigi Malago', Matteo Matteucci, and Giovanni Pistone. "Stochastic Natural Gradient Descent by estimation of empirical covariances". In proceedings of IEEE Congress on Evolutionary Computation (CEC) pp. 949 -956, 2011. **Class 2 (A-)**
- D84. Luigi Malago', Matteo Matteucci, and Gabriele Valentini. "Introducing l1-regularized logistic regression in Markov Networks based EDAs". In proceedings of IEEE Congress on Evolutionary Computation (CEC) pp. 1581 -1588, 2011. **Class 2 (A-)**
- D83. Luigi Malago', Matteo Matteucci, and Giovanni Pistone. "Optimization of pseudo-Boolean Functions by Stochastic Natural Gradient Descent". In Proceedings of 9th Metaheuristics International Conference (MIC), 2011.
- D82. Lorenzo Mussone, Matteo Matteucci, Marco Bassani, and Davide Rizzi. "A comparative analysis of vehicular flow image surveys for roundabout performance evaluation". In proceedings of 3rd International Conference on Roundabouts pp. 1-21, Accepted.

- D81. Malago', Luigi and Matteucci, Matteo and Pistone, Giovanni. *"Towards the geometry of estimation of distribution algorithms based on the exponential family"*. Proceedings of the 11th workshop on Foundations of genetic algorithms pp. 230--242, New York, NY, USA, 2011. **Class 2 (A-)**
- D80. G Ferrigno, G Baroni, A Pedrocchi, F Casolo, E De Momi, G Gini, M Matteucci. *"Medical Robotics and Computer Aided Therapy"*. BioMedPOLIMI Proc 1st Workshop on the Life Sciences at Politecnico di Milano pp. 240-243, November, 2010.
- D79. Andrea Bonarini, Maurizio Garbarino, Matteo Matteucci, Simone Tognetti. *"Affective Evaluation of Robotic Rehabilitation of Upper Limbs in Post-Stroke Subjects"*. BioMedPOLIMI Proc 1st Workshop on the Life Sciences at Politecnico di Milano pp. 290-293, November, 2010.
- D78. Andrea Bonarini, Matteo Matteucci, Simone Ceriani, Luca T Mainardi. *"User-Tailored Shared Autonomy by a Robotic Wheelchair with Multimodal Interface"*. BioMedPOLIMI Proc 1st Workshop on the Life Sciences at Politecnico di Milano pp. 260-263, November, 2010.
- D77. Simone Tognetti, Maurizio Garbarino, Andrea T Bonanno, Matteo Matteucci, Andrea Bonarini. *"Enjoyment Recognition From Physiological Data in a Car Racing Game"*. In Proceedings of 3rd International Workshop on Affective Interaction in Natural Environments (AFFINE) pp. 3-8, October, 2010.
- D76. Davide Eynard, Fabio Marfia, Matteo Matteucci. *"On the Use of Correspondence Analysis to Learn Seed Ontologies from Text"*. In Proceedings of International Conference on Knowledge Engineering and Ontology Development (KEOD) pp. 430-439, October, 2010.
- D75. Sara Mariani, Anna M Bianchi, Elena Manfredini, Valentina Rosso, Martin O Mendez, Liborio Parrino, Matteo Matteucci, Andrea Grassi, Sergio Cerutti, and Mario G Terzano. *"Automatic detection of A phases of the Cyclic Alternating Pattern during sleep"*. In Proceedings of IEEE Engineering in Medicine and Biology Conference (EMBC) pp. 1-4, September, 2010. **Class 3 (B)**
- D74. Simone Tognetti, Maurizio Garbarino, Andrea Bonarini, Matteo Matteucci. *"Modeling enjoyment preference from physiological responses in a car racing game"*. In Proceedings of IEEE Conference on Computational Intelligence and Games pp. 321-328, August, 2010. **Class 3 (B-)**
- D73. Gabriele Valentini, Luigi Malago', Matteo Matteucci. *"Evoptool: an Extensible Toolkit for Evolutionary Optimization Algorithms Comparison"*. In Proceedings of IEEE World Congress on Computational Intelligence pp. 2475-2482, July, 2010. **Class 2 (A-)**
- D72. Luigi Malago', Matteo Matteucci, Giovanni Pistone. *"Stochastic Relaxation as a Unifying Approach in 0/1 Programming"*. NIPS 2009 Workshop on Discrete Optimization in Machine Learning: Submodularity, Sparsity & Polyhedra (DISCML) , December, 2009.
- D71. Daniele Marzorati, Matteo Matteucci, Davide Migliore, Domenico G Sorrenti. *"On the use of inverse scaling in monocular SLAM"*. In proceedings of ICRA '09. IEEE International Conference on Robotics and Automation pp. 2030 – 2036, May, 2009. **Class 2 (A)**
- D70. Davide Migliore, Roberto Rigamonti, Daniele Marzorati, Matteo Matteucci, Domenico G Sorrenti. *"Use a Single Camera for Simultaneous Localization And Mapping with Mobile Object Tracking in dynamic environments"*. In proceedings of International workshop on Safe navigation in open and dynamic environments application to autonomous vehicles , May, 2009.
- D69. Sathya Costagliola Di Polidoro, Bernardo Dal Seno, Matteo Matteucci. *"Recognition and Classification of P300s in EEG Signals by Means of Feature Extraction Using Wavelet Decomposition"*. In proceedings of 2009 IEEE International Joint Conference on Neural Networks (IJCNN) pp. 597-603, June, 2009. **Class 3 (B)**
- D68. Matteo Matteucci, Lorenzo Mussone, Ahmed Ghazia. *"Transportation network user equilibrium assignment by ant colony systems with a variable trail decay coefficient"*. In proceedings of 12th IFAC Symposium on Control in Transportation Systems (CTS09) pp. 430-427, September, 2009.
- D67. Luigi Sportiello, Bernardo Dal Seno, Matteo Matteucci. *"A Comparison of Three Methods with Implicit Features for Automatic Identification of P300s in a BCI"*. In proceedings of 19th International Conference on Artificial Neural Networks (ICANN 2009) pp. 515-524, September, 2009. **Class 3 (B)**
- D66. Bernardo Dal Seno, Matteo Matteucci, Luca T Mainardi. *"On-line Detection of P300 and Error Potentials in a BCI Speller"*. In proceedings of 7th International Symposium on Noninvasive Functional Source Imaging of the Brain and Heart & 7th International Conference on Bioelectromagnetism (7th

NFSI & ICBEM 2009) , May, 2009.

- D65. Simone Tognetti, Cristiano Alessandro, Andrea Bonarini, Matteo Matteucci. *"Fundamental issues on the recognition of autonomic patterns produced by visual stimuli"*. In proceedings of Affective Computing and Intelligent Interaction (ACII 2009) , 2009. **Class 3 (B-)**
- D64. Martin O Mendez, Matteo Matteucci, Sergio Cerutti, F Alletti, Anna M Bianchi. *"Sleep staging classification based on HRV: Time-variant analysis"*. In proceedings of Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2009) pp. 9 - 12, September, 2009. **Class 3 (B)**
- D63. Martin O Mendez, Matteo Matteucci, Sergio Cerutti, Anna M Bianchi, Juha M Kortelainen. *"Automatic detection of sleep macrostructure based on bed sensors"*. In proceedings of Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC 2009) pp. 5555 - 5558, September, 2009. **Class 3 (B)**
- D62. Andrea Bonarini, Axel Furlan, Luigi Malagò, Daniele Marzorati, Matteo Matteucci, Davide Migliore, Marcello Restelli, Domenico G Sorrenti. *"Milan Robocup Team 2009"*. Robocup International Symposium 2009 pp. 1-8, July, 2009.
- D61. Daniele Marzorati, Matteo Matteucci, Davide Migliore, Domenico G Sorrenti. *"Monocular SLAM with Inverse Scaling Parametrization"*. Proceedings of 2008 British Machine Vision Conference pp. 945--954, September, 2008. **Class 2 (A)**
- D60. Daniele Marzorati, Matteo Matteucci, Davide Migliore, Domenico G Sorrenti. *"Data Fusion by Uncertain Projective Geometry in 6DoF Visual SLAM"*. Proceedings of VISAPP International Workshop on Robotic Perception pp. 3--12, 2008.
- D59. Pier Paolo Campari, Matteo Matteucci, Davide Migliore. *"Adding Color to Geodesic Invariant Features"*. Proceedings of VISAPP - International Conference on Computer Vision Theory and Applications pp. 227--234, January, 2008.
- D58. Andrea Bonarini, Luca T Mainardi, Matteo Matteucci, Simone Tognetti, Roberto Colombo. *"Stress Recognition in a Robotic Rehabilitation Task"*. Proceedings of Robotic Helpers: User Interaction, Interfaces and Companions in Assistive and Therapy Robotics pp. 41--48, March, 2008.
- D57. Bernardo Dal Seno, Matteo Matteucci and Luca T Mainardi. *"A genetic algorithm for automatic feature extraction in P300 detection"*. International Joint Conference on Neural Networks pp. 3145--3152, June, 2008. **Class 3 (B)**
- D56. Davide Migliore, Rossella Blatt, Matteo Matteucci, Simone Ceriani, Giulio Fontana, and Bernardo Dal Seno. *"Brain Control of a Smart Wheelchair"*. Proceedings of Intelligent Autonomous Systems pp. 221--228, JULY, 2008.
- D55. Bernardo Dal Seno, Matteo Matteucci, Luca T Mainardi, F Piccione, and Stefano Silvoni. *"Single-trial P300 detection in healthy and ALS subjects by means of a genetic algorithm"*. Proceedings of the 4th International Brain-Computer Interface Workshop & Training Course pp. 104--109, September, 2008.
- D54. Gianmaria Visconti, Bernardo Dal Seno, Matteo Matteucci, and Luca T Mainardi. *"Automatic recognition of error potentials in a P300-based brain-computer interface"*. Proceedings of the 4th International Brain-Computer Interface Workshop & Training Course pp. 238--243, September, 2008.
- D53. Bernardo Dal Seno, Luca T Mainardi, and Matteo Matteucci. *"Assessing the performance of a BCI: A task-oriented approach"*. Proceedings of the 4th International Brain-Computer Interface Workshop & Training Course pp. 274--279, September, 2008.
- D52. Vincenzo Rana, Matteo Matteucci, Daniele Caltabiano, Roberto Sannino, and Andrea Bonarini. *"Low cost smartcam design"*. Proceedings of 6th IEEE Workshop on Embedded Systems for Real-time Multimedia (ESTIMedia 2008) pp. 27--32, Los Alamitos, CA, USA -- USA, October, 2008.
- D51. Rossella Blatt, Andrea Bonarini, Elisa Calabró, Matteo Matteucci, Matteo Della Torre, Ugo Pastorino. *"Pattern Classification Techniques for Early Lung Cancer Diagnosis using an Electronic Nose"*. Proceedings of ECAI – Prestigious Applications of Artificial Intelligence (PAIS) @1:693--697, JULY, 2008.
- D50. Luigi Malago', Matteo Matteucci, and Bernardo Dal Seno. *"An Information Geometry Perspective on Estimation of Distribution Algorithms: Boundary Analysis"*. pp. 2081--2088, July, 2008.
- D49. Giulio Fontana, Matteo Matteucci, Domenico G Sorrenti. *"The RAWSEEDS Proposal for*

- Representation-Independent Benchmarking of SLAM*". GEMBENCH 2008, pp. 1--6, 2008.
- D48. Andrea Bonarini, Matteo Matteucci, and Simone Tognetti. "*Chest expansion reconstruction from respiration sound by using artificial neural networks*". Proceedings of 4th IET International Conference on Advances in Medical, Signal and Information Processing (MEDSIP 2008). pp. 126--129, LOS ALAMITOS CA -- USA, July, 2008.
- D47. Paolo Cremonesi, Roberto Turrin, Eugenio Lentini, and Matteo Matteucci. "*An Evaluation Methodology for Collaborative Recommender Systems*". International Conference on Automated solutions for Cross Media Content and Multi-channel Distribution pp. 224--231, November, 2008.
- D46. Daniele Marzorati, Matteo Matteucci, Domenico Giorgio Sorrenti. "*Particle-based Sensor Modeling for 3D-Vision SLAM*". In proceedings of IEEE International Conference on Robotics and Automation (ICRA) pp. 4801-4806, April, 2007. **Class 2 (A)**
- D45. Daniele Marzorati, Matteo Matteucci, Davide Antonio Migliore, Domenico Giorgio Sorrenti. "*Integration of 3D Lines and Points in 6DoF Visual SLAM by Uncertain Projective Geometry*". In proceedings of European Conference on Mobile Robotics (ECMR) pp. 96-101, 2007.
- D44. Fabio Antonacci, Matteo Matteucci, Davide Antonio Migliore, D Riva, Augusto Sarti, Marco Tagliasacchi, and Stefano Tubaro. "*Tracking Multiple Acoustic Sources in Reverberant Environments using Regularized Particle Filter*". In proceedings of 15th International Conference on Digital Signal Processing pp. 99-102, 2007.
- D43. Rossella Blatt, Andrea Bonarini, Elisa Calabró, Matteo Della Torre, Matteo Matteucci, and Ugo Pastorino. "*Fuzzy k-NN Lung Cancer Identification by an Electronic Nose*". Applications of Fuzzy Sets Theory LNCS 4578/2007:261-268, August, 2007.
- D42. Rossella Blatt, Andrea Bonarini, Elisa Calabro, Matteo Della Torre, Matteo Matteucci, and Ugo Pastorino. "*Lung Cancer Identification by an Electronic Nose based on an Array of MOS Sensors*". In proceedings of International Joint Conference on Neural Networks (IJCNN) pp. 1423-1428, August, 2007. **Class 3 (B)**
- D41. Lorenzo Mussone, and Matteo Matteucci. "*An application of ant colony systems for DUE and SUE assignment in congested transportation networks*". In proceedings of 11th of World Conference on Transportation Research, 2007.
- D40. Martin O Mendez, Davide D Ruini, Omar P Villantieri, Matteo Matteucci, Thomas Penzel, Sergio Cerutti, and Anna Maria Bianchi. "*Detection of Sleep Apnea from surface ECG based on features extracted by an Autoregressive Model*". In proceedings of 29th Annual International Conference of the IEEE Engineering in Medicine and Biology Society pp. 6105-6108, October, 2007 **Class 3 (B)**
- D39. Andrea Bonarini, and Matteo Matteucci. "*FIXCS: a Fuzzy Implementation of XCS*". In proceedings of IEEE International Fuzzy Systems Conference pp. 1-6, July, 2007. **Class 2 (A-)**
- D38. Andrea Bonarini, Wolfram Burgard, Giulio Fontana, Matteo Matteucci, Domenico Giorgio Sorrenti and Juan Domingo Tardos. "*RAWSEEDS: Robotics Advancement through Web-publishing of Sensorial and Elaborated Extensive Data Sets*". In proceedings of IROS'06 Workshop on Benchmarks in Robotics Research On line, 2006.
- D37. Alberto Colombo, Matteo Matteucci, and Domenico G Sorrenti. "*Calibration of General Non Single Viewpoint Catadioptric Sensors*". Proceedings of Robocup Symposium, 2006.
- D36. Andrea Bonarini, Davide A Migliore, Matteo Matteucci, Matteo Naccari. "*A revaluation of frame difference in fast and robust motion detection*". In proceedings of the 4th ACM international workshop on video surveillance and sensor networks pp. 215 - 218, New York, NY, USA, 2006.
- D35. Daniele Marzorati, Matteo Matteucci, Domenico Giorgio Sorrenti. "*3D-6DoF Hierarchical SLAM with 3D vision*". In proceedings of Workshop on Robotic 3D Environment Cognition pp. 53-58, September, 2006.
- D34. Daniele Marzorati, Matteo Matteucci, Domenico Giorgio Sorrenti. "*Multi-criteria data association in 3D-6DoF hierarchical SLAM with 3D segments*". In Proceedings of 3rd International Workshop on Advances in Service Robotics, 2006.
- D33. F Archetti, Cristina E Manfredotti, Matteo Matteucci, Vincenza Messina, Domenico Giorgio Sorrenti. "*Parallel First-Order Markov Chain for On-Line Anomaly Detection in Traffic Video Surveillance*". In proceedings of Institution of Engineering and Technology Conference on Crime and Security pp.

582-587, June, 2006.

- D32. Davide Aliprandi, Alex Mancastroppa, Matteo Matteucci. *"A Bayesian Approach to Learning Classifier Systems in Uncertain Environments"*. In proceedings of the 8th annual conference on Genetic and Evolutionary Computation Conference (GECCO) pp. 1537 - 1544, New York, NY, USA, 2006. **Class 2 (A)**
- D31. Lorenzo Mussone, and Matteo Matteucci. *"OD matrices estimation from link flows by neural networks and PCA"*. In proceedings of IFAC Symposium on Control in Transportation Systems 11:165-170, 2006.
- D30. Lorenzo Mussone, Matteo Matteucci. *"Ant Colony Optimization Technique for Equilibrium Assignment in Congested Transportation Networks"*. In proceeding of the 8th annual conference on Genetic and Evolutionary Computation Conference (GECCO) pp. 87-88, July, 2006. **Class 2 (A)**
- D29. Caterina Melchiorre, Matteo Matteucci, and J Remondo. *"Artificial Neural Networks and Robustness Analysis in Landslide Susceptibility Zonation"*. In proceedings of International Joint Conference on Neural Networks (IJCNN) pp. 4375-4381, 2006. **Class 3 (B)**
- D28. Federico Anzani, Daniele Bosisio, Matteo Matteucci, Domenico Giorgio Sorrenti. *"On-Line Color Calibration in Non-Stationary Environments"*. In proceedings of Robocup Symposium, 2005.
- D27. Andrea Bonarini, Daniele Lavatelli, and Matteo Matteucci. *"A Composite System for Real-Time Robust Whistle Recognition"*. In proceedings of Robocup Symposium, 2005.
- D26. Andrea Bonarini, Matteo Matteucci, Carlo Pincirolì. *"Robust Detection of 3D Scene Horizontal and Vertical Lines in Conical Catadioptric Sensors"*. In Proceedings of the 6th Workshop on Omnidirectional Vision, Camera Networks and Non-classical cameras (OMNIVIS) On line publication, 2005.
- D25. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"Automatic Error Detection and Reduction for an Odometric Sensor based on Two Optical Mice"*. In proceedings of IEEE International Conference on Robotics and Automation (ICRA) pp. 1675- 1680, April, 2005. **Class 2 (A)**
- D24. Daniele Marzorati, Matteo Matteucci, Domenico Giorgio Sorrenti. *"3D-6DOF Hierarchical SLAM with Trinocular Vision"*. In proceedings of 2nd European Conference on Mobile ROBots (ECMR) pp. 56-61, 2005.
- D23. Andrea Bonarini, Matteo Matteucci, Davide Antonio Migliore, Matteo Naccari. *"A Robust Approach to Motion Detection and Tracking in Indoor Video Surveillance"*. In proceedings of 2nd AIXIA Workshop on Ambient Intelligence. pag. 1-12, 2005.
- D22. Nicola Gatti, Matteo Matteucci, Licia Sbattella. *"User Linguistic Model Adaptivity for Prediction in AAC Message Composition"*. In Proceedings of the 11th International Conference on Human-Computer Interaction (HCI) , 2005.
- D21. Valentina DA Corino, Luca Mainardi, Matteo Matteucci. *"Analysis of Heart Rate Variability to Predict Patient Age in Healthy Population"*. In proceedings of 5th BSI Conference pp. 147-150, 2005.
- D20. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"A kinematic-independent dead-reckoning sensor for indoor mobile robotics"*. In proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 4:3750- 3755, September, 2004. **Class 1 (A+)**
- D19. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"A model to manage data reliability in behavior-based robotics"*. In proceedings of Intelligent autonomous vehicles (IAV) , July, 2004.
- D18. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"A novel model to rule behavior interaction"*. In proceedings of Intelligent Autonomous Systems 8 (IAS) pp. 199-206, 2004.
- D17. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"Dead reckoning for mobile robots using two optical mice"*. In proceedings of the 1st International Conference on Informatics in Control, Automation and Robotics (ICINCO) pp. 87-94, 2004.
- D16. Nicola Gatti, Matteo Matteucci, Licia Sbattella. *"An ICT Aid for Verbal Impaired People: Bliss2003"*. In proceedings of International Conference on Computers Helping People with Special Needs (ICHP) pp. 983-990, July, 2004.
- D15. Nicola Gatti, and Matteo Matteucci. *"CABA2L a Bliss Predictive Composition Assistant for AAC Communication Software"*. In Proceedings of the 6th ACM/AAAI International Conference on Enterprise Information Systems (ICEIS) 5:89-96, April, 2004.

- D14. Matteo Matteucci, and Dario Spadoni. *"Bayesian evolution of rich neural networks"*. In proceedings of International Joint Conference on Neural Networks 1:229-234, July, 2004. **Class 3 (B)**
- D13. Erio Grillo, Matteo Matteucci, Domenico Giorgio Sorrenti. *"Getting the most from your color camera in a color-coded world"*. In proceedings of Robocup Symposium, 2004.
- D12. Erio Grillo, Matteo Matteucci, Domenico Giorgio Sorrenti. *"On the optimization of the settings of a color camera for a color-coded world"*. In proceedings of Convegno del Gruppo Italiano Ricercatori in Pattern Recognition (GIRPR), 2004.
- D11. Luca Mainardi, Matteo Matteucci, R Rossi. *"On predicting the spontaneous termination of atrial fibrillation episodes using linear and nonlinear parameters of ECG signal and RR series"*. Computers in Cardiology pp. 665-668, September, 2004.
- D10. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"Concepts and Fuzzy Models for Control and Coordination in Robotics"*. In proceedings of Multiagent Robotic Systems: Trends and Industrial Applications, 2003.
- D9. Andrea Bonarini, Matteo Matteucci, and Marcello Restelli. *"Concepts and Fuzzy Models for Behavior-Based Robotics"*. In proceedings of 6th WILF, 2003.
- D8. Andrea Bonarini, Matteo Matteucci, Marcello Restelli, and Domenico Giorgio Sorrenti. *"Filling the Gap among Coordination, Planning, and Reaction Using a Fuzzy Cognitive Model"*. In proceedings of 7th RoboCup Symposium, 2003.
- D7. Cesare Alippi, Giuseppe D'Angelo, Matteo Matteucci, Giorgio Pasquettaz, Vincenzo Piuri, and Fabio Scotti. *"Composite techniques for quality analysis in automotive laser welding"*. In proceedings of IEEE International Symposium on Computational Intelligence for Measurement Systems and Applications (CIMSA) pp. 72- 77, July, 2003.
- D6. Elena Eneva, Krishna Kumaraswamy, Matteo Matteucci. *"WEKKEM: A Study in Fractal Dimension and Dimensionality Reduction"*. In Proceedings First Workshop on Fractals and Self-similarity in Data Mining , 2002.
- D5. Riccardo Cassinis, Paolo Meriggi, Andrea Bonarini, Matteo Matteucci. *"Device communities development toolkit: an introduction"*. In Proceedings of EUROBOT'01 pp. 155-161, 2001.
- D4. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"Anchoring: do we need new solutions to an old problem or do we have old solutions for a new problem"*. Proceedings of the AAI Fall Symposium on Anchoring Symbols to Sensor Data in Single and Multiple Robot Systems pp. 79-86, 2001.
- D3. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"A framework for robust sensing in multi-agent systems"*. Proceedings of Robocup Symposium, 2001.
- D2. Andrea Bonarini, Giovanni Invernizzi, Fabio Marchese, Matteo Matteucci, Marcello Restelli, Domenico Sorrenti. *"Fun2Mas: The Milan Robocup Team"*. Proceedings of Robocup Symposium, 2001.
- D1. Andrea Bonarini, Matteo Matteucci. *"Learning context motivation in coordinated behaviors"*. Proceedings of Intelligent Autonomous Systems 6 (IAS-6) 6:519-526, 2000.

E. Articles in Collections

- E5. Daniele Marzorati, Matteo Matteucci, Davide Migliore, Domenico G Sorrenti. *"Inverse Scaling Parametrization for Visual SLAM"*. Emerging Paradigms in Informatics, Systems and Communication; QD Quaderni - Department of informatics, systems and communication Research Report n.1 pp. 29--34, June, 2009.
- E4. Antonio Bianchi, Bernardo Dal Seno, Nicola Gatti, Matteo Matteucci, and Licia Sbattella. *"Computational linguistics for learning and language impairments"*. Disabilità e ricerca avanzata al Politecnico di Milano 10:54-59, October, 2006.
- E3. Lorenzo Mussone, and Matteo Matteucci. *"Ritaratura matrici O/D con l'utilizzo di reti neurali"*. I sistemi stradali di trasporto nella societa' dell'informazione. pp. 95-134, 2005.
- E2. Lorenzo Mussone, and Matteo Matteucci. *"Predizione spazio-temporale del flusso veicolare"*. I sistemi stradali di trasporto nella societa' dell'informazione pp. 135-174, 2005.
- E1. Lorenzo Mussone, Matteo Matteucci, Marco Ponzi. *"Ant colony optimization for stochastic user equilibrium"*. I sistemi stradali di trasporto nella societa' dell'informazione pp. 175-199, 2005.

F. Articles in National Journals

- F4. Luca Iocchi, Emanuele Menegatti, Andrea Bonarini, Matteo Matteucci, Enrico Pagello, Luigia Carlucci Aiello, Daniele Nardi, Fulvio Mastrogiovanni, Antonio Sgorbissa, Renato Zaccaria, Rosario Sorbello, Antonio Chella, Marcello Giardina, Primo Zingaretti, Emanuele Frontoni, Adriano Mancini, Grazia Ciciirelli, Alessandro Farinelli, Domenico G. Sorrenti. *"Development of intelligent service robots"*. *Intelligenza Artificiale*, 7(2):139-152, IOS Press, 2013.
- F3. Alberto Ceriani, Lorenzo Mussone, Aldo Colombo, Paola Vigo, and Matteo Matteucci. *"Analisi Ex-Post di un campione di rotatorie per una revisione normativa (prima parte)"*. *Strade & Autostrade*, 3:126-130, EDI-CEM Srl. 2012.
- F2. Alberto Ceriani, Lorenzo Mussone, Aldo Colombo, Paola Vigo, and Matteo Matteucci. *"Analisi Ex-Post di un campione di rotatorie per una revisione normativa (seconda parte)"*. *Strade & Autostrade*, 4:132-135, EDI-CEM Srl. 2012.
- F1. Lorenzo Mussone, Marco Bassani, Matteo Matteucci, Davide Rizzi. *"L'analisi delle prestazioni di una rotatoria con tecniche di analisi delle immagini"*. *Strade & Autostrade* 3(81):164-170, EDI-CEM Srl, 2010.

G. Technical Reports

- G10. Christophe Leroux, Roberto Labruto, Chiara Boscarato, Franco Caroleo, Jan-Philipp Günther, Severin Löffler, Florian Münch, Susanne Beck, Elisa May, Corinne Huebert-Saintot, Madeleine de Cock Buning, Lucky Belder, Roeland de Bruin, Andrea Bonarini, Matteo Matteucci, Pericle Salvini, Burkhard Schafer, Amedeo Santosuosso, Eric Hilgendorf. ["Suggestion for a green paper on legal issues in robotics"](#). Contribution to Deliverable D3.2.1 on ELS issues in robotics. 2012.
- G9. Matteo Matteucci, Davide Rizzi, and Andrea Romanoni. *"3D reconstruction of vehicular trajectory from single and multiple camera images"*. Department of Electronics and Information, Politecnico di Milano, Technical Report 2012.10, 2012.
- G8. Nicola Gatti, Matteo Matteucci. *"CABA²L a Bliss Predictive Composition Assistant for AAC Communication Software"*, Technical Report N 2003.44, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 2003.
- G7. D. Danzi, G. Gatti, Nicola Gatti, Matteo Matteucci, Licia Sbattella. *"BLISS 2003: an Assistive Technology Aid for Verbal Impaired People"*, Technical Report N 2003.43, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 2003.
- G6. Andrea Bonarini, Matteo Matteucci, Marcello Restelli. *"A novel model to rule behavior interaction"*, Technical Report N 2003.42, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 2003.
- G5. Matteo Matteucci. *"Publish/Subscribe Middleware for Robotics: Requirements and State of the Art"*, Technical Report N 2003.3, Dipartimento di Elettronica e Informazione - Politecnico di Milano, 2003.
- G4. Matteo Matteucci. *"Compound techniques for quality analysis in automotive laser welding"*, Technical Report N 2002.40, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 2002.
- G3. Matteo Matteucci. *"ELeaRNT: Evolutionary Learning of Rich Neural Network Topologies"*, Technical Report N CMU-CALD-02-103, Carnegie Mellon University, Pittsburgh PA, 2002.
- G2. Andrea Bonarini, Claudio Bonacina, Matteo Matteucci. *"A framework to support the reinforcement function design in real world, agent-based, application"*, Technical Report N 99.73, Dipartimento di Elettronica e Informazione, Politecnico di Milano, 1999.
- G1. Matteo Matteucci. *"Fuzzy Learning Classifier System: Issues and Architecture"*, Technical Report N 99.71, Dipartimento di Elettronica e Informazione, Politecnico di Milano, Milano I, 1999.

H. Accepted, Under Review, and arXiv preprints

- H19. Paolo Cudrano, Simone Mentasti, Matteo Matteucci, Mattia Bersani, Stefano Arrigoni, Federico Cheli. *"Advances in centerline estimation for autonomous lateral control"*. In IEEE Intelligent Vehicles Symposium (IV). 2020 - Accepted

- H18. Mattia Bersani, Simone Mentasti, Paolo Cudrano, Michele Vignati, Matteo Matteucci, Federico Cheli. *"Robust vehicle pose estimation from vision and INS fusion"*. In IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC) . 2020 - Accepted.
- H17. Alessandro Brusaferri, Matteo Matteucci, Danial Ramin, Stefano Spinelli, Andrea Vitali. *"Probabilistic Day-Ahead Energy Price Forecast by a Mixture Density Recurrent Neural Network"*. In 7th International Conference on Control, Decision and Information Technologies (CoDIT).2020 - Accepted.
- H16. Alessandro Brusaferri, Matteo Matteucci, Stefano Spinelli, Andrea Vitali. *"Extracting Finite State Representations from Recurrent Models of Industrial Cyber Physical Systems"*. In 7th International Conference on Control, Decision and Information Technologies (CoDIT). 2020 - Accepted.
- H15. Alessandro Brusaferri, Matteo Matteucci, Pietro Portolani, Stefano Spinelli, Andrea Vitali. *"Hybrid System Identification Using a Mixture of NARX Experts with LASSO-Based Feature Selection"*. In 7th International Conference on Control, Decision and Information Technologies (CoDIT).2020 - Accepted.
- H14. Aida Brankovic, Matteo Matteucci, Marcello Restelli, Luca Ferrarini, Luigi Piroddi, Andrea Spelta, and Fabrizio Zausa. *"A data-based approach for the prediction of stuck-pipe events in oil drilling operations"*. Abu Dhabi International Petroleum Exhibition & Conference (ADIPEC). Abu Dhabi. 2020 - Accepted
- H13. Francesco Pinto, Andrea Romanoni, Philip Torr, Matteo Matteucci. *"SECI-GAN: Semantic and Edge Conditioned Inpainting for dynamic object removal"*. In British Machine Vision Conference. 2020 - Under review.
- H12. Eugenio Lomurno, Andrea Romanoni, Matteo Matteucci. *"Improving Multi-View Stereo Completeness via Super-Resolution"*. In 25th Intl. Conference on Pattern Recognition (ICPR). 2020 - Under Review
- H11. Chiara Plizzari, Marco Cannici, Matteo Matteucci. *"Spatial Temporal Transformer Network for Skeleton-Based Action Recognition"*. European Conference on Computer Vision (ECCV). 2020 - Under review.
- H10. Marco Cannici, Marco Ciccone, Andrea Romanoni, Matteo Matteucci. *"A Differentiable Recurrent Surface for Asynchronous Event-Based Data"*. European Conference on Computer Vision (ECCV). 2020 - Accepted
- H9. Linda Greta Dui, Francesca Lunardini, Cristiano Termine, Matteo Matteucci, Simona Ferrante. *"A Tablet-Based App to Discriminate Children at Potential Risk of Handwriting Alterations in a Preliteracy Stage"*. In 42nd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC). 2020 - Accepted
- H8. Linda Greta Dui, Francesca Lunardini, Cristiano Termine, Matteo Matteucci, Natale Adolfo Stucchi, Nunzio Alberto Borghese, Simona Ferrante. *"A Tablet App to Anticipate Handwriting Skills Screening at Preliteracy: Instrument Validation Study"*. JMIR Serious Games. 2020 - Under Review
- H7. Luca Cavalli, Gianpaolo Di Pietro, Matteo Matteucci. *"Learning Affordance Prediction via Task-Oriented Grasp Quality Metrics"*. Frontiers Neurorobotics. 2020 - Under review.
- H6. Mattia Bersani, Simone Mentasti, Pragyan Dahal, Stefano Arrigoni, Michele Vignati, Federico Cheli, Matteo Matteucci. *"An integrated algorithm for ego-vehicle and obstacles state estimation for autonomous driving"*. Robotics and Autonomous Systems. 2020 - Accepted.
- H5. Hafeez Husain Cholakkal, Simone Mentasti, Mattia Bersani, Stefano Arrigoni, Matteo Matteucci, Federico Cheli. *"LiDAR - stereo camera fusion for accurate depth estimation"*. In AEIT International Conference of Electrical and Electronic Technologies for Automotive (AUTOMOTIVE). 2020 - Under Review
- H4. Simone Mentasti, Mattia Bersani, Matteo Matteucci, Federico Cheli. *"Multi-state end-to-end learning for autonomous vehicle lateral control"*. In AEIT International Conference of Electrical and Electronic Technologies for Automotive (AUTOMOTIVE). 2020 - Under Review.
- H3. Luca Bascetta, Marcello Farina, Alessandro Gabrielli, Matteo Matteucci. *"A feedback linearisation algorithm for single-track models with structural stability properties"*. arXiv preprint arXiv:2003.13793
- H2. Luca Cavalli, Gianpaolo Di Pietro, Matteo Matteucci. *"Towards Affordance Prediction with Vision via Task Oriented Grasp Quality Metrics"*. arXiv preprint arXiv:1907.04761. 2019
- H1. Francesco Lattari, Marco Ciccone, Jonathan Masci, Francesco Visin. *"ReConvNet: Video Object Segmentation with Spatio-Temporal Features Modulation"* arXiv preprint arXiv:1806.05510. 2018

Autorizzo il trattamento dei miei dati personali presenti nel cv ai sensi del Decreto Legislativo 30 giugno 2003, n. 196 “Codice in materia di protezione dei dati personali” e del GDPR (Regolamento UE 2016/679).

Date: 20/06/2020

In compliance with the Italian legislative Decree no. 196 “Codice in materia di protezione dei dati personali” dated 30/06/2003, and the GDPR (EU 2016/679). I hereby authorize you to use and process my personal details contained in this document.”

Date: 20/06/2020

CV SUMMARY - MATTEO MATTEUCCI

This document is a Summary of Matteo Matteucci Curriculum Vitae and Studiorum (CV) for applications which require a shorter resume; it complements the CV with a selection by the applicant of the information provided by the full CV for a fast access and consultation, but it does not replace the full CV version for a more complete view of Matteo Matteucci teaching, research and professional activities. In addition, with respect to the CV, a detailed report of Matteo Matteucci funding track is reported.

DATE OF BIRTH 23/04/1974

EDUCATION

2003 **PhD Degree** in Computer Engineering and Automation, Politecnico di Milano, Milan
 2002 **Master of Science Degree (MS)** in Knowledge Discovery and Data Mining, Center for Automatic Learning & Discovery, Carnegie Mellon University, Pittsburgh (PA - USA)
 1999 **Master of Science Degree (Laurea)** in Computer Engineering, Politecnico di Milano, Milan (Final Grade 100/100 *cum Laude*)

CAREER

2015 – Present **Associate Professor**, Politecnico di Milano, Milan
 2008 – 2015 **Assistant Professor**, Politecnico di Milano, Milan
 2003 – 2015 **Post-Doc Researcher** (Assegnista di Ricerca), Politecnico di Milano, Milan
 2001 – 2002 **Visiting Student**, Carnegie Mellon University, Pittsburgh (PA), for 1 year
 2000 – 2003 **Ph.D. Student**, Politecnico di Milano, Milan

PARENTAL/ILLNESS LEAVE PERIODS

None

HABILITATION

Type of habilitation	Country	SSD (if Italian habilitation) or topic area	Date of achievement
Full Professor	Italy	ING-INF/05 (09/H1)	04/04/2017
Associate Professor	Italy	INF-01 (01/B1)	29/01/2014
Associate Professor	Italy	ING-INF/05 (09/H1)	03/12/2013

RESEARCH INTERESTS

- **Robotics:** Robot Benchmarking, Simultaneous Localization and Mapping (SLAM), Computer Vision, Sensor Fusion
- **Artificial Intelligence:** Artificial Neural Networks, Deep Learning, Applied Machine Learning

LEADERSHIP IN COMPETITIVE RESEARCH PROJECTS

Project Acronym	Time Period	Funding Institution	Funding Scheme	Role of the applicant	Budget for applicant's institution	Budget managed by applicant
AGRIFOODT EF	2023-2027	European Commission	DIGITAL EU	Participant	€ 1.600.000	€ 1.600.000
AI-SPRINT	2021-2023	European Commission	H2020	Co-coordinator	€ 1.047.000	€ 523.500
LoViCoSpec	2020	EIT Manufacturing	EIT	Principal Investigator	€ 75.645	€ 75.645

METRICS	2020-2022	European Commission	H2020	Principal Investigator	€ 166.865	€ 166.865
DEEPFIELD	2019-2022	European Commission	H2020	Principal Investigator	€ 129.250	€ 129.250
L4MS	2017-2021	European Commission	H2020	Task Leader	€ 423.750	€ 85.625*
BEAST	2019-2020	European Commission	H2020 (Cascade funding)	Principal Investigator	€ 48.750	€ 48.750
MADROB	2019-2020	European Commission	H2020 (Cascade funding)	Principal Investigator	€ 48.750	€ 48.750
MATTCH	2019-2020	European Space Agency	ESA	Principal Investigator	€ 50.500	€ 50.500
Sciroc	2018-2021	European Commission	H2020	Principal Investigator	€ 58.750	€ 58.750
Plug & Bench	2018-2019	European Commission	H2020 (Cascade funding)	Principal Investigator	€ 129.375	€ 129.375
TEINVEIN	2018-2019	Regione Lombardia	FESR	Task Leader	€1.026.468	€ 77.825*
Drone112	2018	EIT Digital	EIT	Principal Investigator	€ 58.500	€ 58.500
RockEU2	2016-2018	European Commission	FP7	Principal Investigator	€ 133.000	€ 133.000
Grape	2016-2018	European Commission	FP7 (Cascade funding)	Task Leader	€ 134.400	€ 65.000*
Clud4Drones	2017	EIT Digital	EIT	Principal Investigator	€ 92.700	€ 92.700
RoCKIn	2013-2015	European Commission	FP7	Principal Investigator	€ 228.980	€ 228.980
ALMA	2013-2015	European Commission	AAL Joint Program	Principal Investigator	€ 284.390	€ 284.390
SHELL	2014-2015	Regione Lombardia	FESR	Task Leader	€ 869.600	€ 68.400*
SINOPIAE	2012-2014	Regione Lombardia / MIUR	Progetti Cluster	Task Leader	€ 119.000	€ 35.000*
ROAMFREE	2011-2013	MIUR	PRIN	Project Coordinator	€ 94.215	€ 94.215
RAWSEEDS	2006-2009	European Commission	FP6	Project Coordinator	€ 192.980	€ 192.980
BCI-EA	2006-2009	Regione Lombardia / POLIMI	Grant "Avvio alla ricerca"	Principal Investigator	€ 30.000	€ 30.000
					TOTAL	€ 2.678.000

*The budget handled by the applicant refers to the applicant unit activities only.

LEADERSHIP IN INDUSTRIAL RESEARCH PROJECTS

Project Acronym	Time Period	Funding Company	Role of the applicant	Budget for the applicant's institution	Budget handled by the applicant
Tigheder Prototipo	2021-2022	WebRatio S.r.l.	Principal Investigator	€ 6.000	€ 6.000
SNAM Fase 1	2021 - 2022	SNAM SpA	Principal	€ 30.000	€ 30.000

			Investigator		
Thales Tram	2021	Thales Italia SpA	Co-Principal Investigator	€ 30.000	€ 15.000
MADE - MdS	2021-2022	MADE S.C.A.R.L.	Co-Principal Investigator	€ 54.000	€ 27.000
Antemotion	2021-2022	Antemotion S.r.l.	Principal Investigator	€ 60.000	€ 60.000
Navigazione Jobot	2020-2021	EUTRONICA S.r.l	Principal Investigator	€ 26.548	€ 26.548
Tracetoo	2020	Tracetoo S.r.l.	Principal Investigator	€ 18.000	€ 18.000
Tecnomare Localizzazione	2019-2020	ENI SpA	Co-Principal Investigator	€ 72.230	€ 42.615*
EB-MIMO	2019-2020	Huawei	Co-Principal Investigator	€ 200.000	€ 65.000*
MOXOFF	2019	Moxoff SpA	Co-Principal Investigator	€ 15.000	€ 5.000*
AGS Train	2019	AGS	Co-Principal Investigator	€ 21.800	€ 10.900*
SDL-ABB	2018-2019	ABB SpA	Principal Investigator	€ 65.000	€ 65.000
Tecnomare DJI Guidance	2018-2019	ENI SpA	Principal Investigator	€ 20.000	€ 20.000
Tecnomare ROS	2018	ENI SpA	Principal Investigator	€ 12.000	€ 12.000
Tecnomare Drone	2017-2018	ENI SpA	Workpackage Leader	€ 32.100	€ 16.050*
ENI-NPT	2017-2018	ENI SpA	Task Leader	€ 230.000	€ 53.375*
ASTER	2017-2018	Elettronica ASTER	Principal Investigator	€ 130.000	€ 130.000
Niguarda	2013-2018	ARCA - Regione Lombardia	Principal Investigator	€ 57.600	€ 57.600
COMAU	2016	COMAU SpA	Co-Principal Investigator	€ 20.000	€ 10.000*
STELE	2015-2016	Infosolution SpA	Principal Investigator	€ 22.000	€ 22.000
Gaiotto	2012-2013	Gaiotto Automation SpA	Principal Investigator	€ 10.300	€ 10.300
Progetto ON	2010-2013	Infosolution SpA	Principal Investigator	€ 70.000	€ 70.000
				TOTAL	€ 772.388

*The reported budget refers to the task/workpackags activities only.

SCIENTIFIC PRODUCTION AND METRICS

- **Scientific Productivity:** 256 publications (184 entries on Scopus, 256 co-authors according to Scopus, no single-author papers):
 - Author/Co-author of 24 top-ranked Q1 journal papers (including IEEE Transactions on Systems, Man, and Cybernetics - Part B; International Journal of Approximate Reasoning, IEEE Transactions on Dependable and Secure Computing, ACM Transactions on Computer-Human Interaction, Pattern Recognition Letters, Bioinformatics, IEEE Robotics and Automation Magazine, IEEE Transactions on Knowledge and Data Engineering, Remote Sensing, Briefings in Bioinformatics, BMC Bioinformatics, Information Sciences, Robotics and Autonomous Systems, Methods of Information in Medicine, Geomorphology, Information Fusion, Computational Intelligence and Neuroscience, Clinical Neurophysiology, Journal of Advanced Transportation, Journal of Neural Engineering, Control Engineering Practice, PloS ONE, Applied Energy)
 - Author/Co-author of 161 scientific publications on peer-reviewed conferences including 7 top-level A++/A+ conferences (including: International Conference on computer Vision (ICCV), IEEE/RSJ International

Conference on Intelligent Robots and Systems (IROS), ACM SIGCHI Conference on Human Factors in Computing Systems (CHI))

- Inventor/Co-inventor of **8 patent applications** (3 Italian + 1 European + 3 EU extensions + 1 US extension) with, among others, Centro Ricerche FIAT, Politecnico di Torino, Infosolution SpA, Università degli Studi Milano-Bicocca, ABB SpA (**2 Italian patents** and **2 EU Extensions** out of **8 patent applications** already granted)

- **Publication Impact:** Based on Google Scholar: h-index **34** citations **4778**
Based on Scopus: h-index **24 (23 w/o self citations)** citations **2298**

AWARDS AND RECOGNITIONS (SELECTION OF PAST 10-YEARS)

- 2019 **Best Paper Award** for the paper "Asynchronous Convolutional Networks for Object Detection in Neuromorphic Cameras" at Second International Workshop on Event-based Vision and Smart Cameras in conjunction with IEEE CVPR 2019
- 2018 Awarded with a 20K\$ **Microsoft Azure sponsorship grant** as a support for the PhD Course on "Deep Learning: Theory Techniques and Applications"
- 2017 Awarded with a 20K\$ **Microsoft Azure sponsorship grant** for the experimentation on "Deep Learning machine on the cloud"
- 2016 **Best Paper Award** for the paper "ReSeg: A Recurrent Neural Network-based Model for Semantic Segmentation" at DeepVision 2016, International Workshop on Deep Learning in Computer Vision in conjunction with IEEE CVPR 2016
- 2016 **Best Paper Award** for the paper "Automatic 3D Reconstruction of Manifold Meshes via Delaunay Triangulation and Mesh Sweeping" at 2016 IEEE Winter Conference on Applications of Computer Vision
- 2015 Awarded by NVIDIA with a Titan X GPU within the **Academic Hardware Grant**
- 2014 Awarded, with Dott. Martino Migliavacca and Prof. Andrea Bonarini, of "The Business Grant and The Acceleration Program" with the project R2P in the **Working Capital Accelerator Program (WCAP 2014)** by Telecom Italia
- 2011 Winner, with the Industrial Partner Infosolution SpA, of "**Premio Antonio D'Auria** 2010 per progetti e prototipi di dispositivi meccatronici innovativi di ausilio a disabili motori" from Società Italiana di Robotica e Automazione
- 2011 Winner, with the Industrial Partner Infosolution SpA, of the Intesa San Paolo "**Startup Initiative**" and the Italian round of the "Global Social Venture Competition" with the RobyWheelChair project

INVITED TALKS AND SEMINARS (SELECTION OF PAST 10-YEARS)

Academic context

- 13/12/2019 Invited talk "*Robotics and Farming: Technologies & Innovations for an Agriculture 4.0*" at the 2nd Annual Automation & Robotics for Non-Road Applications, Gothia Towers, Gothenburg
- 20/09/2019 Invited talk "*Intelligent robots are at the door ... but can they open it?*" at the Sciroc-AI4EU joint Workshop on Evaluation and Benchmarking of Human-Centered AI Systems, Milton Keynes
- 21/07/2017 Invited talk "*The RAWSEEDS Project: How we did it!*" at the CHIST-ERA Conference on "Object recognition and manipulation by robots: Data sharing and experiment reproducibility", Krakov
- 28/09/2015 Invited talk "*Do we really need to replicate experiments?*" in the "Open forum on evaluation of replication of experiments and benchmarking in robotics research" at the 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2015), Hamburg
- 15/09/2015 Invited lecture on "*The RAWSEEDS Project: How we did it!*" at the 2015 IEEE-RAS Summer School on Experimental Methodology, Performance Evaluation and Benchmarking in Robotics, Benicassim
- 15/09/2015 Invited lecture on "*Benchmarking and Competitions*" at the 2015 IEEE-RAS Summer School on Experimental Methodology, Performance Evaluation and Benchmarking in Robotics, Benicassim
- 04/06/2014 Invited talk "RoCKIn: Driving Robotics Forward through Collaboration, Benchmarking and Competition" in the Forum on "Robotics Challenges and Competitions" at the 2014 IEEE International Conference on Robotics and Automation (ICRA 2014), Hong Kong
- 12/04/2014 Invited talk titled "Benchmarking in Robotics: Challenges and Visions" in the Workshop "Shared infrastructures, benchmarking and performance evaluation" at the 2014 European Robotics Forum (ERF), Rovereto
- 23/01/2014 Invited talk "Benchmarking through competitions" to the EuRoC Challenge Design Workshop, Munich

26/01/2012 Invited seminar "Integrating P300 and Error Potentials in a Single BCI: Algorithms, Techniques & Performance Assessment" in the seminar program "Bioengineering Seminar Series" at the Department of Bioengineering of Politecnico di Milano, Milan

Industrial context

18/11/2019 Invited talk "AI applied to 5G: Learning Methods for Channel Estimation" at the Huawei Workshop on High Frequency Technologies for 5G, Politecnico di Milano, Milan

15/10/2019 Invited talk "Dai Big Data ai Sistemi di Autoapprendimento" hosted by Fondazione Faraggiana, Novara

04/02/2019 Invited talk "Intelligenza artificiale: ambiti applicativi e impatto sui sistemi industriali" hosted by APITech as part of the event "Intelligenza Artificiale e Sistemi Industriali: Applicazioni nel contesto Impresa 4.0", Lecco

18/09/2018 Invited talk "Cosa intendiamo per IA ... tutto ciò che avreste voluto sapere e non avete mai osato chiedere." at Convegno "Intelligenza Artificiale e Ricerca Clinica", Milan

05/07/2018 Invited talk "A che punto siamo con le tecnologie dell'AI e il loro futuro possibile" at Convegno "AI Tech Conference: Artificial Intelligence per le imprese e per un mondo migliore", Milan

30/11/2017 Invited talk "Sul senso (comune) dei Big Data" at Convegno "Big Data e Robotica nella fabbrica digitale", Milan

29/04/2017 Invited talk "AI and Cognitive Technology: Why, What, How ..." at the Cognitech Conference, Milan

21/04/2016 Invited talk "Modular development of service robots: from Rapid Robot Prototyping to Nova Core" in the Sessione Specialistica "Robot & Co." at A&T 2016, Torino

18/04/2016 Invited talk "Do Big Data make (common) sense?" at Convegno "L'automazione nell'era dei Big Data: scenari e prospettive", at MECSPE, Parma

TEACHING EXPERIENCE (SELECTION OF PAST 5-YEARS)³

Institution name	Course name	Credits or Hours	No. of students	Reference Study Course	Time period	Students Eval. ⁴
POLIMI UNIMI	Machine Learning	6 CFU	27	Bioinformatics for Computational Genomics	2019/2020	--
POLIMI	Artificial Neural Networks and Deep Learning	5 CFU	256	Computer Science Engineering	2019/2020	--
POLIMI (Polo di Como)	Machine Learning	5 CFU	13	Computer Science Engineering	2018/2019	3.9/4.0
POLIMI	Robotics	5 CFU	214	Computer Science Engineering	2018/2019	3.2/4.0
POLIMI (Poli di Como)	Cognitive Robotics	5 CFU	14	Computer Science Engineering	2018/2019	3.0/4.0
POLIMI (Polo di Como)	Machine Learning	5 CFU	43	Computer Science Engineering	2017/2018	3.5/4.0
POLIMI	Robotics	5 CFU	93	Computer Science Engineering	2017/2018	3.1/4.0
POLIMI (Polo di Como)	Cognitive Robotics	5 CFU	14	Computer Science Engineering	2016/2017	--
POLIMI (Polo di Como)	Machine Learning	5 CFU	58	Computer Science Engineering	2016/2017	High
POLIMI	Robotics	5 CFU	133	Computer Science Engineering	2016/2017	Medium
POLIMI (Polo di Como)	Cognitive Robotics	5 CFU	44	Computer Science Engineering	2016/2017	High
POLIMI	Robotics	5 CFU	119	Computer Science Engineering	2015/2016	High

³ Courses reported here have been selected as those for which I have been responsible or co-responsible for. A complete report of teaching experience can be found in my complete Curriculum Vitae.

⁴ Institution mean and percentiles for the years 2018/2019 and 2017/2018 are: Mean = 3.1, 75% = 3.6, 25% = 2.6.

POLIMI (Polo di Como)	Data Analysis and Retrieval	10 CFU	66	Computer Science Engineering	2015/2016	High
POLIMI	Advances in Deep Learning with Applications in Text and Image Processing	3/5 CFU*	72	PhD School	2018/2019	--
POLIMI	Integration and Computation Analysis of Genomic Information	1/5 CFU*	5	PhD School	2017/2018	--
POLIMI	Deep Learning: Theory, Techniques and Applications	3/5 CFU*	116	PhD School	2017/2018	--
POLIMI	Soft Computing: Theory, Techniques, and Applications	2/5 CFU*	31	PhD School	2016/2017	--
POLIMI	Biosignal Processing 4 All: Affective Computing, Biometrics Neuromarketing	1/5 CFU*	19	PhD School	2016/2017	--
POLIMI	Big Data Technologies	1/5 CFU*	24	PhD School	2016/2017	--
CEFRIEL	Statistical Machine Learning	24 h	22	Master in Alto Apprendistato	2019	--
CEFRIEL	Advanced Machine Learning	32 h	20	Master in Alto Apprendistato	2018	--
CEFRIEL [Nokia]	Machine Learning	14 h	15	Company Training	2020	
CEFRIEL [Nokia]	Machine Learning for Big Data	12 h	15	Company Training	2020	
POLIMI [AGS]	Deep learning per l'analisi di immagini	24 h	12	Company Training	2019	--
CEFRIEL [Nokia]	Artificial Intelligence and Machine Learning	36 h	15	Company Training	2019	--
MIP [COFIDIS]	Machine Learning	8 h	18	Company Training	2019	4.6/5.0
CEFRIEL [Edison]	Artificial Intelligence	24 h	20	Company Training	2018	--
CEFRIEL [Bosch]	Data Mining and Machine Learning	16 h	24	Company Training	2018	--

*The reported credits refer to the part of the course taught by the applicant over the total number of credits.

INSTITUTIONAL RESPONSIBILITIES (SELECTION OF PAST 10-YEARS)

- 2019 - present **Member** of Technical Board on “Robust and Efficient mmWave Communication” in the Joint Research Center (JRC) between Huawei and Politecnico di Milano
- 2018 - present **Scientific Director** of the “Master di I livello in Artificial Intelligence and Machine Learning” offered by Politecnico di Milano in conjunction with CEFRIEL
- 2018 - present Member of the **Faculty Board** of the “Data Analytics and Decision Sciences” PhD Program at Politecnico di Milano
- 2016 - present **Segretario Consiglio Corso di Studi** in Computer Science and Engineering of Politecnico di Milano
- 2015 - present **Member** of “REPRISE: Register of Expert Peer Reviewers for Italian Scientific Evaluation” for Ministero dell’Istruzione dell’Università e della Ricerca
- 2015 - present **Coordinator** (and co-founder in 2014) of the Interaction between Driver, Road Infrastructure, Vehicle, and Environment (I.DRIVE) interdepartmental laboratory of Politecnico di Milano (<http://idrive.polimi.it>)
- 2013 - present **Member** (and co-founder in 2012) of the Assistive Technology Group of Politecnico di Milano - Polo Regionale di Como (<http://atg.deib.polimi.it/>)
- 2009 - present **Member** of “Commissione Orari” for the Computer Engineering track of Politecnico di Milano - Polo Regionale di Como
- 2008 - present **Delegato SAT** (Struttura Accademica dei Tirocini) for the Computer Engineering track at Politecnico di Milano - Campus Leonardo

SUPERVISION OF MASTER, DOCTORAL STUDENTS AND POSTDOCTORAL RESEARCHERS

- 2003 – present **Advisor/Co-advisor** of **25** Doctoral Students at Politecnico di Milano, **2** Doctoral Students at Università degli Studi di Milano-Bicocca, and **1** Doctoral Student at Politecnico di Torino
- 2014 – present **Opponent Member** of **9** Doctoral Examination Committees at Politecnico di Milano, Politecnico di Torino, Università degli Studi di Roma “La Sapienza”, Università della Svizzera Italiana, Università della Calabria
- 2010 – present **Advisor/Co-advisor** of **100+** Master Students in Computer Engineering, Politecnico di Milano, Milan

ORGANIZATION OF SCIENTIFIC MEETINGS (SELECTION OF PAST 10-YEARS)

- 2020 **Local Chair** for the 25th International Conference on Pattern Recognition (ICPR 2020), Milan, Italy, January 2021
- 2018 **Co-Chair** of ERF 2018 Workshop on “Research Reproducibility in Robotics” (part of the Competitions and Benchmarking session) co-organized with Fabio Bonsignorio, Tampere, Finland, 12th March 2018
- 2017 **Co-Chair** of ERF 2017 Workshop on “Robotics Competitions and Challenges” Co-organized with Agostino De Santis, Edinburgh, Scotland, 24th March 2017
- 2012 **Co-Chair** of IROS 2012 Workshop on “Progress, Challenges and Future Perspectives in Navigation and Manipulation Assistance for Robotic Wheelchairs”, Vilamoura, Algarve, Portugal, October 7th to 12th, 2012.

COMMISSIONS OF TRUST (SELECTION OF PAST 10-YEARS)

- 2017 - present IEEE RAS Standards Development **Working Group** “3D-MDR - Robot 3D Map Data Representation”
- 2014 - present IFAC **Technical Committee** “7.5. Transportation and Vehicle Systems - Intelligent Autonomous Vehicles”
- 2007 - present **Program Committee** of 20+ International Conferences and Workshops
- 2020 **Senior Program Committee Member** of the European Conference on Artificial Intelligence
- 2020 **Chair of Evaluation Committee** for a PhD Scholarship from Atlantic International Research Center (AIR Center) and the Portuguese Foundation for Science and Technology (FCT)
- 2020 **Member of Evaluation Committee** for a PhD Scholarship from Atlantic International Research Center (AIR Center) and the Portuguese Foundation for Science and Technology (FCT)
- 2019 **Member of Evaluation Committee** for a 09/H1 (ING-INF/05) “Professore II fascia” position at Università degli studi di Roma “La Sapienza”
- 2018 **Associate Editor** for the Special Session on “Research Reproducibility and Benchmarking of Intelligent Robots” of 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2018).
- 2017 **Associate Editor** of 2017 IEEE International Conference on Robotics and Automation (ICRA 2017).
- 2017 **Member of Evaluation Committee** for the 09/H1 (ING-INF/05) RTDa position funded by “Progetto H2020 INTCATCH”, Dipartimento di Informatica, Università degli Studi di Verona
- 2015 - 2016 IEEE RAS **Standardization Committee** for the “1873-2015 - IEEE Standard for Robot Map Data Representation for Navigation”
- 2014 **Project Reviewer and rapporteur** for the SIR 2014 (Scientific Independence of young Researchers) funding program by the Italian Ministry of University and Research (MIUR)
- 2011 **Project Reviewer** for the PRIN 2010-2011 (Programmi di Ricerca Scientifica di Rilevante Interesse Nazionale) funding program by the Italian Ministry of University and Research (MIUR)

SPIN-OFF

- **Nova Labs S.r.l.:** founded in November 2015 by Martino Migliavacca (CEO), Davide Rizzi (CTO), Andrea Bonarini, Matteo Matteucci, and Industrio Venture. Registered as Politecnico di Milano spin-off. Year 2019 income € 746.000.

EVIDENCE OF TECHNICAL ACCOMPLISHMENT

- **RAWSEEDS Benchmarking Toolkit:** from 2006 to 2009 I coordinated the FP6FP6-045144 European project RAWSEEDS (www.rawseeds.org) out of which the first European Benchmarking Toolkit for Simultaneous

Localization and Mapping was developed. It consists of 11 multi-sensor datasets (~600GB), validated ground truth, and evaluation metrics. The project and its results have been recognized as the first European effort in Autonomous Robot benchmarking and evaluation. The RAWSEEDS benchmarking toolkit is still downloaded, used, and referenced in scientific publications after the end of the project in 2009; the project website has been visited by 28,325 new users since May 2009 (after the end of the project) for a total of 200.000+ pageviews.

- **Rapid Robot Prototyping (Nova Core):** HW/SW framework for the development of robots, started in 2013, as part of the PhD thesis of Martino Migliavacca, it has been awarded by Telecom Italia with “The Business Grant and The Acceleration Program” in the Working Capital Accelerator Program (WCAP 2014). Later, under the name of Nova Core, Rapid Robot Prototyping has become the technological asset of Politecnico di Milano spin-off Nova Labs S.r.l. to develop products such as the COMAU robot E.Do (<https://www.comau.com/en/our-competences/robotics/edo>).
- **Unmanned vehicles for the Italian Ministero della Difesa:** Between 2010 and 2013 I participated to the QUADRIVIO project (Scientific directors Prof. Gianantnio Magnani and Prof. Luca Bascetta) for the development of an autonomous system composed of an Unmanned Ground Vehicle (UGV) and an Unmanned Aerial Vehicle (UAV) for search and rescue operations. I have been responsible for the development of the UGV autonomous navigation software system. The vehicle has been successfully demonstrated by Aero Sekur under the name UxV to the officers of the Ministero della Difesa (<http://www.cybernaia.it/rubriche/rubricadett.php?idnews=3993>). Based on the success of the UxV demonstration, between 2017 and 2018 I have been the Principal Investigator in the research project which has led to the development of the IDRA UGV (<http://www.elaster.it/default.php?idm=57&idp=64>) with Elettronica ASTER SPA. This time I have been responsible for the development of the entire unmanned vehicle autonomous navigation software. The outcome of the project, i.e., an electrical autonomous vehicle for CBRN defence, has been demonstrated successfully in front of the officers of the Ministero della Difesa (TERRARM).
- **Bed Mover:** Between 2013 and 2018 I have worked on the development of an autonomous hospital bed mover as part of the Regione Lombardia Pre-Commercial Procurement call by the Azienda Regionale Centrale Acquisti (ARCA SpA) in Associazione Temporanea di Scopo (ATS) with Università degli Studi di Milano-Bicocca and Infosolution SpA. The project has led to the application for and Italian patent with European and US extensions to protect the idea under the commercial product “BeN - Bed Navigator” commercialized by Infosolution SpA under the brand Kedos M.I.T (https://www.kedos-mit.com/images/brochure/KMIT_BeN_Bed_Navigator_ITA_19_V02-min.pdf).
- **LURCH Autonomous Wheelchair:** In 2007 I started the LURCH project for the development of an autonomous wheelchair, which could help people unable to move autonomously. This effort has resulted in an autonomous wheelchair, which sports the “shared autonomy” design paradigm: depending on the needs of the user, the robot supplies the needed autonomy from simple safety enhancer up to fully autonomous vehicle. The project has reached the maturity of an industrial prototype and this has been recognized by the two prizes (*Antonio D’Auria SIRI Prize* and Intesa *San Paolo “Startup Initiative” award*) obtained under the commercial name of RobyWheelChair. The competences built during the development of LURCH prototype have been the base for the project “ALMA: Aging without Losing Mobility and Autonomy” which led to the deployment of an autonomous wheelchair at Casa Anziani Giardino in Chiasso. The LURCH prototype has been demonstrated in public events such as Robotica Fair, and MeetMeTonight, and it has been shown on the national tv (<http://it.youtube.com/watch?v=IRP-ae4iaZA>).
- **P300 Brain-Computer Interface:** While developing the autonomous wheelchair LURCH, lot of effort has been directed to the development of effective human robot interfaces, the most challenging one being an Event Related Brain-Computer Interface. From 2007 to 2009 I developed with my PhD student Bernardo Dal Seno an interface able to capture the user intent from brain activity so to transfer this intent to the autonomous wheelchair or to enable people suffering severe pathologies (e.g., Amyotrophic Lateral Sclerosis) to communicate. In the winter of 2009 this BCI was used to control the autonomous wheelchair and this had quite resonance on the press (<http://airwiki.elet.polimi.it/index.php/MediaCoverage>). After that the lab prototype has been tested with some Amyotrophic Lateral Sclerosis patients within the two projects ON and STELE funded by FILAS SpA.

FIFTEEN MOST RELEVANT PUBLICATIONS⁵

⁵ For journal paper the SJR (Scimago Journal & Country Rank <https://www.scimagojr.com/>) quartiles for the year of publication of the paper, or the year before if not yet available, is reported. When the journal has multiple subjects the quartile of the subject appropriate for the paper topic is reported. For conference papers the class from the GII-GRIN-SCIE Rating (<http://valutazione.unibas.it/gii-grin-scie-rating/ratingSearch.jsf>) for the venue of the paper is

1. Andrea Romanoni, Matteo Matteucci. "Mesh-based camera pairs selection and occlusion-aware masking for mesh refinement". Pattern Recognition Letters, vol. 125, pp. 364-372. Elsevier. 2019. [doi:10.1016/j.patrec.2019.05.006](https://doi.org/10.1016/j.patrec.2019.05.006) **SJR: Q1**
I supervised the Post-Doc who did the actual implementation of the method, I discussed with him and suggested improvements on the technique, I supervised and revised the writing of the paper as well as the rebuttal process.
2. Vahid Jalili, Matteo Matteucci, Jeremy Goecks, Yashar Deldjoo, Stefano Ceri. "Next Generation Indexing for Genomic Intervals". IEEE Transactions on Knowledge and Data Engineering, vol. 30, issue 10, p. 2008-2021, IEEE. 2018. [doi:10.1109/TKDE.2018.2871031](https://doi.org/10.1109/TKDE.2018.2871031) **SJR: Q1**
I supervised the two PhD students who did the work, I contributed to the definition of the original idea and data structure, I co-supervised the writing and co-reviewed the paper as well as the rebuttal process. Data structure is called Di4 (One-dimensional Intervals Incremental Inverted Index) and it is an improvement of Di3 adding a novel incremental encoding and adaptive indexes for gene expression.
3. Vahid Jalili, Matteo Matteucci, Marco Masseroli, Stefano Ceri. "Indexing Next-Generation Sequencing data". Information Sciences, vol. 384, pp. 90-109, Elsevier, 2017. [doi:10.1016/j.ins.2016.08.085](https://doi.org/10.1016/j.ins.2016.08.085) **SJR: Q1**
I supervised the PhD student doing the work, I contributed to the definition of the original idea and data structure, co-supervised the writing and co-reviewed the paper as well as the rebuttal process. The data structure is called Di3 (1D Interval Inverted Index).
4. Vahid Jalili, Matteo Matteucci, Marco J. Morelli, Marco Masseroli. "MuSERA: Multiple sample enriched region assessment". Briefings in Bioinformatics, vol.18, issue 3, pp. 367-381, Oxford Academics, 2017. [doi:10.1093/bib/bbw029](https://doi.org/10.1093/bib/bbw029) **SJR: Q1**
I co-supervised the PhD student doing the work, I contributed to the definition of the idea, I co-supervised the writing and co-reviewed the paper as well as the rebuttal process.
5. Jalili Vahid, Matteo Matteucci, Marco Masseroli, Marco J. Morelli. "Using combined evidence from replicates to evaluate ChIP-seq peaks". Bioinformatics, vol. 31, issue 17, pp. 2761-2769, Oxford Academics, 2015. [doi:10.1093/bioinformatics/btv293](https://doi.org/10.1093/bioinformatics/btv293) **SJR: Q1**
I co-supervised the PhD student doing the work, I contributed to the definition of the idea, I co-supervised the writing and co-reviewed the paper as well as the rebuttal process.
6. Francesco Amigoni, Emanuele Bastianelli, Jakob Berghofer, Andrea Bonarini, Giulio Fontana, Nico Hochgeschwender, Luca Iocchi, Gerhard K. Kraetzschmar, Pedro Lima, Matteo Matteucci, Pedro Miraldo, Daniele Nardi, Viola Schiaffonati. "Competitions for Benchmarking. Task and Functionality Scoring Complete Performance Assessment". IEEE Robotics and Automation Magazine, vol. 22, issue 3, pp. 53-61, IEEE, 2015. [doi:10.1109/MRA.2015.2448871](https://doi.org/10.1109/MRA.2015.2448871) **SJR: Q1**
I co-developed the proposed framework and co-invented the original idea (with co-authors Amigoni, Bonarini, Fontana, and Schiaffonati), I collaborated to the writing and reviewing of the paper as well as the rebuttal process. Names are in alphabetical order.
7. Andrea Bonarini, Matteo Matteucci, Martino Migliavacca, Davide Rizzi. "R2P: An open source hardware and software modular approach to robot prototyping". Robotics and Autonomous Systems, vol. 62, issue 7, pp. 1073-1084, Elsevier, 2014. [doi: 10.1016/j.robot.2013.08.009](https://doi.org/10.1016/j.robot.2013.08.009) **SJR: Q1**
I co-developed the proposed framework, I co-supervised the Bachelor (D. Rizzi) and PhD (M. Migliavacca) students designing, implementing, and evaluating the system, I participated in the writing and reviewing of the paper as well as to the rebuttal process. Names are in alphabetical order.
8. Tiziano D'Albis, Rossella Blatt, Roberto Tedesco, Licia Sbattella, Matteo Matteucci. "A predictive speller controlled by a brain-computer interface based on motor imagery". ACM Transactions on Computer-Human Interaction, vol. 19, issue 3, pp. 1-25, ACM New York, 2012. [doi:10.1145/2362364.2362368](https://doi.org/10.1145/2362364.2362368) **SJR: Q1**
I coordinated the whole research work this paper originated from, I co-supervised the Master (T. D'Albis) and PhD (R. Blatt) students developing and evaluating the system, I participated to the writing and reviewing of the paper as well as to the rebuttal process.
9. Federico Maggi, Matteo Matteucci, Stefano Zanero. "Detecting Intrusions through System Call Sequence and

reported.

Argument Analysis”. IEEE Transactions on Dependable and Secure Computing, vol. 7, issue 4, pp. :381-395, IEEE, 2010. [doi:10.1109/TDSC.2008.69](https://doi.org/10.1109/TDSC.2008.69) **SJR: Q1**

I co-supervised the PhD Student developing and evaluating the proposed framework and participated in the development of the original idea, I collaborated to the writing and reviewing of the paper as well as during the rebuttal process.

10. Federico Maggi, Matteo Matteucci and Stefano Zanero. “Reducing false positives in anomaly detectors through fuzzy alert aggregation”. Information Fusion, vol. 10, issue 4, pp. 300-311, Elsevier, 2009. [doi:10.1016/j.inffus.2009.01.004](https://doi.org/10.1016/j.inffus.2009.01.004) **SJR: Q1**

I co-supervised the PhD Student developing and evaluating the proposed framework, I collaborated to the writing and reviewing of the paper as well as the rebuttal process.

11. Andrea Romanoni, Matteo Matteucci. “TAPA-MVS: Textureless-Aware PATCHMATCH Multi-View Stereo” In Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10413-10422, Seoul, Korea, IEEE, 2019. [doi:10.1109/ICCV.2019.01051](https://doi.org/10.1109/ICCV.2019.01051) **GII-GRIN-SCIE Class 1 (A++)**

I supervised the Post-Doc doing the implementation of the method, discussed and suggested improvements on the technique, supervised and revised the writing of the paper as well as the rebuttal process (this conference has a rebuttal phase).

12. Gianluca Bardaro, Mohamed El-Shamouly, Giulio Fontana, Ramez Awad, Matteo Matteucci. “Toward model-based benchmarking of robot components”. In Proceedings of 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 1682-1687, IEEE, Macau, China, 2019. [doi:10.1109/IROS40897.2019.8967682](https://doi.org/10.1109/IROS40897.2019.8967682) **GII-GRIN-SCIE Class 1 (A+)**

I coordinated the whole research work this paper originated from, I co-invented the original idea, I supervised the PhD student developing and evaluating the system, I participated in the writing and reviewing of the paper.

13. Andrea Romanoni, Daniele Fiorenti, Matteo Matteucci. “Mesh-based 3D Textured Urban Mapping”. In Proceedings of 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 3460-3466, IEEE, Vancouver, Canada, 2017. [doi:10.1109/IROS.2017.8206186](https://doi.org/10.1109/IROS.2017.8206186) **GII-GRIN-SCIE Class 1 (A+)**

I co-supervised, together with the PhD student, the Master student doing the implementation of the method, I co-invented the original idea, I participated in the writing and reviewing of the paper.

14. G. Postica, A. Romanoni, M. Matteucci. “Robust Moving Objects Detection in Lidar Data Exploiting Visual Cues”. In Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 1093-1098, Daejeon, Korea, 2016. [doi:10.1109/IROS.2016.7759185](https://doi.org/10.1109/IROS.2016.7759185) **GII-GRIN-SCIE Class 1 (A+)**

I co-supervised, together with the PhD student, the Master student doing the implementation of the method, I co-invented the original idea, I participated in the writing and reviewing of the paper.

15. Andrea Romanoni, Matteo Matteucci. “Reconstruction of Urban Environments by Edge-Points Delaunay Triangulation”. Proceedings of the 2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 4473-4479, Hamburg, IEEE, 2015. [doi:10.1109/IROS.2015.7354012](https://doi.org/10.1109/IROS.2015.7354012) **GII-GRIN-SCIE Class 1 (A+)**

I supervised the PhD student doing the implementation of the method, I co-invented the original idea, I participated to the writing and reviewing of the paper.

LIST OF FIVE PEERS WHO COULD PROVIDE A REFERENCE LETTER

- **Prof. Daniele Nardi**, Full Professor of Artificial Intelligence at Sapienza Università di Roma, Facoltà di Ingegneria dell'Informazione, Informatica e Statistica, nardi@dis.uniroma1.it, Tel. +39-06-77274113 (int. 35113)
- **Prof. Pedro Lima**, Full Professor of Electrical and Computer Engineering at the Instituto Superior Técnico (IST) - Universidade de Lisboa (IST/UL), Instituto de Sistemas e Robótica, Instituto Superior Técnico, Lisboa, pedro.lima@tecnico.ulisboa.pt, Tel.: +351-218418274
- **Prof. Alessandro Saffiotti**, Full professor of Computer Science at the University of Örebro, Sweden, head of the AASS Cognitive Robotic Systems Laboratory, alessandro.saffiotti@oru.se, Tel.+46 19 303794

- **Prof. Herman Bruyninckx**, Full Professor (“Gewoon Hoogleraar”) at KU Leuven, and part time Full Professor at Eindhoven University of Technology, herman.bruyninckx@kuleuven.be, Tel. +32 16 32 80 56
- **Dr. Anne Bajart**, Deputy Head of Unit European Commission DG CONNECT.G2 Interactive Technologies, Digital for Culture and Education, anne.bajart@ec.europa.eu.